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Hybrid manufacturing with modified via-last process

Abstract

Microelectronic assemblies fabricated using hybrid manufacturing with modified via-last process are disclosed. The fabrication approach is based on using hybrid manufacturing to bond first and second IC structures originally provided on different dies but filling at least portions of vias that are supposed to couple across a bonding interface between the first and second IC structures with electrically conductive materials after the IC structures have been bonded. A resulting microelectronic assembly that includes the first and second IC structures bonded together may have vias extending through all of the first IC structure and into the second IC structure, thus providing electrical coupling between one or more components of the first IC structure and those of the second IC structure, where an electrically conductive material in the individual vias is continuous through the first IC structure and at least a portion of the second IC structure.

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Background/Summary

TECHNICAL FIELD

(1) This disclosure relates generally to the field of semiconductor devices, and more specifically, to microelectronic assemblies, as well as related devices and methods, fabricated using hybrid manufacturing.

BACKGROUND

(2) For the past several decades, the scaling of features in integrated circuits has been a driving force behind an ever-growing semiconductor industry. Scaling to smaller and smaller features enables increased densities of functional units on the limited real estate of semiconductor chips. For example, shrinking transistor size allows for the incorporation of an increased number of memory or logic devices on a chip, lending to the fabrication of products with increased capacity. The drive for the ever-increasing capacity, however, is not without issue. The necessity to optimize the performance of each integrated circuit (IC) die and each IC package that includes one or more dies becomes increasingly significant.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments will be readily understood by the following detailed description in conjunction with the accompanying drawings. To facilitate this description, like reference numerals designate

like structural elements. Embodiments are illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings.

- (2) FIGS. 1A-1D illustrate cross-sectional side views of a microelectronic assembly with a first and a second IC structures bonded using front-to-front (f2f) bonding with modified via-last process, in accordance with some embodiments.
- (3) FIGS. 2A-2D illustrate cross-sectional side views of a microelectronic assembly with a first and a second IC structures bonded using front-to-back (f2b) bonding with modified via-last process, in accordance with some embodiments.
- (4) FIGS. 3A-3D illustrate cross-sectional side views of a microelectronic assembly with a first and a second IC structures bonded using back-to-back (b2b) bonding with modified via-last process, where the first IC structure retains at least a portion of its substrate, but the substrate of the second IC structure is removed, in accordance with some embodiments.
- (5) FIG. 4 illustrates a cross-sectional side view of a microelectronic assembly with a first and a second IC structures bonded using b2b bonding with modified via-last process, where the second IC structure retains at least a portion of its substrate, but the substrate of the first IC structure is removed, in accordance with some embodiments.
- (6) FIG. 5 illustrates a cross-sectional side view of a microelectronic assembly with a first and a second IC structures bonded using b2b bonding with modified via-last process, where each of the first IC structure and the second IC structure retains at least a portion of its substrate, in accordance with some embodiments.
- (7) FIG. 6 illustrates a cross-sectional side view of a microelectronic assembly with a first and a second IC structures bonded using b2b bonding with modified via-last process, where substrates of both the first IC structure and the second IC structure are removed, in accordance with some embodiments.
- (8) FIGS. 7A-7I illustrate cross-sectional side views during a first method for fabricating a microelectronic assembly with a first and a second IC structures bonded using f2f bonding with modified via-last process, in accordance with some embodiments.
- (9) FIGS. 8A-8I illustrate cross-sectional side views during a second method for fabricating a microelectronic assembly with a first and a second IC structures bonded using f2f bonding with modified via-last process, in accordance with some embodiments.
- (10) FIGS. 9A-9C illustrate cross-sectional side views of a microelectronic assembly that is an f2f-bonded assembly with modified via-last process and with an after-bonding via formed from the top, in accordance with some embodiments.
- (11) FIGS. 10A-10C illustrate cross-sectional side views of a microelectronic assembly that is an f2b-bonded assembly with modified via-last process and with an after-bonding via formed from the top, in accordance with some embodiments.
- (12) FIGS. 11A-11C illustrate cross-sectional side views of a microelectronic assembly that is an b2b-bonded assembly with modified via-last process and with an after-bonding via formed from the top, in accordance with some embodiments.
- (13) FIGS. 12A-12H illustrate IC packages that may include one or more microelectronic assemblies fabricated using hybrid manufacturing with modified via-last process, in accordance with some embodiments.
- (14) FIGS. 13A and 13B are top views of, respectively, a wafer and dies that may include one or more microelectronic assemblies fabricated using hybrid manufacturing with modified via-last process, in accordance with various embodiments.
- (15) FIG. 14 is a cross-sectional side view of an IC device that may be included in a microelectronic assembly fabricated using hybrid manufacturing with modified via-last process in accordance with any of the embodiments disclosed herein.
- (16) FIG. 15 is a cross-sectional side view of an IC device assembly that may include one or more microelectronic assemblies fabricated using hybrid manufacturing with modified via-last process,

in accordance with various embodiments.

(17) FIG. 16 is a block diagram of an example computing device that may include one or more microelectronic assemblies fabricated using hybrid manufacturing with modified via-last process or IC devices or packages with such microelectronic assemblies, in accordance with various embodiments.

DETAILED DESCRIPTION

(18) Overview

(19) The systems, methods and devices of this disclosure each have several innovative aspects, no single one of which is solely responsible for all desirable attributes disclosed herein. Details of one or more implementations of the subject matter described in this specification are set forth in the description below and the accompanying drawings.

(20) For purposes of illustrating hybrid manufacturing with modified via-last process for IC devices and assemblies as described herein it might be useful to first understand phenomena that may come into play during IC fabrication. The following foundational information may be viewed as a basis from which the present disclosure may be properly explained. Such information is offered for purposes of explanation only and, accordingly, should not be construed in any way to limit the broad scope of the present disclosure and its potential applications.

(21) IC dies are conventionally coupled to a package substrate for mechanical stability and to facilitate connection to other components, such as circuit boards. The performance that can be realized by such conventional IC packages is constrained by the performance of the die, manufacturing, materials, and thermal consideration, among others. A given die typically is configured for certain functionality and has certain performance characteristics. Enhancements in functionality or performance of a single die may be achieved by including multiple dies in an IC package. However, routing large numbers of signals between two or more dies in a multi-die IC package, typically through a package substrate or a circuit board, is challenging due to the increasingly small size of such dies, thermal constraints, and power delivery constraints, among others. Therefore, recently, hybrid manufacturing approaches have been explored as an alternative, where the term “hybrid manufacturing” refers to fabricating a microelectronic assembly by bonding together two or more different dies (or, more specifically, two or more IC structures that have been fabricated on different dies), thus enhancing functionality and/or performance compared to a single die, while reducing or eliminating the need to route signals between these dies through a package substrate or a circuit board. Advantageously, hybrid manufacturing approach may be used to combine/bond dies that may be fabricated by different manufacturers, using different materials, or different manufacturing techniques.

(22) While hybrid manufacturing approach has the potential to dramatically improve functionality and performance of microelectronic assemblies, inventors of the present disclosure realized that there are some challenges that need to be addressed. One challenge relates to electrically coupling interconnects (e.g., electrically conductive vias or trenches) of an IC structure included in/on a first die to interconnects of an IC structure includes in/on a second die when these two dies are hybrid-bonded together. In particular, achieving adequate electrical coupling (e.g., coupling with sufficiently low resistance) between the interconnects of two different dies in a way that does not cause delamination of the dies or other electrical problems over the lifetime of the assembly is not trivial. As the foregoing illustrates, further improvements are desirable in order to fully realize the potential of hybrid manufacturing.

(23) Embodiments of the present disclosure relate to microelectronic assemblies fabricated using an approach that may be described to as “hybrid manufacturing with modified via-last process,” as well as related devices and methods. In the following, hybrid manufacturing with modified via-last process is described with reference to forming a microelectronic assembly by bonding a first IC structure (i.e., an IC structure fabricated on/in a first die) and a second IC structure (i.e., an IC structure fabricated on/in a second die). The approach is based on using hybrid manufacturing to

bond a first and a second IC structures originally provided on different dies (hence, the approach may be referred to as “hybrid manufacturing”), but at least portions of vias that are supposed to couple across a bonding interface between the first and second IC structures with electrically conductive materials after the IC structures have been bonded (hence, the approach may be referred to as a “via-last process”). A resulting microelectronic assembly that includes the first and second IC structures bonded together may have vias extending through all of the first IC structure and into the second IC structure, thus providing electrical coupling between one or more IC components of the first IC structure (the components including, e.g., any of one or more of interconnects, transistors, resistors, capacitors, and any other IC devices and circuits of the first IC structure) and one or more IC components of the second IC structure (the components including, e.g., any of one or more of interconnects, transistors, resistors, capacitors, and any other IC devices and circuits of the second IC structure), where an electrically conductive material in the individual vias is continuous through substantially all of the first IC structure and at least a portion of the second IC structure. Other features characteristic of the use of hybrid manufacturing with modified via-last process as described herein are possible, which characteristic features may depend on the orientation of the first and second IC structures with respect to one another, locations and manner of applying a bonding material for bonding the first and second IC structures, or whether a barrier liner is present and at which point in the fabrication process the barrier liner in the via extending through the first IC structure and into the second IC structure is provided. Such characteristic features will be described in greater detail below. Hybrid manufacturing with modified via-last process as described herein may provide improvements in terms of electrical coupling between interconnects of the first and second IC structures and/or in terms of stability of the bonding over time (e.g., in terms of reducing the risk of delamination).

(24) For purposes of explanation, specific numbers, materials and configurations are set forth in order to provide a thorough understanding of the illustrative implementations. However, it will be apparent to one skilled in the art that the present disclosure may be practiced without the specific details or/and that the present disclosure may be practiced with only some of the described aspects. In other instances, well known features are omitted or simplified in order not to obscure the illustrative implementations.

(25) Further, references are made to the accompanying drawings that form a part hereof, and in which is shown, by way of illustration, embodiments that may be practiced. It is to be understood that other embodiments may be utilized, and structural or logical changes may be made without departing from the scope of the present disclosure. Therefore, the following detailed description is not to be taken in a limiting sense. For convenience, if a collection of drawings designated with different letters are present, e.g., FIGS. 1A-1D, such a collection may be referred to herein without the letters, e.g., as “FIG. 1.”

(26) In the drawings, some schematic illustrations of example structures of various devices and assemblies described herein may be shown with precise right angles and straight lines, this is simply for ease of illustration, and embodiments of these assemblies may be curved, rounded, or otherwise irregularly shaped as dictated by, and sometimes inevitable due to, the manufacturing processes used to fabricate semiconductor device assemblies. Therefore, it is to be understood that such schematic illustrations may not reflect real-life process limitations which may cause the features to not look so “ideal” when any of the structures described herein are examined using e.g., scanning electron microscopy (SEM) images or transmission electron microscope (TEM) images. In such images of real structures, possible processing defects could also be visible, e.g., not-perfectly straight edges of materials, tapered vias or other openings, inadvertent rounding of corners or variations in thicknesses of different material layers, occasional screw, edge, or combination dislocations within the crystalline region, and/or occasional dislocation defects of single atoms or clusters of atoms. There may be other defects not listed here but that are common within the field of device fabrication. Furthermore, although a certain number of a given element

may be illustrated in some of the drawings (e.g., a certain number and type of interconnects in the first and second IC structures of the microelectronic assemblies illustrated in some drawings, a certain number of dies in the IC packages illustrated in other drawings, etc.), this is simply for ease of illustration, and more, or less, than that number may be included in microelectronic assemblies and related devices according to various embodiments of the present disclosure. Still further, various views shown in some of the drawings are intended to show relative arrangements of various elements therein. In other embodiments, various microelectronic assemblies fabricated using hybrid manufacturing with modified via-last process, or portions thereof, may include other elements or components that are not illustrated (e.g., transistor portions, various components that may be in electrical contact with any of the illustrated components of the microelectronic assemblies, etc.). Inspection of layout and mask data and reverse engineering of parts of a device to reconstruct the circuit using e.g., optical microscopy, TEM, or SEM, and/or inspection of a cross-section of a device to detect the shape and the location of various device elements described herein using e.g., physical failure analysis (PFA) would allow determination of presence of one or more microelectronic assemblies fabricated using hybrid manufacturing with modified via-last process as described herein.

(27) Various operations may be described as multiple discrete actions or operations in turn, in a manner that is most helpful in understanding the claimed subject matter. However, the order of description should not be construed as to imply that these operations are necessarily order dependent. These operations may not be performed in the order of presentation. Operations described may be performed in a different order from the described embodiment. Various additional operations may be performed, and/or described operations may be omitted in additional embodiments.

(28) For the purposes of the present disclosure, the phrase “A and/or B” means (A), (B), or (A and B). For the purposes of the present disclosure, the phrase “A, B, and/or C” means (A), (B), (C), (A and B), (A and C), (B and C), or (A, B, and C). The term “between,” when used with reference to measurement ranges, is inclusive of the ends of the measurement ranges. Although certain elements may be referred to in the singular herein, such elements may include multiple sub-elements. For example, “an electrically conductive material” may include one or more electrically conductive materials.

(29) The description uses the phrases “in an embodiment” or “in embodiments,” which may each refer to one or more of the same or different embodiments. The terms “comprising,” “including,” “having,” and the like, as used with respect to embodiments of the present disclosure, are synonymous. The disclosure may use perspective-based descriptions such as “above,” “below,” “top,” “bottom,” and “side” to explain various features of the drawings, but these terms are simply for ease of discussion, and do not imply a desired or required orientation. The accompanying drawings are not necessarily drawn to scale. Unless otherwise specified, the use of the ordinal adjectives “first,” “second,” and “third,” etc., to describe a common object, merely indicate that different instances of like objects are being referred to, and are not intended to imply that the objects so described must be in a given sequence, either temporally, spatially, in ranking or in any other manner.

(30) In the following detailed description, various aspects of the illustrative implementations will be described using terms commonly employed by those skilled in the art to convey the substance of their work to others skilled in the art.

(31) For example, the term “interconnect” may be used to describe any element formed of an electrically conductive material for providing electrical connectivity to one or more components associated with an IC or/and between various such components. In general, the “interconnect” may refer to both trenches (also sometimes referred to as “lines” or “conductive traces”) and conductive vias. In general, in context of interconnects, the term “trench” may be used to describe an electrically conductive element isolated by an insulator material (e.g., a low-k dielectric material)

that is provided within the plane of an IC die. Trenches are typically stacked into several levels, or several layers, of metallization stacks. On the other hand, the term “via” may be used to describe an electrically conductive element that interconnects two or more trenches of different levels. To that end, a via may be provided substantially perpendicularly to the plane of an IC die and may interconnect two trenches in adjacent levels or two trenches in not adjacent levels. A term “metallization stack” may be used to refer to a stack of one or more interconnects for providing connectivity to different circuit components of an IC chip. Sometimes, trenches and vias may be referred to as “conductive trenches” and “conductive vias”, respectively, to highlight the fact that these elements include electrically conductive materials such as metals.

(32) Interconnects as described herein, in particular interconnects of the IC structures of microelectronic assemblies fabricated using hybrid manufacturing with modified via-last process as described herein, may be used for providing electrical connectivity to one or more components associated with an IC or/and between various such components, where, in various embodiments, components associated with an IC or IC structure described herein may include, for example, transistors, diodes, power sources, resistors, capacitors, inductors, sensors, transceivers, receivers, antennas, etc. Components associated with an IC may include those that are mounted on IC or those connected to an IC. The IC may be either analog or digital and may be used in a number of applications, such as microprocessors, optoelectronics, logic blocks, audio amplifiers, etc., depending on the components associated with the IC. The IC may be employed as part of a chipset for executing one or more related functions in a computer.

(33) In another example, the terms “package” and “IC package” are synonymous, as are the terms “die” and “IC die,” the term “insulating” means “electrically insulating,” the term “conducting” means “electrically conducting,” unless otherwise specified. Furthermore, the term “connected” may be used to describe a direct electrical or magnetic connection between the things that are connected, without any intermediary devices, while the term “coupled” may be used to describe either a direct electrical or magnetic connection between the things that are connected, or an indirect connection through one or more passive or active intermediary devices. The term “circuit” may be used to describe one or more passive and/or active components that are arranged to cooperate with one another to provide a desired function.

(34) In yet another example, if used, the terms “oxide,” “carbide,” “nitride,” etc. refer to compounds containing, respectively, oxygen, carbon, nitrogen, etc., the term “high-k dielectric” refers to a material having a higher dielectric constant than silicon oxide, while the term “low-k dielectric” refers to a material having a lower dielectric constant than silicon oxide.

(35) The terms “substantially,” “close,” “approximately,” “near,” and “about,” generally refer to being within $\pm 20\%$ of a target value based on the context of a particular value as described herein or as known in the art. Similarly, terms indicating orientation of various elements, e.g., “coplanar,” “perpendicular,” “orthogonal,” “parallel,” or any other angle between the elements, generally refer to being within $\pm 5\text{-}20\%$ of a target value based on the context of a particular value as described herein or as known in the art.

(36) Example Microelectronic Assemblies and Methods of Manufacturing

(37) In general, hybrid manufacturing with modified via-last process is described herein with reference to a first IC structure **110** and a second IC structure **120** bonded to one another using a bonding material **130**. In some embodiments, the IC structures **110** and **120** may be fabricated by different manufacturers, using different materials, or different manufacturing techniques. In other embodiments, the IC structures **110** and **120** may be fabricated by the same manufacturers, or using the same materials, or using the same manufacturing techniques, but in both sets of these embodiments, the IC structure **110** is fabricated on a first die and the IC structure **120** is fabricated on a second die, separate from the first die, after which these two IC structures are bonded as described herein. Various examples of microelectronic assemblies that include the IC structures **110** and **120**, and methods of manufacturing thereof, will now be described with reference to FIGS. 1-

11.

(38) A number of elements referred to in the description of FIGS. **1-11** with reference numerals are illustrated in these figures with different patterns, with a legend showing the correspondence between the reference numerals and patterns being provided at the bottom of each drawing page containing FIGS. **1-11**. For example, the legend illustrates that FIGS. **1-11** use different patterns to show first interconnects **112**, second interconnects **122**, a bonding material **130**, etc.

(39) A number of elements are illustrated in FIGS. **1-11** as included in various embodiments of the microelectronic assembly **100**, but a number of these elements may not be present in other embodiments of the microelectronic assembly **100**. For example, while FIGS. **1** and **2** illustrate a support structure **132** of the first IC structure **110**, in other embodiments, the support structure **132** may not be included. Further, unless stated otherwise, any features of any embodiments of the microelectronic assemblies described with reference to one of the FIGS. **1-11** may be combined with any features of any embodiments of the microelectronic assemblies described with reference to another one or more of the FIGS. **1-11**. In some embodiments, individual ones of the microelectronic assemblies **100** disclosed herein may serve as a system-in-package (SiP) in which multiple IC structures **110**, **120**, or multiple pairs of such IC structures, having different functionality are included. In such embodiments, the microelectronic assembly **100** may be referred to as a SiP.

(40) FIGS. **1A-1D** illustrate cross-sectional side views of a microelectronic assembly **100** with a first IC structure **110** and a second IC structure **120** bonded using f2f bonding with modified via-last process, in accordance with some embodiments. In particular FIGS. **1A-1D** show different embodiments of the microelectronic assembly **100** with f2f bonding with modified via-last process, with the differences resulting from using four different fabrication methods of f2f hybrid manufacturing with modified via-last process. The cross-sections of FIGS. **1A-1D** are cross-sections in a plane that is substantially perpendicular to a bonding interface between the first and second IC structures **110**, **120** (e.g., substantially perpendicular to the plane of a bonding material **130** which is also the plane of the bonding interface).

(41) FIG. **1A** illustrates a cross-sectional side view of a microelectronic assembly **100** according to embodiments where f2f bonding is used according to a first fabrication method of hybrid manufacturing with via-last process (e.g., the method illustrated in FIGS. **7A-7I**). Many of the elements of the microelectronic assembly **100** of FIG. **1A** are included in other ones of the accompanying figures; the discussion of these elements is not repeated when discussing these figures, and any of these elements may take any of the forms disclosed herein.

(42) As shown in FIG. **1A**, the IC structure **110** may include one or more, typically a plurality, of interconnects **112**, while the IC structure **110** may include one or more, typically a plurality, of interconnects **122**. In the IC structure **110**, the interconnects **112** may be arranged in one or more, typically a plurality, of layers of a metallization stack, where each layer may include an insulating material **114** (e.g., a dielectric material formed in multiple layers, as known in the art). The interconnects **112** may include one or more conductive trenches and conductive vias, providing one or more conductive pathways through the insulating material **114**. Similarly, in the IC structure **120**, the interconnects **122** may be arranged in one or more, typically a plurality, of layers of a metallization stack, where each layer may include an insulating material **124** (e.g., a dielectric material formed in multiple layers, as known in the art). The interconnects **122** may include one or more conductive trenches and conductive vias, providing one or more conductive pathways through the insulating material **124**. Any of the conductive pathways disclosed herein (e.g., conductive trenches and/or conductive vias) may be formed of any appropriate conductive material, such as copper, tungsten, silver, nickel, gold, aluminum, or other metals or alloys, for example.

(43) In some embodiments, at least one of the insulating material **114** and the insulating material **124** may include a dielectric material, such as silicon dioxide, silicon nitride, oxynitride, polyimide materials, glass reinforced epoxy matrix materials, or a low-k or ultra-low-k dielectric (e.g.,

carbon-doped dielectrics, fluorine-doped dielectrics, porous dielectrics, organic polymeric dielectrics, photo-imageable dielectrics, and/or benzocyclobutene-based polymers). In some embodiments, at least one of the insulating materials **114** and the insulating material **124** may include a semiconductor material, such as silicon, germanium, or a III-V material (e.g., gallium nitride), and one or more additional materials. For example, at least one of the insulating materials **114** and the insulating material **124** may include silicon oxide or silicon nitride. The conductive pathways formed by the interconnects **112** may include conductive trenches and/or conductive vias and may be connected to one another in any suitable manner. Similarly, the conductive pathways formed by the interconnects **122** may include conductive trenches and/or conductive vias and may be connected to one another in any suitable manner. Although FIGS. **1-11** illustrate a specific number and arrangement of conductive pathways formed by the interconnects **112** and by the interconnects **122**, these are simply illustrative, and any suitable number and arrangement may be used.

(44) In some embodiments, any of the interconnects **112** and the interconnects **122** may include conductive pathways to route power, ground, and/or signals to/from various components of the IC structure **110** and the IC structure **120**. In general, any of the interconnects **112** and the interconnects **122** may include conductive pathways in the form of through-substrate vias (TSVs), blind vias, trenches, or other electrically conductive structures, at least partially filled with one or more electrically conductive materials, and isolated from the surrounding silicon or other semiconductor materials or from other conductive pathways by the insulating materials **114**, **124**. For example, any of the IC structure **110** and the IC structure **120** may include TSVs through which power, ground, and/or signals may be transmitted between a package substrate or a circuit board to which the microelectronic assembly **100** may be coupled (e.g., a package substrate **202**, shown in FIG. **12**) and one or more dies or IC structures “on top” or “at the bottom of” the microelectronic assembly **100**. For example, if an IC package is such that the IC structure **110** is at the bottom and is coupled to a package substrate and the IC structure **120** is at the top, then the IC structure **110** may include conductive pathways to route power, ground, and/or signals between the package substrate and the IC structure **120** and/or between the package substrate and different ones of the dies or IC structures that may be “on top” of the IC structure **120**. In another example, if an IC package is such that the IC structure **120** is at the bottom and is coupled to a package substrate and the IC structure **110** is at the top (i.e., the microelectronic assembly **100** is flipped upside down compared to what is shown in FIG. **1A**), then the IC structure **120** may include conductive pathways to route power, ground, and/or signals between the package substrate and the IC structure **110** and/or between the package substrate and different ones of the dies or IC structures that may be “on top” of the IC structure **110**. In some embodiments, the IC structure **110** or the IC structure **120** may be the source and/or destination of signals communicated between the package substrate and the other IC structure and/or other dies or IC structures included in an IC package. In some embodiments, the IC structure **110** may not route power and/or ground to the IC structure **120**; instead, IC structure **120** may couple directly to power and/or ground lines in a package substrate of an IC package. By allowing the IC structure **120** to couple directly to power and/or ground lines in the package substrate, such power and/or ground lines need not be routed through the IC structure **110**, allowing the IC structure **110** to be made smaller or to include more active circuitry or signal pathways.

(45) In some embodiments, any of the IC structure **110** and IC structure **120** may only include conductive pathways formed by the interconnects **112**, **114**, respectively, and may not contain active or passive circuitry. In other embodiments, any of the IC structure **110** and IC structure **120** may include active or passive circuitry (e.g., transistors, diodes, resistors, inductors, and capacitors, among others), schematically illustrated in the example of FIG. **1A** as device circuitry **116** that may be included in the IC structure **110** and device circuitry **126** that may be included in the IC structure **120**. For example, if an IC package is such that the IC structure **110** is at the bottom and is coupled

to a package substrate and the IC structure **120** is at the top and include the device circuitry **126**, power and/or ground signals may be routed from/through the package substrate and to the device circuitry **126** of the IC structure **120** through the interconnects **112** and the interconnects **122**. In various such embodiments, the IC structure **110** may or may not include the device circuitry **116**. In another example, if an IC package is such that the IC structure **120** is at the bottom and is coupled to a package substrate and the IC structure **110** is at the top and include the device circuitry **116**, power and/or ground signals may be routed from/through the package substrate and to the device circuitry **116** of the IC structure **110** through the interconnects **122** and the interconnects **112**. In various such embodiments, the IC structure **120** may or may not include the device circuitry **126**. In general, the microelectronic assembly **100** may include the device circuitry **116** in some embodiments while exclude it in others and, similarly, the microelectronic assembly **100** may include the device circuitry **126** in some embodiments while exclude it in others.

(46) The device circuitry **116** and **126** may include semiconductor material systems including, for example, N-type or P-type materials systems, as active materials (e.g., as channel materials of transistors).

(47) In some embodiments, active materials of the device circuitry **116** and/or **126** may include substantially monocrystalline semiconductors, such as silicon or germanium.

(48) In some embodiments, active materials of the device circuitry **116** and/or **126** may include compound semiconductors, e.g., compound semiconductors with a first sub-lattice of at least one element from group III of the periodic table (e.g., Al, Ga, In), and a second sub-lattice of at least one element of group V of the periodic table (e.g., P, As, Sb). In some embodiments, active materials of the device circuitry **116** and/or **126** may include a binary, ternary, or quaternary III-V compound semiconductor that is an alloy of two, three, or even four elements from groups III and V of the periodic table, including boron, aluminum, indium, gallium, nitrogen, arsenic, phosphorus, antimony, and bismuth.

(49) In some embodiments, active materials of the device circuitry **116** and/or **126** may be/include an intrinsic IV or III-V semiconductor material or alloy, not intentionally doped with any electrically active impurity. In alternate embodiments, nominal impurity dopant levels may be present within the active materials of the device circuitry **116** and/or **126**, for example to set a threshold voltage V_t , or to provide halo pocket implants, etc. In such impurity-doped embodiments however, impurity dopant level within the active materials may be relatively low, for example below about 10^{15} cm⁻³, and advantageously below 10^{13} cm⁻³.

(50) For exemplary P-type transistor embodiments, active materials of the device circuitry **116** and/or **126** may advantageously be group IV materials having a high hole mobility, such as, but not limited to, Ge or a Ge-rich SiGe alloy. For some exemplary embodiments, such active materials may have a Ge content between 0.6 and 0.9, and advantageously is at least 0.7.

(51) For exemplary N-type transistor embodiments, active materials of the device circuitry **116** and/or **126** may advantageously be a III-V material having a high electron mobility, such as, but not limited to InGaAs, InP, InSb, and InAs. For some such embodiments, the active material may be a ternary III-V alloy, such as InGaAs or GaAsSb. For some In_xGa_{1-x}As fin embodiments, In content in the such active material may be between 0.6 and 0.9, and advantageously at least 0.7 (e.g., In_{0.7}Ga_{0.3}As).

(52) In some embodiments, active materials of the device circuitry **116** and/or **126** may be thin-film materials, in which embodiments the devices **116** and/or **126** could be thin-film transistors (TFTs). A TFT is a special kind of a field-effect transistor (FET), made by depositing a thin film of an active semiconductor material, as well as a dielectric layer and metallic contacts, over a support structure that may be a non-conducting (and non-semiconducting) support structure. During operation of a TFT, at least a portion of the active semiconductor material forms a channel of the TFT, and, therefore, the thin film of such active semiconductor material is referred to herein as a "TFT channel material." This is different from conventional, non-TFT, transistors where the active

semiconductor channel material is typically a part of a semiconductor substrate, e.g., a part of a silicon wafer. In various such embodiments, active materials of the device circuitry **116** and/or **126** may include a high mobility oxide semiconductor material, such as tin oxide, antimony oxide, indium oxide, indium tin oxide, titanium oxide, zinc oxide, indium zinc oxide, indium gallium zinc oxide (IGZO), gallium oxide, titanium oxynitride, ruthenium oxide, or tungsten oxide.

(53) In some embodiments, active materials of the device circuitry **116** and/or **126** may include one or more of tin oxide, cobalt oxide, copper oxide, antimony oxide, ruthenium oxide, tungsten oxide, zinc oxide, gallium oxide, titanium oxide, indium oxide, titanium oxynitride, indium tin oxide, indium zinc oxide, nickel oxide, niobium oxide, copper peroxide, IGZO, indium telluride, molybdenite, molybdenum diselenide, tungsten diselenide, tungsten disulfide, N- or P-type amorphous or polycrystalline silicon, germanium, indium gallium arsenide, silicon germanium, gallium nitride, aluminum gallium nitride, indium phosphide, and black phosphorus, each of which may possibly be doped with one or more of gallium, indium, aluminum, fluorine, boron, phosphorus, arsenic, nitrogen, tantalum, tungsten, and magnesium, etc.

(54) Although FIG. **1A** illustrates the device circuitry **116** and **126** as front end of line (FEOL, or simply “frontend”) devices in the IC structures **110**, **120**, respectively, in general, the device circuitry **116** and/or **126** may include any combination of FEOL devices, back end of line (BEOL) devices (also referred to as “backend devices”), TFTs, etc.

(55) FIG. **1A** illustrates that the device circuitry **116** may be provided over a support structure **132**, so that the device circuitry **116** is between the support structure **132** and at least some of the interconnects **112**. In general, the support structure **132** may include any material that may serve as a foundation upon which the IC structures are described herein may be formed. In some embodiments, the support structure **132** may be a semiconductor substrate, composed of semiconductor material systems including, for example, N-type or P-type materials systems. In one implementation, the semiconductor substrate may be a crystalline substrate formed using a bulk silicon or a silicon-on-insulator substructure. In other implementations, the semiconductor substrate may be formed using alternate materials, which may or may not be combined with silicon, that include but are not limited to germanium, indium antimonide, lead telluride, indium arsenide, indium phosphide, gallium arsenide, indium gallium arsenide, gallium antimonide, or other combinations of group III-V, group II-VI, or group IV materials.

(56) Although not shown in FIG. **1A**, the device circuitry **126** may also be provided over a support structure similar to the support structure **132**, so that the device circuitry **126** is between such a support structure and at least some of the interconnects **122**. FIG. **1A** does not illustrate a support structure of the IC structure **120** in order to illustrate that, in some embodiments, after the IC structure **120** has been bonded, in an f2f bonding, with the IC structure **130**, the support structure of the IC structure **120** may no longer be needed (e.g., because the support structure **132** may provide sufficient mechanical stability to the microelectronic assembly **100**) and may be removed, e.g., using a suitable thinning/polishing process as described with reference to FIG. **7E**. Such embodiments may advantageously decrease the overall height of the microelectronic assembly **100** (i.e., the dimension measured along a vertical axis for the examples shown in the present drawing). In other embodiments, the IC structure **120** may be provided over the IC structure **110** using layer transfer, in which case as the IC structure **120** is placed over the IC structure **110** for bonding, its' support structure has already been substantially removed.

(57) Although not specifically shown in the present drawings, in some embodiments, one or both of the IC structures **110**, **120** may include, or be a part of, one or more of a central processing unit, a memory device, e.g., a high-bandwidth memory device, a logic circuit, input/output circuitry, a transceiver such as a field programmable gate array transceiver, a gate array logic such as a field programmable gate array logic, of a power delivery circuitry, a III-V or a III-N device such as a III-N or III-N amplifier (e.g., GaN amplifier), Peripheral Component Interconnect Express circuitry, or Double Data Rate transfer circuitry.

(58) Furthermore, although also not specifically shown in the present drawings, in some embodiments, any of the IC structures described herein (e.g., the first IC structure **110** or the second IC structure **120**) may include backend devices, e.g., backend memory cells, backend logic transistors, etc., implemented in the BEOL layer, above the frontend devices of the FEOL layer. For example, the backend memory cells may be embedded dynamic random-access memory (DRAM) cells, because this type of memory has been introduced in the past to address the limitation in density and standby power of other types of memory. However, embodiments of the present disclosure are equally applicable to the first IC structure **110** or the second IC structure **120** including backed memory implemented using other technologies. Thus, in general, backend memory described herein may be implemented as eDRAM cells, spin-transfer torque random-access memory (STTRAM) cells, cross-point memory, NAND memory, static random-access memory (SRAM), resistive switching memory, or any other memory types.

(59) DRAM and in particular, embedded DRAM (eDRAM), has been introduced in the past to address the limitation in density and standby power of other types of memory. As an example, a DRAM cell may include a capacitor for storing a bit value, or a memory state (e.g., logical “1” or “0”) of the cell, and an access transistor controlling access to the cell (e.g., access to write information to the cell or access to read information from the cell). Such a memory cell may be referred to as a “1T-1C memory cell,” highlighting the fact that it uses one transistor (i.e., “1T” in the term “1T-1C memory cell”) and one capacitor (i.e., “1C” in the term “1T-1C memory cell”). The capacitor of a 1T-1C memory cell may be coupled to one S/D region of the access transistor (e.g., to the source region of the access transistor), while the other S/D region of the access transistor (e.g., to the drain region) may be coupled to a bit-line (BL), and a gate terminal of the transistor may be coupled to a word-line (WL). Since such a memory cell can be fabricated with as little as a single access transistor, it can provide higher density and lower standby power versus some other types of memory in the same process technology.

(60) Implementing memory in the backend of an IC structure allows significantly increasing density of memory cells in a memory array having a given footprint area (the footprint area being defined as an area in a plane of the substrate, or a plane parallel to the plane of the substrate), or, conversely, allows significantly reducing the footprint area of the memory array with a given memory cell density.

(61) As an example, a DRAM cell may include a capacitor for storing a bit value, or a memory state (e.g., logical “1” or “0”) of the cell, and an access transistor controlling access to the cell (e.g., access to write information to the cell or access to read information from the cell). Such a memory cell may be referred to as a “1T-1C memory cell,” highlighting the fact that it uses one transistor (i.e., “1T” in the term “1T-1C memory cell”) and one capacitor (i.e., “1C” in the term “1T-1C memory cell”). The capacitor of a 1T-1C memory cell may be coupled to one S/D region of the access transistor (e.g., to the source region of the access transistor), while the other S/D region of the access transistor (e.g., to the drain region) may be coupled to a bit-line (BL), and a gate terminal of the transistor may be coupled to a word-line (WL). Since such a memory cell can be fabricated with as little as a single access transistor, it can provide higher density and lower standby power versus some other types of memory in the same process technology.

(62) Backend memory such as backend DRAM may be implemented using TFTs as access transistors of the memory cells embedded in the BEOL layer. Using TFTs as access transistors of memory cells provides several advantages and enables unique architectures that were not possible with conventional, FEOL logic transistors. For example, one advantage is that a TFT may have substantially lower leakage than a logic transistor, allowing to relax the demands on the large capacitance placed on a capacitor of a 1T-1C memory cell. In other words, using a lower leakage TFT in a 1T-1C memory cell allows the memory cell to use a capacitor with lower capacitance and smaller aspect ratio while still meeting the same data retention requirements of other approaches, alleviating the scaling challenges of capacitors.

(63) Additionally, or alternatively, to TFT-based memory, backend memory may be implemented using layer transfer to form access transistors of the memory cells embedded in the BEOL layer. Layer transfer may include epitaxially growing a layer of a highly crystalline semiconductor material on another substrate and then transferring the layer, or a portion thereof, to embed it in the BEOL layer provided over a second substrate. Channel regions of backend transistors then include at least portions of such transferred semiconductor material layer. Performing layer transfer may advantageously allow forming non-planar transistors, such as FinFETs, nanowire transistors, or nanoribbon transistors, in the BEOL layer. In some embodiments, transistors, or portions thereof (e.g., S/D regions) may be formed on the first substrate (i.e., on the substrate on which a layer of a highly crystalline semiconductor material is grown) before the layer transfer takes place, and then a layer with such transistors, or portions thereof, is transferred.

(64) Layer transfer approach for providing backend memory may be particularly suitable for forming access transistors with channel regions formed of substantially single-crystalline semiconductor materials. On the other hand, TFT-based backend memory may be seen as an example of a monolithic integration approach because the semiconductor materials for the channel regions are deposited in a BEOL layer of an IC device, as opposed to being epitaxially grown elsewhere and then transferred, which may be particularly suitable for forming access transistors with channels formed of polycrystalline, polymorphous, or amorphous semiconductor materials, or various other thin-film channel materials. Whether a semiconductor material of a channel region for a given backend device (e.g., a backend transistor) has been provided by monolithic integration approach or by layer transfer can be identified by inspecting grain size of active semiconductor material of the device (e.g., of the semiconductor material of the channel region of a backend transistor). An average grain size of the semiconductor material being between about 0.5 and 1 millimeters (in which case the material may be considered to be polycrystalline) or smaller than about 0.5 millimeter (in which case the material may be considered to be polymorphous) may be indicative of the semiconductor material having been deposited in the BEOL layer of the device (i.e., monolithic integration approach), e.g., to form a TFT. On the other hand, an average grain size of the semiconductor material being equal to or greater than about 1 millimeter (in which case the material may be considered to be a single-crystal material) may be indicative of the semiconductor material having been included in the BEOL layer of the device by layer transfer. The discussions of monolithic integration vs. layer transfer approaches for forming backend memory are equally applicable to backend transistors that are not part of a memory array (e.g., if backend transistors are implemented in an IC device to serve as logic transistors, switches, or for any other purposes or in any other circuits).

(65) Moving access transistors to the BEOL layer of an advanced complementary metal oxide semiconductor (CMOS) process, either by monolithic integration (e.g., using TFTs) or by layer transfer, means that their corresponding capacitors can be implemented in the upper metal layers with correspondingly thicker interlayer dielectric (ILD) and larger metal pitch to achieve higher capacitance. This eases the integration challenge introduced by embedding the capacitors. Furthermore, when at least some access transistors are implemented as backend transistors, at least portions of different memory cells may be provided in different layers of a BEOL layer above a substrate, thus enabling a stacked architecture of memory arrays.

(66) Although descriptions of the present disclosure may refer to logic devices (e.g., implemented using frontend transistors of a FEOL Layer) or memory cells provided in a given layer of an IC device, each layer of the IC devices described herein may also include other types of devices besides logic or memory devices described herein. For example, in some embodiments, FEOL layers with logic transistors may also include memory cells and/or BEOL layers with memory cells may also include logic transistors. In general, a FEOL layer may include one or more layers, each including frontend components and/or interconnects, and a BEOL layer may include one or more layers, each including backend components (e.g., backend memory) and/or interconnects.

(67) In general, the hybrid manufacturing to form various embodiments of the microelectronic assemblies **100** may be performed as follows (different variations of this approach to realize hybrid manufacturing with modified via-last process will be described in greater detail below). First, the IC structures **110** and **120** may be fabricated individually, e.g., at different manufacturing facilities, by different companies, etc. After that, one face of the IC structure **110** and one face of the IC structure **120** may be bonded. In some embodiments, bonding of the faces of the IC structures **110** and **120** may be performing using insulator-insulator bonding, e.g., as oxide-oxide bonding, where the insulating material **114** is bonded to the insulating material **124**. In some embodiments, the bonding material **130** may be present in between at least portions of the faces of the IC structures **110** and **120** that are bonded together. To that end, the bonding material **130** may be applied to at least portions of the one or both faces of the IC structures **110** and **120** that should be bonded and then the IC structures **110** and **120** are put together, possibly while applying a suitable pressure and heating up the assembly to a suitable temperature (e.g., to moderately high temperatures, e.g., between about 50 and 200 degrees Celsius) for a duration of time. In some embodiments, the bonding material **130** may be an adhesive material that ensures attachment of the IC structures **110** and **120** to one another. In some embodiments, the bonding material **130** may be an etch-stop material. In some embodiments, the bonding material may be both an etch-stop material and have suitable adhesive properties to ensure attachment of the IC structures **110** and **120** to one another. In some embodiments, no bonding material **130** may be used, in which case the layer labeled “**130**” in FIG. 1A and other drawings represents a bonding interface resulting from the bonding of the IC structures **110** and **120** to one another. Such a bonding interface may be recognizable as a seam or a thin layer in the microelectronic assembly **100**, using, e.g., selective area diffraction (SED), even when the specific materials of the insulators of the IC structures **110** and **120** that are bonded together may be the same, in which case the bonding interface would still be noticeable as a seam or a thin layer in what otherwise appears as a bulk insulator (e.g., bulk oxide) layer. As used herein, unless specified otherwise, references to the “bonding material **130**” are applicable to a “bonding interface” for the embodiments where no deliberately added adhesive material is used to bond the IC structures **110** and **120**.

(68) Although not specifically shown in the present drawings, any embodiments of the microelectronic assembly **100** as described herein may further include one or more etch-stop materials that may be included in the IC structure **110**, e.g., between some or all pairs of metal layers of a metallization stack of the IC structure **110**, and/or in the IC structure **120**, e.g., between some or all pairs of metal layers of a metallization stack of the IC structure **120**. Such layers of etch-stop materials are commonly used in the field of semiconductor manufacturing and may be provided at different locations of the IC structures **110**, **120**, the locations being dependent on, e.g., specific processing techniques used to manufacture portions of these IC structures. What is unique about the etch-stop materials in context of hybrid manufacturing is that because the IC structures **110**, **120** may be fabricated by different manufacturers, using different materials, or different manufacturing techniques, the material compositions of their etch-stop materials may be different. For example, the etch-stop material included in the IC structure **110** may include a material with silicon and nitrogen (e.g., silicon nitride), while the etch-stop material included in the IC structure **120** may include a material with silicon and carbon (e.g., silicon carbide), or one of the etch-stop materials included in the IC structures **110**, **120** may include a material with aluminum and oxygen (e.g., aluminum oxide). Furthermore, the bonding material **130** at the interface between the IC structures **110** and **120** may have a material composition different from one or both of the etch-stop material included in the IC structure **110** and the etch-stop material included in the IC structure **120**. For example, in some embodiments, the bonding material **130** may include silicon, nitrogen, and carbon, where the atomic percentage of any of these materials may be at least 1%, e.g., between about 1% and 50%, indicating that these elements are added deliberately, as opposed to being accidental impurities which are typically in concentration below about 0.1%. Having both

nitrogen and carbon in these concentrations in addition to silicon is not typically used in conventional semiconductor manufacturing processes where, typically, either nitrogen or carbon is used in combination with silicon, and, therefore, would be a characteristic feature of the hybrid manufacturing as described herein. Using an etch-stop material at the interface between the IC structures **110** and **120** that includes include silicon, nitrogen, and carbon, where the atomic percentage of any of these materials may be at least 1%, e.g., SiOCN, may be advantageous in terms that such a material may act both as an etch-stop material, and have sufficient adhesive properties to bond the IC structures **110** and **120** together. In addition, an etch-stop material at the interface between the IC structures **110** and **120** that includes include silicon, nitrogen, and carbon, where the atomic percentage of any of these materials may be at least 1%, may be advantageous in terms of improving etch-selectivity of this material with respect to the etch-stop materials of the IC structures **110** and **120**.

(69) For each IC structure, the terms “bottom face” or “backside” of the structure may refer to the back of the IC structure, e.g., bottom of the support structure **132** for the IC structure **110**, while the terms “top face” or “frontside” of the structure may refer to the opposing other face. FIG. **1A** indicates a backside **134-1** and a frontside **134-2** for each of the IC structures **110** and **120**. As can be seen in FIG. **1A**, the microelectronic assembly **100** is an f2f-bonded assembly because the frontside **134-2** of the IC structure **120** is bonded to the frontside **134-2** of the IC structure **110**. Thus, in an f2f-bonded assembly, one of the IC structures **110**, **120** is flipped upside down for bonding so that the top face of the flipped IC structure is facing and is bonded to the top face of the IC structure that is not flipped.

(70) As a result of the hybrid manufacturing of the IC structures **110** and **120**, the microelectronic assembly **100** may exhibit several characteristic features.

(71) One such characteristic feature of the hybrid manufacturing of the IC structures **110** and **120** may be revealed by examining the material composition of the interconnects **112** and **122**. In general, each of the interconnects **112** and **122** may include a liner and an electrically conductive fill material. If included, the liner may be an adhesion liner and/or a barrier liner. FIG. **1A** illustrates an embodiment in which at least some of the interconnects **112** may include an electrically conductive fill material **118** and a liner **119**, while at least some of the interconnects **122** may include an electrically conductive fill material **128** and a liner **129** (insets within the dashed contours on the left side of FIG. **1A** illustrate such examples of one of the interconnects **112** and one of the interconnects **122**). In some such embodiments, the material composition of the liners **119** and **129** may be different. For example, the liner **119** of the interconnects **112** may be a liner having one or more of tantalum, tantalum nitride, titanium nitride, and tungsten carbide, while the liner **129** of the interconnects **122** may be a liner having one or more of tantalum, tantalum nitride, and cobalt, or vice versa. In the liners **119** and **129**, any of the individual materials (e.g., any of the examples listed above) may be included in the amount of between about 1% and 75%, e.g., between about 5% and 50%, indicating that these materials are included by intentional alloying of materials, in contrast to potential accidental doping or impurities being included, which would be less than about 0.1% for any of these metals. Thus, any differences in the material composition of the liners **119** and **129** that are beyond the unintentional doping and/or impurities levels of about 0.1% of a given material could be indicative of the hybrid manufacturing used to provide the microelectronic assembly **100**. Similarly, in some embodiments, material composition of the electrically conductive fill materials **118** and **128** may be different. For example, the electrically conductive fill material **118** of the interconnects **112** may include copper (Cu), while the electrically conductive fill material **128** of the interconnects **122** may include tungsten (W), aluminum (Al), ruthenium (Ru), cobalt (Co), or AlCu (e.g., in proportions of between 1:1 to 1:100), or vice versa. In the electrically conductive fill materials **118** and **128**, any of the individual materials (e.g., any of the examples listed above) may be included in the amount of between about 1% and 75%, e.g., between about 5% and 50%, indicating that these materials are included by

intentional alloying of materials, in contrast to potential accidental doping or impurities being included, which would be less than about 0.1% for any of these metals. Thus, any differences in the material composition of the electrically conductive fill materials **118** and **128** that are beyond the unintentional doping/impurities levels of about 0.1% of a given material could be indicative of the hybrid manufacturing used to provide the microelectronic assembly **100**. In some embodiments, the liner **119** may be excluded in at least some of the interconnects **112**, in which case those interconnects **112** may only include the electrically conductive fill material **118**. In some embodiments, the liner **129** may be excluded in at least some of the interconnects **122**, in which case those interconnects **122** may only include the electrically conductive fill material **128**.

(72) Another characteristic feature of the hybrid manufacturing of the IC structures **110** and **120** may be revealed by examining the geometry of the cross-sections of the interconnects **112** and **122**. For example, in some embodiments, a thickness of the liner **119** may be different from a thickness of the liner **129**, e.g., at least about 5% different, at least about 10% different, or at least 5-50% different. For example, the liner **119** may have a thickness between about 1 and 6 nanometers, including all values and ranges therein, while the liner **129** may have a thickness between about 4 and 10 nanometers, including all values and ranges therein, or vice versa.

(73) Yet another characteristic feature of the hybrid manufacturing of the IC structures **110** and **120** may be revealed by examining the material composition of the insulating materials **114** and **124** in that, in some such embodiments, the material composition of the insulating materials **114** and **124** may be different. For example, any of the individual materials of the insulating materials **114** and **124** may be included in the amount of between about 1% and 75%, e.g., between about 5% and 50%, indicating that these materials are included intentional, in contrast to potential accidental doping or impurities being included, which would be less than about 0.1% for any of the materials. Thus, any differences in the material composition of the insulating materials **114** and **124** that are beyond the unintentional doping/impurities levels of about 0.1% of a given material could be indicative of the hybrid manufacturing used to provide the microelectronic assembly **100**.

(74) For certain manufacturing processes, cross-sectional shapes of interconnects in the plane such as that of FIG. 1A may be trapezoidal, i.e., a cross-section of an interconnect may have two parallel sides, one of which is a short side and another one of which is a long side, with the long side being closer to the top face of an IC structure than the short side (i.e., the long side faces the front of the IC structure, while the short side faces the back of the IC structure). For example, dual-Damascene or single-Damascene processes for manufacturing interconnects could result in such trapezoidal cross-sections. Examining the trapezoidal cross-sectional shapes of the interconnects **112**, **122** may reveal still another characteristic feature of the hybrid manufacturing of the IC structures **110** and **120**, namely, that, when comparing the conductive trenches of the interconnects **112** and **122**, the heights of the trapezoids of the conductive trenches of the interconnects **112** and **122** may be different, e.g., at least about 5% different, at least about 10% different, or at least 5-50% different. In some embodiments, the angles of the trapezoidal cross-sections of the interconnects **112** and **122** may be different (e.g., the angles between the long sides and one of the sidewalls of the trapezoids) e.g., at least about 5% different, at least about 10% different, or at least 5-50% different. In some embodiments, a pitch of at least some of the interconnects **112** may be different from a pitch of at least some of the interconnects **122**, e.g., at least about 5% different, or at least about 10% different.

(75) Besides the characteristic features described above, which may be indicative of hybrid manufacturing for various embodiments of bonding the IC structures **110** and **120**, some features may be particular to some relative arrangements of the IC structures **110** and **120** (e.g., one of f2f, f2b, or b2b arrangements as described herein) but not the others.

(76) One example of a feature characteristic of hybrid manufacturing with f2f arrangement of the IC structure **110** and **120** is that the cross-sectional shapes of the interconnects **112** may be inverted with respect to those of the interconnects **122**, e.g., as shown in FIG. 1A. In particular, FIG. 1A illustrates that the long sides of the trapezoidal cross-sections of the interconnects **112** are closer to

the frontside **134-2** of the IC structure **110**, and, therefore, to the bonding interface **130** to the IC structure **120**, than their short sides. Such manufacturing processes used in manufacturing the IC structure **120** would, similarly, result in that the long sides of the trapezoidal cross-sections of the interconnects **122** are closer to the frontside **134-2** of the IC structure **120**, and, therefore, to the bonding interface **130** to the IC structure **110**, than their short sides. Because in an f2f bonding arrangement, the IC structure **120** is flipped over, this means that the long sides of the trapezoidal cross-sections of the interconnects **122** are closer to the bonding interface to the IC structure **110** than the short sides, as is shown in FIG. 1A.

(77) Other features characteristic of hybrid manufacturing of the IC structure **110** and **120**, in particular, features characteristic of hybrid manufacturing with modified via-last process, are revealed by examining conductive via structures that extend though at least portions of the IC structures **110** and **120** to route power and/or ground and/or signals to, from, or between the IC structure **110** and **120** (e.g., to route power and/or ground and/or signals to, from, or between various components or device circuitry of the IC structure **110** and **120**). Two examples of such conductive via structures are shown in FIG. 1A as conductive via structures **150-1** and **150-2**, but, in general, any of the embodiments of the microelectronic assembly **100** as described herein may include any number of conductive via structures **150** that extend through all of the top IC structure (i.e., the IC structure **120** for all examples of the present drawings), though the bonding interface **130**, and into at least a portion of the bottom IC structure (i.e., the IC structure **110** for all examples of the present drawings).

(78) In particular, the conductive via structures **150** shown in FIG. 1A illustrate features characteristic of hybrid manufacturing with modified via-last process applied to the f2f arrangement of the IC structures **110** and **120**. As shown in FIG. 1A, with f2f bonding, a conductive via structure **150** may extend from the backside **134-1** of the IC structure **120**, through the bonding interface **130**, and at least partially into the IC structure **110**. While FIG. 1A illustrates that the conductive via structures **150** do not extend all the way down to the backside **134-1** of the IC structure **110**, this may be the case in other embodiments of the microelectronic assembly **100**.

(79) For purposes of discussions, a conductive via structure **150** may be seen as including four portions, continuous with one another in that they refer to consecutive portions of the conductive via structure **150** going from the top to the bottom. In this context, the “top” refers to the end of the conductive via structure **150**, at the top face of the IC structure **120**, from which the electrically conductive fill material in at least some of the portions of the conductive via structure **150** will be deposited, and the “bottom” is then the end of the conductive via structure **150** that is opposite to the top. The four portions of the conductive via structure **150** include a first portion **152-1**, a second portion **152-2**, a third portion **152-3**, and a fourth portion **152-4**. As shown in FIG. 1A, the first portion **152-1** extends from the top of the conductive via structure **150** (i.e., from the face of the IC structure **120** that is the backside **134-1** for the f2f arrangement), through the IC structure **120**, to the bonding interface **130**. The second portion **152-2** is the portion that extends through the bonding interface **130**. The third portion **152-3** then extends, in the IC structure **110** (from the frontside **134-2** of the IC structure **110**), from the bonding interface **130** to a portion of the conductive via structure **150** within the IC structure **110** that is at a first distance **140-1** from the bonding interface **130**. The fourth portion **152-4** extends, in the IC structure **110**, from the plane at the first distance **140-1** from the bonding interface **130** to a portion of the conductive via structure **150** within the IC structure **110** at a second distance **140-2** from the bonding interface **130**, the second distance **140-2** being greater than the first distance **140-1**. For example, the fourth portion **152-4** may be a portion of the conductive via structure **150** that extends in the IC structure **110** from the third portion **152-3** to the bottom of the conductive via structure **150**, as labeled in FIG. 1A.

(80) As will be illustrated in FIG. 7 with reference to the description of the first fabrication method that may be used to fabricate the microelectronic assembly **100** as shown in FIG. 1A, the first

portion **152-1** is based on a via opening that is formed in the IC structure **120** prior to bonding the IC structure **120** and the IC structure **110**. Therefore, the cross-sectional trapezoidal shape of the first portion **152-1** is similar to that of the interconnects **122** in the IC structure **120** (e.g., for the f2f arrangement as shown in FIG. 1A, the long side of the trapezoidal cross-section of the first portion **152-1** is closer to the frontside **134-2** of the IC structure **120**, and, therefore, to the bonding interface **130** to the IC structure **110**, than its short side). Similarly, together, the third portion **152-3** and the fourth portion **152-4** are based on a via opening that is formed in the IC structure **110** prior to bonding the IC structure **120** and the IC structure **110**. Therefore, the cross-sectional trapezoidal shapes of the third portion **152-3** and the fourth portion **152-4** are similar to that of the interconnects **112** in the IC structure **110** (e.g., for the f2f arrangement as shown in FIG. 1A, the long sides of the trapezoidal cross-sections of the third portion **152-3** and the fourth portion **152-4** are closer to the frontside **134-2** of the IC structure **110**, and, therefore, to the bonding interface **130** to the IC structure **120**, than their short sides).

(81) In general, the modified via-last process, proposed herein, is based on, first, forming a via opening for the first portion **152-1** in the IC structure **120** and, separately, forming a via opening for the third portion **152-3** and the fourth portion **152-4** in the IC structure **110**, where, at that time, each of the via opening for the first portion **152-1** and a portion of the via opening for the third portion **152-3** is filled with one or more sacrificial materials (in general, the first portion **152-1** and the third portions **152-3** may be filled with different sacrificial materials), and a portion of the via opening for the fourth portion **152-4** is filled with an electrically conductive fill material **144**. The modified via-last process may then continue with bonding the IC structures **110** and **120**, aligning the IC structures **110** and **120** so that said via openings overlap as much possible (e.g., limited by the resolution and capabilities of the equipment). The modified via-last process may conclude with replacing, after the bonding of the IC structures **110** and **120**, the sacrificial material in the first portion **152-1** and the third portion **152-3** with an electrically conductive fill material. As a result, the same electrically conductive fill material (i.e., a material of substantially the same material composition) will be present in the second portion **152-2**, i.e., in a portion of the bonding interface **130** that is between the first portion **152-1** and the third portion **152-3**. This is illustrated in FIG. 1A with each of the first portion **152-1**, the second portion **152-2**, and the third portion **152-3** including a first electrically conductive fill material **154**, while the fourth portion **152-4** includes a second electrically conductive fill material **144**. Thus, the first electrically conductive fill material **154** may be continuous across/through the first portion **152-1**, the second portion **152-2**, and the third portion **152-3**.

(82) In various embodiments, the electrically conductive fill materials **144**, **154** may include any of the materials described with reference to the electrically conductive fill materials **118**, **128**. In some embodiments, the electrically conductive fill materials **144**, **154** may be different (i.e., a material composition of the second electrically conductive fill material **144** may be different from a material composition of the first electrically conductive fill material **154**). For example, the first electrically conductive fill material **154** may include copper, while the second electrically conductive fill material **144** may include tungsten, aluminum, ruthenium, cobalt, or AlCu (e.g., in proportions of between 1:1 to 1:100), or vice versa. In other embodiments, the electrically conductive fill materials **144**, **154** may be substantially the same, but even in such embodiments, in some cases, a grain boundary may be detectable (e.g., by means of TEM) at the interface between the third portion **152-3** and the fourth portion **152-4**, indicative of their electrically conductive materials being deposited at different times and, possibly, under different conditions. In other cases, such a grain boundary may not be present, e.g., if the microelectronic assembly **100** undergoes application of sufficient heat or pressure after it is formed, in which case the grain boundary between the third portion **152-3** and the fourth portion **152-4** may disappear. Furthermore, although not specifically illustrated in the present drawings, in any embodiments of the microelectronic assemblies **100** described herein, the fourth portion **152-4** may also be first filled with one or more sacrificial

materials, the same as the third portion **152-3**, and, in a later part of the modified via-last process, the sacrificial material in the fourth portion **152-4** is replaced with the second electrically conductive fill material **144** (i.e., in some such embodiments, the first electrically conductive fill material **154** may be absent in the conductive via structure **150**).

(83) In some embodiments, besides the first electrically conductive fill material **154**, the first portion **152-1** may include a first liner **156**, while the third portion **152-4** and the fourth portion **152-4** may include a second liner **146**, as shown in FIG. 1A. In various embodiments, the liners **146**, **156** may include any of the materials described with reference to the liners **119**, **129**. In some embodiments, the liners **146**, **156** may be different in their material composition (i.e., a material composition of the second liner **146** may be different from a material composition of the first liner **156**) and/or in their thickness (i.e., a thickness of the second liner **146** may be different from a thickness of the first liner **156**). In various embodiments, the thicknesses of the liners **146**, **156** may be as described above for the liners **119**, **129**. Indicative of the hybrid manufacturing with modified via-first process applied to a f2f arrangement, the liner **156** may be present only at the sidewalls of the via opening for the first portion **152-1**, but not at the top of the IC structure **120**, in a portion highlighted in FIGS. 1A-1D with a dotted contour at the backside **134-1** of the IC structure **120**. In other words, at the top face of the IC structure **120** (i.e., at the backside **134-1** for the f2f arrangement as shown in FIG. 1A), the liner **156** may be present only at the sidewalls of the conductive via structure **150** and not extending at the top face of the IC structure **120**. Indicative of the hybrid manufacturing with modified via-first process according to the first fabrication method, which involves providing the bonding material **130** over portions of the faces of the IC structures **110** and **120** where the conductive via structure(s) **150** is(are) to be formed, the second portion **152-2** may include the electrically conductive fill material **154**, but no liner (neither one of the liners **146**, **156**), as is shown in a portion highlighted in FIG. 1A with a dotted contour at the second portion **152-2** of the conductive via structure **150-1**. Thus, in the embodiments of the microelectronic assembly **100** including the f2f arrangement formed according to the first fabrication method, the liners **146** and **156** may be discontinuous with one another and not in contact with one another because they are separated by the bonding interface **130**. Although not specifically shown in the present drawings, in other embodiments of the microelectronic assembly **100** with the f2f arrangement formed according to the first fabrication process, the liner **146** and/or the liner **156** may be absent.

(84) FIG. 1B illustrates a cross-sectional side view of a microelectronic assembly **100** according to embodiments where f2f bonding is used according to a second fabrication method of hybrid manufacturing with via-last process (e.g., the method illustrated in FIGS. 8A-8I). In the first fabrication method, the liners **146**, **156** may be present in the IC structures **110**, **120**, respectively, before these IC structures are bonded. In the second fabrication method, the liner **156** may be deposited after the sacrificial material has been removed from the first and third portions **152-1** and **152-3** (and, therefore, also the bonding material is removed from the second portion **152-2**), prior to deposition of the electrically conductive fill material **154** into the etched opening. As a result, in the microelectronic assembly **100** of FIG. 1B, the liner **156** is not only in the first portion **152-1**, but also in the second portion **152-2** and in the third portion **152-3**, as is shown in FIG. 1B, and the liner **156** is continuous among these three portions and is also present at the bottom of the third portion **152-3** (as indicated in FIG. 1B with a dotted contour around the bottom of the third portion **152-3**) as it is deposited by conformal deposition that provides the liner **156** on all surfaces of the conductive via structure that have been exposed by removing the sacrificial material. Moreover, in the embodiments of FIG. 1B where the liner **146** is present in the fourth portion **152-4**, the liner **156** may be in contact with the liner **146** where the third portion **152-3** interfaces the fourth portion **152-4**, as shown in FIG. 1B, even though, in general the liners **156**, **146** may have different thicknesses and/or material compositions, as described above. In other embodiments of the second fabrication method, the liner **156** may be present as shown in FIG. 1B, but the liner **146** may be

absent (i.e., the fourth portion **152-4** may be implemented only with the electrically conductive fill material **144** but without the liner **146**). In still other embodiments of the second fabrication method, the liner **156** may be absent, while the liner **146** may be present (i.e., each of the first portion **152-1**, the second portion **152-2**, and the third portion **152-3**) may be implemented only with the electrically conductive fill material **154** but without the liner **156**. The rest of the descriptions of the microelectronic assembly **100** that are provided for FIG. **1A** are applicable to FIG. **1B** and, therefore, in the interests of brevity, are not repeated.

(85) FIG. **1C** illustrates a cross-sectional side view of a microelectronic assembly **100** according to embodiments where f2f bonding is used according to a third fabrication method of hybrid manufacturing with via-last process (e.g., the method as illustrated in FIGS. **7A-7I** but modified as described herein). In the first and second fabrication methods, the bonding material **130** may be provided over faces of the IC structure **110** and/or the IC structure **120** so that the bonding material **130** is also present where the vias of these IC structures are to overlap and form the conductive via structure **150** as described herein. The third fabrication method may be substantially as the first fabrication method, except that the bonding material **130** may be absent in the areas of the bonding faces of the IC structure **110** and the IC structure where the vias of these IC structures are to overlap and form the conductive via structure **150**. As a result, in the microelectronic assembly **100** of FIG. **1B**, in contrast to the embodiment of FIG. **1A**, the liner **156** may be present in the second portion **152-2**. In case when the overlap of the vias formed in the IC structures **110** and **120** prior to bonding to form the conductive via structure **150** is sufficiently large and the transverse cross-sectional dimensions of these vias are substantially the same, then at least some of the liner **156** of the second portion **152-2** may be in contact with at least some of the liner **146** of the third portion **152-3**. In other embodiments, not specifically shown in the present drawings, the liner **146** may be present in the second portion **152-2** instead of the liner **156**. In case when the overlap of the vias formed in the IC structures **110** and **120** prior to bonding to form the conductive via structure **150** is sufficiently large and the transverse cross-sectional dimensions of these vias are substantially the same, then at least some of the liner **146** of the second portion **152-2** may be in contact with at least some of the liner **156** of the first portion **152-1**. The rest of the descriptions of the microelectronic assembly **100** that are provided for FIG. **1A** are applicable to FIG. **1C** and, therefore, in the interests of brevity, are not repeated.

(86) FIG. **1D** illustrates a cross-sectional side view of a microelectronic assembly **100** according to embodiments where f2f bonding is used according to a fourth fabrication method of hybrid manufacturing with via-last process (e.g., the method as illustrated in FIGS. **8A-8I** but modified as described herein). The fourth fabrication method may be seen as a combination of the second and third fabrication methods in that it is substantially as the second fabrication method, except that the bonding material **130** may be absent in the areas of the bonding faces of the IC structure **110** and the IC structure where the vias of these IC structures are to overlap and form the conductive via structure **150** as in the third fabrication method. The rest of the descriptions of the microelectronic assembly **100** that are provided for FIGS. **1A-1C** are applicable to FIG. **1D** and, therefore, in the interests of brevity, are not repeated. In the embodiment of FIG. **1D**, similar to that of FIG. **1B**, the liner **156** is present at the bottom of the third portion **152-3** (as indicated in FIG. **1D** with a dotted contour around the bottom of the third portion **152-3**).

(87) FIGS. **2A-2D** illustrate cross-sectional side views of a microelectronic assembly **100** with the first and second IC structures **110**, **120** bonded using f2b bonding, in accordance with some embodiments. The microelectronic assembly **100** shown in FIGS. **2A-2D** is similar to that described with reference to FIGS. **1A-1D**, except that the IC structure **120** is not flipped upside down for bonding, so that the frontside **134-2** of the IC structure **120** is facing up and the backside **134-1** of the IC structure **120** is bonded to the frontside **134-2** of the IC structure **110**. Similar to FIGS. **1A-1D**, FIGS. **2A-2D** show different embodiments of the microelectronic assembly **100** with modified via-last process, with the differences resulting from using four different fabrication

methods of hybrid manufacturing with modified via-last process as described herein, except that in FIGS. 2A-2D the first, second, third, and fourth fabrication methods as described herein are applied to f2b arrangement of the IC structures **120** and **110** as described herein.

(88) In contrast to FIGS. 1A-1D, because the microelectronic assembly **100** shown in FIGS. 2A-2D an f2b-bonded assembly, it is the backside **134-1** of the IC structure **120** that is bonded to the frontside **134-2** of the IC structure **110**. One characteristic feature that is specific to the f2b bonding as shown in FIGS. 2A-2D is that the cross-sectional trapezoidal shapes of the interconnects **112** and of the interconnects **122** may be oriented in the same manner (i.e., not inverted with respect to one another). This is illustrated in FIGS. 2A-2D with the long sides of the trapezoidal cross-sections of the interconnects **112** being closer to the front of the IC structure **110**, and, therefore, to the bonding interface **130** to the IC structure **120**, than their short sides, and with the short sides of the trapezoidal cross-sections of the interconnects **122** being closer to the bonding interface **130** to the IC structure **110** than their long sides. Characteristic features of hybrid manufacturing in general described with reference to FIG. 1 besides the feature specific to the f2f bonding of FIG. 1 are applicable to various embodiments of the microelectronic assembly of FIG. 2 and, therefore, in the interests of brevity, are not repeated for FIG. 2.

(89) Similar to the embodiments of FIG. 1, other features characteristic of hybrid manufacturing of the IC structure **110** and **120**, in particular, features characteristic of hybrid manufacturing with modified via-last process applied to the f2b arrangement of the IC structures **110** and **120** of FIG. 2, are revealed by examining the conductive via structures **150**. Each of FIGS. 2A-2D illustrates one such conductive via structure **150**, but, in various embodiments, any number of such conductive via structures **150** may be included in the f2b arrangements as described herein.

(90) As shown in FIG. 2A, with f2b bonding, a conductive via structure **150** may extend from the frontside **134-2** of the IC structure **120**, through the bonding interface **130**, and at least partially into the IC structure **110**. While FIG. 2A illustrates that the conductive via structures **150** do not extend all the way down to the backside **134-1** of the IC structure **110**, this may be the case in other embodiments of the microelectronic assembly **100** with the f2b arrangement.

(91) The conductive via structure **150** of FIG. 2A has four portions **152**, as described above, except that, for the f2b arrangement, the first portion **152-1** extends towards the bonding interface **130** from the frontside **134-2** of the IC structure **120**. Since the microelectronic assembly **100** of FIG. 2A may be fabricated according to the first fabrication method described with reference to FIG. 1A but adapted for the f2b arrangement, then, similar to FIG. 1A, the first portion **152-1** is based on a via opening that is formed in the IC structure **120** prior to bonding the IC structure **120** and the IC structure **110** and, therefore, the cross-sectional trapezoidal shape of the first portion **152-1** is similar to that of the interconnects **122** in the IC structure **120** (e.g., for the f2b arrangement as shown in FIG. 2A, the short side of the trapezoidal cross-section of the first portion **152-1** is closer to the bonding interface **130** to the IC structure **110**, than its long side). The orientations of the trapezoids of the third portion **152-3** and the fourth portion **152-4** remain the same as in FIG. 1A, i.e., the cross-sectional trapezoidal shapes of the third portion **152-3** and the fourth portion **152-4** are similar to that of the interconnects **112** in the IC structure **110** (e.g., for the f2b arrangement as shown in FIG. 2A, the long sides of the trapezoidal cross-sections of the third portion **152-3** and the fourth portion **152-4** are closer to the frontside **134-2** of the IC structure **110**, and, therefore, to the bonding interface **130** to the IC structure **120**, than their short sides).

(92) Indicative of the hybrid manufacturing with modified via-first process applied to a f2b arrangement, the liner **156** may be present only at the sidewalls of the via opening for the first portion **152-1**, but not at the bottom of the IC structure **120**, in a portion highlighted in FIGS. 2A-2D with a dotted contour at the backside **134-1** of the IC structure **120**, because the liner **156** in that portion that was present before bonding is etched away as a part of the modified via-last process of the first fabrication method. In other words, at the bottom face of the IC structure **120** (i.e., at the backside **134-1** for the f2b arrangement as shown in FIG. 2A), the liner **156** may be present only at

the sidewalls of the conductive via structure **150** and not extending along the second portion **152-2** of the conductive via structure **150**. In some embodiments of the f2b arrangement of the microelectronic assembly **100**, the liner **156** may be absent in at least about 70% (e.g., in at least about 80% or in at least about 90%) of each of an interface between the first portion **152-1** and the second portion **152-2**, and an interface between the second portion **152-2** and the third portion **152-3**. Similar to the embodiment of FIG. **1A**, indicative of the hybrid manufacturing with modified via-first process according to the first fabrication method, which involves providing the bonding material **130** over portions of the faces of the IC structures **110** and **120** where the conductive via structure(s) **150** is(are) to be formed, the second portion **152-2** may include the electrically conductive fill material **154**, but no liner (neither one of the liners **146**, **156**), as is shown in a portion highlighted in FIG. **2A** with a dotted contour at the second portion **152-2** of the conductive via structure **150-1**. Thus, in the embodiments of the microelectronic assembly **100** including the f2b arrangement formed according to the first fabrication method, similar to the f2f arrangement, the liners **146** and **156** may be discontinuous with one another and not in contact with one another because they are separated by the bonding interface **130**. Although not specifically shown in FIG. **2A**, in other embodiments of the microelectronic assembly **100** with the f2b arrangement formed according to the first fabrication process, the liner **146** and/or the liner **156** may be absent. The rest of the descriptions of the microelectronic assembly **100** that are provided for FIG. **1A** are applicable to FIG. **2A** and, therefore, in the interests of brevity, are not repeated.

(93) FIG. **2B** illustrates a cross-sectional side view of a microelectronic assembly **100** according to embodiments where f2b bonding is used according to the second fabrication method of hybrid manufacturing with via-last process (e.g., the method illustrated in FIGS. **8A-8I** but adapted to the f2b arrangement of the IC structures **110**, **120**). The descriptions of the microelectronic assembly **100** that are provided for FIG. **113** are applicable to FIG. **2B**. In particular, the descriptions of how the use of the second fabrication method causes certain differences from the microelectronic assembly fabricated using the first fabrication method, provided for the f2f arrangement of FIG. **1B**, are applicable to the differences between the f2b arrangement of FIG. **2B** and that of FIG. **2A**. Therefore, in the interests of brevity, these descriptions are not repeated. In the embodiment of FIG. **2B**, similar to that of FIG. **1B**, the liner **156** is present at the bottom of the third portion **152-3** (as indicated in FIG. **2B** with a dotted contour around the bottom of the third portion **152-3**).

(94) FIG. **2C** illustrates a cross-sectional side view of a microelectronic assembly **100** according to embodiments where f2b bonding is used according to the third fabrication method of hybrid manufacturing with via-last process (e.g., the method as illustrated in FIGS. **7A-7I** but modified as described with reference to FIG. **1C** and further adapted to the f2b arrangement of the IC structures **110**, **120**). The descriptions of the microelectronic assembly **100** that are provided for FIG. **1C** are applicable to FIG. **2C**. In particular, the descriptions of how the use of the third fabrication method causes certain differences from the microelectronic assembly fabricated using the first fabrication method, provided for the f2f arrangement of FIG. **1C**, are applicable to the differences between the f2b arrangement of FIG. **2C** and that of FIG. **2A**. Therefore, in the interests of brevity, these descriptions are not repeated.

(95) FIG. **2D** illustrates a cross-sectional side view of a microelectronic assembly **100** according to embodiments where f2b bonding is used according to the fourth fabrication method of hybrid manufacturing with via-last process (e.g., the method as illustrated in FIGS. **8A-8I** but modified as described with reference to FIG. **1D** and further adapted to the f2b arrangement of the IC structures **110**, **120**). The descriptions of the microelectronic assembly **100** that are provided for FIG. **1D** are applicable to FIG. **2D**. In particular, the descriptions of how the use of the fourth fabrication method causes certain differences from the microelectronic assembly fabricated using the second or third fabrication methods, provided for the f2f arrangement of FIG. **1D**, are applicable to the differences between the f2b arrangement of FIG. **2D** and that of FIG. **2B** or FIG. **2C**. Therefore, in the interests of brevity, these descriptions are not repeated. In the embodiment of FIG. **2D**, similar

to that of FIG. 2B, the liner **156** is present at the bottom of the third portion **152-3** (as indicated in FIG. 2D with a dotted contour around the bottom of the third portion **152-3**).

(96) FIGS. 3A-3D illustrate cross-sectional side views of a microelectronic assembly **100** with the first and second IC structures **110**, **120** bonded using b2b bonding, where the first IC structure **110** retains at least a portion of its substrate but the substrate of the second IC structure **120** is removed, in accordance with some embodiments. The microelectronic assembly **100** shown in FIGS. 3A-3D is similar to that described with reference to FIGS. 1A-1D, except that the IC structure **120** is not flipped upside down for bonding but the IC structure **110** is. In this manner, the frontside **134-2** of the IC structure **120** is facing up and the backside **134-1** of the IC structure **120** is bonded to the backside **134-1** of the IC structure **110**. Similar to FIGS. 1A-1D, FIGS. 3A-3D show different embodiments of the microelectronic assembly **100** with modified via-last process, with the differences resulting from using four different fabrication methods of hybrid manufacturing with modified via-last process as described herein, except that in FIGS. 3A-3D the first, second, third, and fourth fabrication methods as described herein are applied to b2b arrangement of the IC structures **120** and **110** as described herein.

(97) In contrast to FIGS. 1A-1D, because the microelectronic assembly **100** shown in FIGS. 3A-3D an b2b-bonded assembly, the backside **134-1** of the IC structure **120** is bonded to the backside **134-1** of the IC structure **110**. One characteristic feature that is specific to the b2b bonding as shown in FIGS. 3A-3D is that the cross-sectional trapezoidal shapes of the interconnects **112** and of the interconnects **122** may be inverted with respect to one another but be oriented in a way that is the opposite of FIGS. 1A-1D. This is illustrated in FIGS. 3A-3D with the short sides of the trapezoidal cross-sections of the interconnects **112** being closer to the bonding interface **130** to the IC structure **120**, than their long sides, and with the short sides of the trapezoidal cross-sections of the interconnects **122** being closer to the bonding interface **130** to the IC structure **110** than their long sides. Characteristic features of hybrid manufacturing in general described with reference to FIG. 1 besides the feature specific to the f2f bonding of FIG. 1 are applicable to various embodiments of the microelectronic assembly of FIG. 3 and, therefore, in the interests of brevity, are not repeated for FIG. 3.

(98) Similar to the embodiments of FIG. 1, other features characteristic of hybrid manufacturing of the IC structure **110** and **120**, in particular, features characteristic of hybrid manufacturing with modified via-last process applied to the b2b arrangement of the IC structures **110** and **120** of FIG. 3, are revealed by examining the conductive via structures **150**. Each of FIGS. 3A-3D illustrates one such conductive via structure **150**, but, in various embodiments, any number of such conductive via structures **150** may be included in the b2b arrangements as described herein.

(99) As shown in FIG. 3A, with b2b bonding, a conductive via structure **150** may extend from the frontside **134-2** of the IC structure **120**, through the bonding interface **130**, and at least partially into the IC structure **110** (in contrast to the embodiments of FIGS. 1 and 2, this time the conductive via structure **150** enters the IC structure **110** through the backside **134-1** of the IC structure **110**). While FIG. 3A illustrates that the conductive via structures **150** do not extend all the way down to the frontside **134-2** of the IC structure **110**, this may be the case in other embodiments of the microelectronic assembly **100** with the b2b arrangement.

(100) The conductive via structure **150** of FIG. 3A has four portions **152**, as described above for the f2f arrangement, except that, for the b2b arrangement, the first portion **152-1** extends towards the bonding interface **130** from the frontside **134-2** of the IC structure **120**, and the third portion **152-3** extends into the IC structure **110** from the backside **134-1** of the IC structure **110**. Since the microelectronic assembly **100** of FIG. 3A may be fabricated according to the first fabrication method described with reference to FIG. 1A but adapted for the b2b arrangement, then, similar to FIG. 1A, the first portion **152-1** is based on a via opening that is formed in the IC structure **120** prior to bonding the IC structure **120** and the IC structure **110** and, therefore, the cross-sectional trapezoidal shape of the first portion **152-1** is similar to that of the interconnects **122** in the IC

structure **120** (e.g., for the b2b arrangement as shown in FIG. 3A, the short side of the trapezoidal cross-section of the first portion **152-1** is closer to the bonding interface **130** to the IC structure **110**, than its long side). The orientations of the trapezoids of the third portion **152-3** and the fourth portion **152-4** are inverted with respect to those of the f2f arrangement of FIG. 1A, i.e., for the b2b arrangement as shown in FIG. 3A, the short sides of the trapezoidal cross-sections of the third portion **152-3** and the fourth portion **152-4** are closer to the bonding interface **130** to the IC structure **120**, than their long sides).

(101) Similar to the f2b arrangement, indicative of the hybrid manufacturing with modified via-first process applied to a b2b arrangement, the liner **156** may be present only at the sidewalls of the via opening for the first portion **152-1**, but not at the bottom of the IC structure **120**, in a portion highlighted in FIGS. 3A-3D with a dotted contour at the backside **134-1** of the IC structure **120**, because the liner **156** in that portion that was present before bonding is etched away as a part of the modified via-last process of the first fabrication method. In other words, at the bottom face of the IC structure **120** (i.e., at the backside **134-1** of the IC structure **120** for the b2b arrangement as shown in FIG. 3A), the liner **156** may be present only at the sidewalls of the conductive via structure **150** and not extending along the second portion **152-2** of the conductive via structure **150**. Similar to the f2b arrangement, in some embodiments of the b2b arrangement of the microelectronic assembly **100**, the liner **156** may be absent in at least about 70% (e.g., in at least about 80% or in at least about 90%) of each of an interface between the first portion **152-1** and the second portion **152-2**, and an interface between the second portion **152-2** and the third portion **152-3**. Different from both the f2f and the f2b arrangements, indicative of the hybrid manufacturing with modified via-first process applied to a b2b arrangement, the liner **146** may be present only at the sidewalls of the via opening for the third portion **152-3**, but not at the top of the IC structure **110** (i.e., at the bonding interface **130**), in a portion highlighted in FIGS. 3A-3D with a dotted contour at the backside **134-1** of the IC structure **110**, because the liner **146** in that portion that was present before bonding is etched away as a part of the modified via-last process of the first fabrication method. In other words, at the top face of the IC structure **110** (i.e., at the backside **134-1** of the IC structure **110** for the b2b arrangement as shown in FIG. 3A), the liner **146** may be present only at the sidewalls of the conductive via structure **150** and not extending along the second portion **152-2** of the conductive via structure **150**. In some embodiments of the b2b arrangement of the microelectronic assembly **100**, the liner **146** may be absent in at least about 70% (e.g., in at least about 80% or in at least about 90%) of each of an interface between the first portion **152-1** and the second portion **152-2**, and an interface between the second portion **152-2** and the third portion **152-3**.

(102) Similar to the embodiment of FIG. 1A, indicative of the hybrid manufacturing with modified via-first process according to the first fabrication method, which involves providing the bonding material **130** over portions of the faces of the IC structures **110** and **120** where the conductive via structure(s) **150** is(are) to be formed, the second portion **152-2** may include the electrically conductive fill material **154**, but no liner (neither one of the liners **146**, **156**), as is shown in FIG. 3A (but, unlike the illustration of FIG. 1A and FIG. 2A, this portion is not highlighted in FIG. 3A with a dotted contour at the second portion **152-2** of the conductive via structure **150-1** in order to not clutter the drawing). Thus, in the embodiments of the microelectronic assembly **100** including the b2b arrangement formed according to the first fabrication method, similar to the f2f and the f2b arrangements, the liners **146** and **156** may be discontinuous with one another and not in contact with one another because they are separated by the bonding interface **130**. Although not specifically shown in FIG. 3A, in other embodiments of the microelectronic assembly **100** with the b2b arrangement formed according to the first fabrication process, the liner **146** and/or the liner **156** may be absent. The rest of the descriptions of the microelectronic assembly **100** that are provided for FIG. 1A are applicable to FIG. 3A and, therefore, in the interests of brevity, are not repeated.

(103) FIG. 3B illustrates a cross-sectional side view of a microelectronic assembly **100** according

to embodiments where b2b bonding is used according to the second fabrication method of hybrid manufacturing with via-last process (e.g., the method illustrated in FIGS. **8A-8I** but adapted to the b2b arrangement of the IC structures **110, 120**). The descriptions of the microelectronic assembly **100** that are provided for FIG. **1B** are applicable to FIG. **3B**. In particular, the descriptions of how the use of the second fabrication method causes certain differences from the microelectronic assembly fabricated using the first fabrication method, provided for the f2f arrangement of FIG. **1B**, are applicable to the differences between the b2b arrangement of FIG. **3B** and that of FIG. **3A**. Therefore, in the interests of brevity, these descriptions are not repeated. In the embodiment of FIG. **3B**, similar to that of FIG. **1B**, the liner **156** is present at the bottom of the third portion **152-3** (as indicated in FIG. **3B** with a dotted contour around the bottom of the third portion **152-3**).

(104) FIG. **3C** illustrates a cross-sectional side view of a microelectronic assembly **100** according to embodiments where b2b bonding is used according to the third fabrication method of hybrid manufacturing with via-last process (e.g., the method as illustrated in FIGS. **7A-7I** but modified as described with reference to FIG. **1C** and further adapted to the b2b arrangement of the IC structures **110, 120**). The descriptions of the microelectronic assembly **100** that are provided for FIG. **1C** are applicable to FIG. **3C**. In particular, the descriptions of how the use of the third fabrication method causes certain differences from the microelectronic assembly fabricated using the first fabrication method, provided for the f2f arrangement of FIG. **1C**, are applicable to the differences between the b2b arrangement of FIG. **3C** and that of FIG. **2A**. Therefore, in the interests of brevity, these descriptions are not repeated.

(105) FIG. **3D** illustrates a cross-sectional side view of a microelectronic assembly **100** according to embodiments where b2b bonding is used according to the fourth fabrication method of hybrid manufacturing with via-last process (e.g., the method as illustrated in FIGS. **8A-8I** but modified as described with reference to FIG. **1D** and further adapted to the b2b arrangement of the IC structures **110, 120**). The descriptions of the microelectronic assembly **100** that are provided for FIG. **1D** are applicable to FIG. **3D**. In particular, the descriptions of how the use of the fourth fabrication method causes certain differences from the microelectronic assembly fabricated using the second or third fabrication methods, provided for the f2f arrangement of FIG. **1D**, are applicable to the differences between the b2b arrangement of FIG. **3D** and that of FIG. **3B** or FIG. **3C**. Therefore, in the interests of brevity, these descriptions are not repeated. In the embodiment of FIG. **3D**, similar to that of FIG. **3B**, the liner **156** is present at the bottom of the third portion **152-3** (as indicated in FIG. **3D** with a dotted contour around the bottom of the third portion **152-3**).

(106) With b2b arrangements some variations are possible as to the support structure **132** of the IC structure **110** or an analogous substrate of the IC structure **120**. FIGS. **3A-3D** illustrate embodiments of the b2b-bonded microelectronic assembly **100** where the IC structure **110** includes the support structure **132**, while a support structure of the IC structure **120** has been removed prior to bonding. In such embodiments, the bonding interface **130** may be between the backside **134-1** of the support structure **132** of the IC structure **110** and the backside **134-1** of the IC structure **120**. FIG. **4** illustrates an embodiment of the b2b-bonded microelectronic assembly **100** where the IC structure **120** includes the support structure **132**, while a support structure of the IC structure **110** has been removed prior to bonding. In such embodiments, the bonding interface **130** may be between the backside **134-1** of the support structure **132** of the IC structure **120** and the backside **134-1** of the IC structure **110**. FIG. **4** is analogous to FIG. **3A**, i.e., illustrating the b2b arrangement of the microelectronic assembly **100** fabricated according to the first fabrication method. Although not shown in the present drawings, embodiments similar to those shown in FIGS. **3B-3D** are possible for further variations of the b2b arrangement shown in FIG. **4** and, therefore, are within the scope of the present disclosure. FIG. **5** illustrates an embodiment of the b2b-bonded microelectronic assembly **100** where both the IC structure **120** and the IC structure **110** include their respective support structures **132**. In such embodiments, the bonding interface **130** may be between the backside **134-1** of the support structure **132** of the IC structure **120** and the backside

134-1 of the support structure **132** of the IC structure **110**. FIG. 5 is analogous to FIG. 3A, i.e., illustrating the b2b arrangement of the microelectronic assembly **100** fabricated according to the first fabrication method. Although not shown in the present drawings, embodiments similar to those shown in FIGS. 3B-3D are possible for further variations of the b2b arrangement shown in FIG. 5 and, therefore, are within the scope of the present disclosure. FIG. 6 illustrates an embodiment of the b2b-bonded microelectronic assembly **100** where the support structures **132** of both the IC structure **120** and the IC structure **110** have been removed prior to bonding, but a carrier support structure **136** may be attached to the IC structure **110** to provide mechanical stability to the microelectronic assembly **100**. In other embodiments, the carrier support structure **136** may be attached to the IC structure **120**, and/or one carrier support structure **136** may be attached to the IC structure **110** and another carrier support structure **136** may be attached to the IC structure **120**. In such embodiments, the bonding interface **130** may be between the backside **134-1** of the IC structure **120** and the backside **134-1** of the IC structure **110** without the support structure in between. FIG. 6 is analogous to FIG. 3A, i.e., illustrating the b2b arrangement of the microelectronic assembly **100** fabricated according to the first fabrication method. Although not shown in the present drawings, embodiments similar to those shown in FIGS. 3B-3D are possible for further variations of the b2b arrangement shown in FIG. 6 and, therefore, are within the scope of the present disclosure.

(107) FIGS. 7A-7I illustrate cross-sectional side views during different processes of the first fabrication method for an example of a first and a second IC structures bonded using f2f bonding with modified via-last process, in accordance with some embodiments. The example first fabrication method shown in FIGS. 7A-7I may include other operations not specifically shown in FIGS. 7A-7I, such as various cleaning or planarization operations as known in the art. For example, in some embodiments, support structures of the IC structures **110**, **120**, as well as layers of various other materials subsequently deposited thereon, may be cleaned prior to, after, or during any of the processes of the first fabrication method described herein, e.g., to remove oxides, surface-bound organic and metallic contaminants, as well as subsurface contamination. In some embodiments, cleaning may be carried out using e.g., a chemical solutions (such as peroxide), and/or with ultraviolet (UV) radiation combined with ozone, and/or oxidizing the surface (e.g., using thermal oxidation) then removing the oxide (e.g., using hydrofluoric acid (HF)). In another example, the arrangements/assemblies described herein may be planarized prior to, after, or during any of the processes of the first fabrication method described herein, e.g., to remove overburden or excess materials. In some embodiments, planarization may be carried out using either wet or dry planarization processes, e.g., planarization be a chemical mechanical planarization (CMP), which may be understood as a process that utilizes a polishing surface, an abrasive, and a slurry to remove the overburden and planarize the surface. The same applies to the second fabrication method shown in FIGS. 8A-8I, as well as to the third and fourth fabrication methods described herein.

(108) The first fabrication method may begin with a process **702**, shown in FIG. 7A, that includes fabricating the IC structures **110** and **120** separately, to include all of the desired components therein, including vias, except that for the vias that are later to form a conductive via structure **150** as described herein are filled with a sacrificial material **160**. FIG. 7A illustrates indications of portions **152-1**, **152-3**, and **152-4** of the future conductive via structure **150**. As shown in FIG. 7A, in the first fabrication method, in the IC structure **120**, the first portion **152-1** is lined with the liner **156**, and then filled with the sacrificial material **160**, while, in the IC structure **110**, a via in which the third and fourth portions **152-3**, **152-4** will be provided is lined with the liner **146**, where the third portion **152-3** then being filled with the sacrificial material **160**, while the fourth portion **152-4** is filled with the electrically conductive fill material **144**. In some embodiments, the liners **146**, **156** may be deposited using any suitable conformal deposition processes, such as atomic layer deposition (ALD). The sacrificial material **160** may include any material that is sufficiently etch-selective with respect to the surrounding materials, where, as known in the art, two materials are

described as being “sufficiently etch-selective” if etchants used to etch one material do not substantially etch the other material, and possibly vice versa. In some embodiments, the sacrificial material **160** may be an insulator material because insulator materials are typically sufficiently etch-selective with respect to liner materials (which are often electrically conductive materials). Although the sacrificial material **160** is shown in both the IC structure **110** and the IC structure **120**, in various embodiments, these sacrificial materials **160** may have different material compositions (or the material compositions may be substantially the same). The sacrificial material **160** may be deposited using any suitable deposition techniques, such as spin-coating, dip-coating, ALD, chemical vapor deposition (CVD), or physical vapor deposition (PVD) (e.g., evaporative deposition, magnetron sputtering, or e-beam deposition). The electrically conductive fill material **144** may be deposited using any suitable deposition techniques, such as ALD, CVD, or PVD.

(109) The first fabrication method may then proceed with a process **704**, shown in the upper portion of FIG. 7B, that includes depositing a bonding material over some or all portions of the face of the IC structure **110** that is to be hybrid-bonded to the IC structure **120**. Although not shown in the present drawings, additionally or alternatively, a bonding material may be deposited over some or all portions of the face of the IC structure **120** that is to be hybrid-bonded to the IC structure **110**. In the first fabrication method, the bonding material may be deposited over the face of the IC structure **110** and/or the face of the IC structure **120** in such a manner that, when the faces to be bonded are facing one another and the IC structures are aligned, then the bonding material may be present in between the first portion **152-1** and the third portion **152-3**. The bonding material may be any material for forming the bonding interface **130** and may be deposited using any suitable deposition techniques, such as spin-coating, dip-coating, ALD, CVD, or PVD.

(110) The lower portion of FIG. 7B illustrates a process **704'**, which may be a process that would replace the process **704** of the first fabrication method when performed as a part of the third fabrication method. As shown in the lower portion of FIG. 7B, in the process **704'**, the bonding material **130** is not deposited over the sacrificial material **160** of the IC structure **110** (and, similarly, would not be deposited over the sacrificial material **160** of the IC structure **120** for the embodiments of the process **704'** where the bonding material **130** is deposited over the IC structure **120**). The rest of the descriptions of the process **704** and applicable to the process **704'**.

Furthermore, the third fabrication method may further include the rest of the processes of the first fabrication method provided with reference to FIGS. 7A-7I, except with the minor modification due to the differences between the process **704** and **704'**.

(111) The first fabrication method may then include a process **706**, shown in FIG. 7C, in which one of the IC structures is flipped over, e.g., the IC structure **120**, so that their faces that are supposed to be bonded are facing each other.

(112) The first fabrication method may then proceed with a process **708**, shown in FIG. 7D, which includes performing hybrid manufacturing of the IC structures **110** and **120**, so that at least a part of the first portion **152-1** overlaps at least a part of the third portion **152-3**, possibly with the bonding material in between. In some embodiments, the overlap between the sacrificial materials **160** of the first portion **152-1** and the third portion **152-3** may be substantially 100% of the surface area (at the bonding interface) of the smallest one of the first portion **152-1** and the third portion **152-3**, as is shown in FIG. 7D for the example of the conductive via structure **150-2**. In other embodiments, the overlap between the sacrificial materials **160** of the first portion **152-1** and the third portion **152-3** may be partial (i.e., less than 100%) of the surface area (at the bonding interface) of the smallest one of the first portion **152-1** and the third portion **152-3**, as is shown in FIG. 7D for the example of the conductive via structure **150-1**. Bonding the IC structures **110** and **120** may include applying a suitable pressure and/or heat (e.g., between about 50 and 200 degrees Celsius) for a duration of time.

(113) The first fabrication method may further include a process **710**, shown in FIG. 7E, in which the support structure **132** of the IC structure **120** is removed, as the support structure **132** may

provide sufficient mechanical stability to the microelectronic assembly **100** and removing the support structure **132** of the IC structure **120** may advantageously decrease the overall height of the microelectronic assembly **100**. To that end, a suitable thinning/polishing process may be used to grind away the support structure **132** of the IC structure **120**. As a result of removing the support structure **132** of the IC structure **120**, the back side of the FEOL layer of the IC structure **120** may be exposed, thus exposing the bottoms (but now facing up) of the vias in which the first portions **152-1** are formed. Although not specifically shown in the present drawings, in other embodiments, the IC structure **120** may be provided over the IC structure **110** using layer transfer, in which case when the IC structure **120** is placed over the IC structure **110** for bonding, its' support structure **132** has already been substantially removed so that the bottoms of the vias with the first portions **152-1** are exposed.

(114) The first fabrication method may then proceed with a process **712**, shown in FIG. 7F, which includes providing an etch mask **162** over the surface of the IC structure **120**, leaving openings **164** for the bottoms of the vias with the first portions **152-1** that have been exposed in the process **710**. The etch mask **162** may include any suitable material to protect areas covered by the mask from being etched by etchants that can remove materials exposed by the openings **164**. The openings **164** may be provided in the etch mask **162** using any suitable etching technique (e.g., a dry etch, such as e.g., radio frequency (RF) reactive ion etch (RIE) or inductively coupled plasma (ICP) RIE) in combination with lithography (e.g., photolithography or electron-beam lithography) to define the locations and the sizes of these openings.

(115) The first fabrication method may further include a process **714**, shown in FIG. 7G, that includes etching the liner **156** that has been exposed by the openings **164**, using any suitable etching process such as a dry etch. In some embodiments, the etches performed in the fabrication methods described herein may include performing an anisotropic etch, using etchants in a form of e.g., chemically active ionized gas (i.e., plasma) using e.g., bromine (Br) and chloride (Cl) based chemistries. In some embodiments, during the etches of the fabrication methods described herein, the microelectronic assemblies may be heated to elevated temperatures, e.g., to temperatures between about room temperature and 200 degrees Celsius, including all values and ranges therein, to promote that byproducts of the etch are made sufficiently volatile to be removed from the surface.

(116) Because the sacrificial material **160** within the first portion **152-1** is sufficiently etch-selective with respect to the liner **156**, the etchants used to remove the liner **156** from the bottoms of the vias exposed by the openings **164** will not substantially etch the sacrificial material **160**. Therefore, the first fabrication method may further proceed with a process **716**, shown in FIG. 7H, in which the etchants are switched to continue etching the sacrificial material **160** that has been exposed in the process **714**. The etchants used in this process may be such that they will not substantially etch the liner **156**, but will etch the bonding interface **130**, and, therefore, continue etching the sacrificial material **160** in the IC structure **110**, until the etch process substantially stops when the electrically conductive fill material **144** (also etch-selective with respect to the sacrificial material **160**) is exposed, as is shown in FIG. 7H. In this manner, openings **166** may be formed within portions of the vias of the IC structures **120** and **110** that were filled with the sacrificial materials **160** and bonded together, the openings **166** extending from the top of the IC structure **120** and into the IC structure **110** until the electrically conductive fill material **144** is exposed.

(117) The first fabrication method may further include a process **718**, shown in FIG. 7I, that includes depositing the electrically conductive fill material **154** into the openings **166** and removing the etch mask **162**. The electrically conductive fill material **154** may be deposited using any of the deposition techniques described above for the electrically conductive fill material **144**. In some embodiments, the electrically conductive fill material **154** may be deposited through the etch mask **162**, and the etch mask **162** be removed after said deposition. In other embodiments, first, the etch mask **162** may be removed and then the electrically conductive fill material **154** may be deposited

in the openings **166**, possibly followed by a suitable polishing process to remove the excess of the electrically conductive fill material **154** over the top surface of the IC structure **120**. As a result of depositing the electrically conductive fill material **154** into the openings **166**, each of the first portion **152-1**, the second portion **152-2**, and the third portion **152-3** will have a continuous electrically conductive fill material **154**, which is expected to reduce the risk of delamination between the IC structures **110** and **120**.

(118) While the first fabrication method has been described with reference to the f2f arrangement of the IC structures **110** and **120**, it can be applied to f2b and b2b arrangements with modifications that would be apparent in view of the descriptions presented herein.

(119) FIGS. **8A-8I** illustrate cross-sectional side views during different processes of the second fabrication method for an example of a first and a second IC structures bonded using f2f bonding with modified via-last process, in accordance with some embodiments.

(120) The second fabrication method may begin with a process **802**, shown in FIG. **8A**, that includes fabricating the IC structures **110** and **120** separately, to include all of the desired components therein, including vias, except that for the vias that are later to form a conductive via structure **150** as described herein are filled with a sacrificial material **160**. The process **802** may be substantially the same as the process **702**, except that no liner is provided in the first portion **152-1** or the third portion **152-3**, as is shown in FIG. **8A**. The fourth portion **152-4** may include the liner **146** and be subsequently filled with the electrically conductive fill material **144**, as illustrated in FIG. **8A**, or the liner **146** may be omitted there as well.

(121) The second fabrication method may then proceed with a process **804**, shown in the upper portion of FIG. **8B**, that includes depositing a bonding material over some or all portions of the face of the IC structure **110** that is to be hybrid-bonded to the IC structure **120**. The process **804** may be substantially the same as the process **704**, except that now it is applied to the resulting IC structure(s) after the process **802**.

(122) The lower portion of FIG. **8B** illustrates a process **804'**, which may be a process that would replace the process **804** of the second fabrication method when performed as a part of the fourth fabrication method. As shown in the lower portion of FIG. **8B**, in the process **804'**, the bonding material **130** is not deposited over the sacrificial material **160** of the IC structure **110** (and, similarly, would not be deposited over the sacrificial material **160** of the IC structure **120** for the embodiments of the process **804'** where the bonding material **130** is deposited over the IC structure **120**). The rest of the descriptions of the process **804** and applicable to the process **804'**.

Furthermore, the fourth fabrication method may further include the rest of the processes of the second fabrication method provided with reference to FIGS. **8A-8I**, except with the minor modification due to the differences between the process **804** and **804'**.

(123) The second fabrication method may then include a process **806**, shown in FIG. **8C**, in which one of the IC structures is flipped over, e.g., the IC structure **120**, so that their faces that are supposed to be bonded are facing each other, and then a process **808**, shown in FIG. **8D**, which includes performing hybrid manufacturing of the IC structures **110** and **120**, so that at least a part of the first portion **152-1** overlaps at least a part of the third portion **152-3**, possibly with the bonding material in between. The processes **806** and **808** may be substantially the same as the processes **706** and **708**, except that now they are applied to the resulting IC structures after the process **804**.

(124) The second fabrication method may further include a process **810**, shown in FIG. **8E**, in which the support structure **132** of the IC structure **120** is removed, followed by a process **812**, shown in FIG. **8F**, which includes providing an etch mask **162** over the surface of the IC structure **120**, leaving openings **164** for the bottoms of the vias with the first portions **152-1** that have been exposed in the process **810**. The processes **810** and **812** may be substantially the same as the processes **710** and **712**, except that now they are applied to the resulting microelectronic assemblies (i.e., IC structures **110** and **120** bonded together) after the process **808**.

(125) The second fabrication method may further include a process **814**, shown in FIG. **8G**, that

includes performing an etch through the openings **164**. The process **814** is different from the process **714** because there is no liner **156** that needs to be removed first, so the process **814** may be substantially the same as the process **716**, except that now applied to the resulting microelectronic assembly after the process **812**. The openings **166**, as described above, are formed as a result of the process **814**.

(126) The second fabrication method may then include a process **816**, shown in FIG. **8H**, that includes depositing the liner **156** on the sidewalls and bottoms of the openings **166** formed in the process **814**, thus forming lined openings **168** as illustrated in FIG. **8H**. Any suitable conformal deposition process may be used to deposit the liner **156** in the process **816**. As a result of depositing the liner **156** in the process **816**, the liner **156** will continuously extend over the inner sidewalls of the first portion **152-1**, the second portion **152-2**, and the third portion **152-3**, and will also be present at the bottom of the third portion **152-3**, thus separating the electrically conductive fill materials **144** and **154** after the electrically conductive fill material **154** is deposited, which will be characteristic of the use of the second fabrication method and different from the first fabrication method where, if used, the liners in the first portion **152-1** and the third portion **152-3** are discontinuous from one another.

(127) The second fabrication method may further include a process **818**, shown in FIG. **8I**, that includes depositing the electrically conductive fill material **154** into the lined openings **168** and removing the etch mask **162**. The process **818** may be substantially the same as the process **718**, except that now applied to the resulting microelectronic assembly after the process **816**.

(128) While the first fabrication method has been described with reference to the f2b arrangement of the IC structures **110** and **120**, it can be applied to f2b and b2b arrangements with modifications that would be apparent in view of the descriptions presented herein.

(129) Furthermore, the third and the fourth fabrication methods may be easily derived from the illustrations of the first and second fabrication methods, as described above, and may also be applied to f2b and b2b arrangements with modifications that would be apparent in view of the descriptions presented herein.

(130) After-Bonding Interconnects

(131) FIGS. **1-8** illustrate interconnects that may be included in the IC structures **110**, **120** before these IC structures are bonded together. In some embodiments of hybrid manufacturing, the microelectronic assembly **100** may further include interconnects, such as conductive vias or conductive trenches, provided in one or both of the IC structures **110**, **120** after these IC structures have been bonded together. In the present disclosure, such interconnects are referred to as “after-bonding interconnects” and may include any combination of one or more after-bonding vias and/or after-bonding trenches. Providing one or more after-bonding interconnects may provide significant advantages in terms of its ability to provide electrical connectivity between various components of the microelectronic assembly **100** and/or reduced resistance. Because of the reduced resistance, the after-bonding interconnects may be particularly suitable for routing power to various components of the microelectronic assembly **100**, although they may also be used to route ground and/or signals to various components of the microelectronic assembly **100**. Also because of the reduced resistance, the after-bonding interconnects may be referred to as “express interconnects” (e.g., an express via) because they may allow routing power, ground, and/or signals to various components of the microelectronic assembly **100** faster than what would be achieved with the interconnects **112** and **122** that were in the individual IC structures **110**, **120** before they were bonded together.

(132) Some embodiments of the microelectronic assembly **100** with after-bonding interconnects are shown in FIGS. **9-11**. In particular, FIGS. **9-11** illustrate some embodiments of the microelectronic assembly **100** with an after-bonding via **170** formed from the top of the microelectronic assembly **100** fabricated according to hybrid manufacturing with modified via-last process for, respectively f2f, f2b, and b2b arrangements. However, in other embodiments of the microelectronic assemblies **100**, not shown in the present drawings, analogous after-bonding interconnects may be formed

from the bottoms of the microelectronic assemblies **100** shown in the present drawings (e.g., by flipping the microelectronic assemblies **100** shown herein upside down so that what is shown as the bottom side is facing up, and processing the side facing up to form the after-bonding vias as described herein). Furthermore, although also not specifically shown in the present drawings, after-bonding interconnects as described herein may extend through multiple pairs of IC structures **110** and **120** bonded together. While FIGS. **9-11** illustrate certain arrangements of various components of the IC structures **110** and **120**, one or more of the after-bonding interconnects as described with reference to the after-bonding via **170** of FIGS. **9-11** may be provided in any embodiments of the microelectronic assembly **100** as described with reference to FIGS. **1-8**. In other words, any embodiments described with reference to FIGS. **9-11** may be combined with any embodiments described with reference to FIGS. **1-8**.

(133) In various embodiments (e.g., for any embodiments of the microelectronic assembly **100** as described with reference to FIGS. **9-11**), such after-bonding interconnects may include at least an electrically conductive fill material and, optionally, a liner material. In some embodiments, one characteristic feature of the hybrid manufacturing of the IC structures **110** and **120** may be that a material composition of the electrically conductive fill material of the after-bonding interconnects may be different from a material composition of the electrically conductive fill materials **118**, **144** and/or a material composition of the electrically conductive fill materials **128**, **154**. Differences in material compositions of the electrically conductive fill materials **118**, **144**, **154**, and **128**, described above, are applicable to differences in material compositions of the electrically conductive fill materials of the after-bonding interconnects and the electrically conductive fill materials **118**, **144**, **128**, and/or **156**, and, therefore, in the interests of brevity, are not repeated. In some embodiments, another characteristic feature of the hybrid manufacturing of the IC structures **110** and **120** may be that a material composition of the liner of the after-bonding interconnects may be different from a material composition of the liners **119**, **146** and/or a material composition of the liners **129**, **156**. Differences in material compositions of the liners **119**, **146**, **129**, and **156**, described above, are applicable to differences in material compositions of the liners of the after-bonding interconnects and the liners **119**, **146**, **129**, **156**, and, therefore, in the interests of brevity, are not repeated.

(134) FIGS. **9-11** illustrate some embodiments of the microelectronic assembly **100** with an after-bonding via **170** for different bonding arrangements/configurations. For any of the embodiments of FIGS. **9-11** (e.g., irrespective of whether f2f, f2b, or b2b bonding is used), the after-bonding via **170** extending from the top (or an analogous after-bonding via extending from the bottom) of the microelectronic assembly **100** may stop at (i.e., may have a bottom that interfaces with) any of 1) an insulating material (e.g., the insulating material **114** or the insulating material **124**), 2) any of the etch-stop materials that may be present in the microelectronic assembly **100** (e.g., any of the etch-stop materials **113**, **123**), 3) the bonding material **130** or the bonding interface between the different IC structures bonded together, and/or 4) any portions of the interconnects **112**, **122** or other components of the IC structures **110**, **120** (e.g., the device circuitry **116**, **126**). In some embodiments, the location/depth of the bottom of the after-bonding via **170** extending from the top or from the bottom of the microelectronic assembly **100** may be based on a particular etch process used to form the after-bonding via **170**. For example, if an opening for the after-bonding via **170** is formed using a selective etch process with etchants that remove the insulating materials **114**, **124**, as well as an etch-stop material that may be present in the IC structure **120** (not shown), but do not remove, e.g., the etch-stop material that may be present in the IC structure **110** (not shown), then the after-bonding via **170** may have a bottom aligned with the top of the etch-stop material in the IC structure **110**. In another example, if an opening for the after-bonding via **170** is formed using a selective etch process with etchants that remove the insulating materials **114**, **124**, as well as the etch-stop material that may be present in the IC structure **120**, but do not remove the bonding material **130**, then the after-bonding via **170** may have a bottom aligned with the top of the bonding material **130**. To ensure this, respective materials are to be selected as being sufficiently etch-

selective. In other embodiments, some materials of the microelectronic assembly **100** may be sufficiently etch-selective with respect to one another and the after-bonding via **170** may still extend through them, which may be realized by using different etchants to form the opening for the after-bonding via **170** through different materials of the microelectronic assembly **100**. In other embodiments, the location/depth of the bottom of the after-bonding via **170** may be based on the time of the etch, where, generally, the longer is the time of the etch, the deeper is the opening for the after-bonding via **170**.

(135) FIGS. **9A-9C** illustrate cross-sectional side views of a microelectronic assembly that is an f2f-bonded assembly with modified via-last process and with an after-bonding via formed from the top, in accordance with some embodiments. The microelectronic assembly **100** shown in FIGS. **9A-9C** is similar to that shown in FIG. **1A**, except that it further includes the after-bonding via **170** formed from the top of the microelectronic assembly **100**. Such an after-bonding via may also be formed in the f2f embodiments of the microelectronic assembly as shown in FIGS. **1B-1D**. Furthermore, an after-bonding via similar to that described with reference to FIGS. **9A-9C** may be formed from the bottom for any of the embodiments of the microelectronic assembly as shown in FIGS. **1A-1D**.

(136) For the embodiments of FIGS. **9A-9C**, the after-bonding via **170** may be referred to as a “top f2f after-bonding via **170**.” For the f2f embodiments shown in FIG. **9**, in accordance with the previous descriptions, the “top” of the microelectronic assembly **100** is considered to be the backside **134-1** of the IC structure **120**, while the “bottom” of the microelectronic assembly **100** is considered to be the backside **134-1** of the IC structure **110**. Different ones of FIGS. **9A-9C** illustrate embodiments of the microelectronic assembly **100** with the top f2f after-bonding vias **170** of different depths.

(137) FIG. **9A** illustrates an embodiment where the top f2f after-bonding via **170** extends from the top of the microelectronic assembly **100** towards the bottom of the microelectronic assembly **100** and stops before reaching the bonding material **130**. Thus, FIG. **9A** illustrates that the top f2f after-bonding via **170** may be such that it is a blind via in the top IC structure, e.g., in the IC structure **120**. Although not specifically shown in the cross-section of FIG. **9A**, the top f2f after-bonding via **170** may be electrically coupled to one or more components of the IC structure **120** (e.g., to one or more of the interconnects **122**, devices of the device circuitry **126**, etc.) to provide power, ground, and/or signal to these components. Furthermore, although also not specifically shown in the cross-section of FIG. **9A**, in some embodiments, the top f2f after-bonding via **170** that is a blind via in the IC structure **120**, e.g., as shown in FIG. **9A**, may be electrically coupled to one or more components of the IC structure **110** (e.g., to one or more of the interconnects **112**, devices of the device circuitry **116**, etc.) to provide power, ground, and/or signal to these components of the IC structure **110**, which may, e.g., be done by the top f2f after-bonding via **170** being electrically coupled to one or more interconnects **122**, and one of more of those interconnects **122** being electrically coupled to one or more components of the IC structure **110**.

(138) FIG. **9B** illustrates an embodiment where the top f2f after-bonding via **170** extends from the top of the microelectronic assembly **100** towards the bottom of the microelectronic assembly **100**, extends through all of the top IC structure of the microelectronic assembly **100** (e.g., the IC structure **120**), the bonding material **130**, and into the bottom IC structure of the microelectronic assembly **100** (e.g., the IC structure **110**), and stops before reaching the bottom of the microelectronic assembly **100**. Thus, FIG. **9B** illustrates that the top f2f after-bonding via **170** may be such that it is a blind via extending through the top IC structure and into the bottom IC structure, e.g., from the IC structure **120** into the IC structure **110**. Although not specifically shown in the cross-section of FIG. **9B**, such top f2f after-bonding via **170** may be electrically coupled to one or more components of the IC structure **120** to provide power, ground, and/or signal to these components. Furthermore, although also not specifically shown in the cross-section of FIG. **9B**, the top f2f after-bonding via **170** that extends all through the IC structure **120** and forms a blind via in

the IC structure **110** as shown in FIG. **9B** may further be electrically coupled to one or more components of the IC structure **110** to provide power, ground, and/or signal to these components of the IC structure **110**.

(139) FIG. **9C** illustrates an embodiment where the top f2f after-bonding via **170** extends from the top of the microelectronic assembly **100** all the way to the bottom of the microelectronic assembly **100**. In this manner, the top f2f after-bonding via **170** is a TSV, extending between the opposing faces of the microelectronic assembly **100**. An electrical connection may then be made to the top f2f after-bonding via **170** from one or both of the top and the bottom of the microelectronic assembly **100**. Although not specifically shown in the cross-section of FIG. **9C**, the top f2f after-bonding via **170** that is a TSV in the microelectronic assembly **100** may be electrically coupled to one or more components of the IC structure **120** to provide power, ground, and/or signal to these components of the IC structure **120** and/or may be electrically coupled to one or more components of the IC structure **110** to provide power, ground, and/or signal to these components of the IC structure **110**.

(140) In some embodiments, one characteristic feature that is specific to the top f2f after-bonding via **170** extending from the top of the f2f-bonded microelectronic assembly as shown in FIGS. **9A-9C** may be that the cross-sectional trapezoidal shape of the top f2f after-bonding via **170** is inverted with respect to the trapezoidal shapes of the interconnects **122**. This is illustrated in FIGS. **9A-9C** with the short side of the trapezoidal cross-section of the top f2f after-bonding via **170** being closer to the bottom of the microelectronic assembly **100**, than its long side, and with the long sides of the trapezoidal cross-sections of the interconnects **122** being closer to the bottom of the microelectronic assembly **100** than their short sides. In some embodiments, another characteristic feature that is specific to the top f2f after-bonding via **170** extending from the top of the f2f-bonded microelectronic assembly as shown in FIGS. **9A-9C** may be that the cross-sectional trapezoidal shape of the top f2f after-bonding via **170** is oriented in the same manner (i.e., not inverted) as the trapezoidal shapes of the interconnects **112**. This is illustrated in FIGS. **9A-9C** with the short side of the trapezoidal cross-section of the top f2f after-bonding via **170** being closer to the bottom of the microelectronic assembly **100**, than its long side, and with the short sides of the trapezoidal cross-sections of the interconnects **112** being closer to the bottom of the microelectronic assembly **100** than their long sides.

(141) FIGS. **10A-10C** illustrate cross-sectional side views of a microelectronic assembly that is an f2b-bonded assembly with modified via-last process and with an after-bonding via formed from the top, in accordance with some embodiments. The microelectronic assembly **100** shown in FIGS. **10A-10C** is similar to that shown in FIG. **2A**, except that it further includes the after-bonding via **170** formed from the top of the microelectronic assembly **100**. Such an after-bonding via may also be formed in the f2b embodiments of the microelectronic assembly as shown in FIGS. **2B-2D**. Furthermore, an after-bonding via similar to that described with reference to FIGS. **10A-10C** may be formed from the bottom for any of the embodiments of the microelectronic assembly as shown in FIGS. **2A-2D**.

(142) For the embodiments of FIGS. **10A-10C**, the after-bonding via **170** may be referred to as a “top f2b after-bonding via **170**.” For the f2b embodiments shown in FIG. **10**, the “top” of the microelectronic assembly **100** is considered to be the frontside **134-2** of the IC structure **120**, while the “bottom” of the microelectronic assembly **100** is considered to be the backside **134-1** of the IC structure **110**. Different ones of FIGS. **10A-10C** illustrate embodiments of the microelectronic assembly **100** with the top f2b after-bonding via **170** of different depths.

(143) FIG. **10A** illustrates an embodiment where the top f2b after-bonding via **170** extends from the top of the microelectronic assembly **100** towards the bottom of the microelectronic assembly **100** and stops before reaching the bonding material **130** (i.e., FIG. **10A** is similar to FIG. **9A** except that FIG. **10A** illustrates the f2b bonding configuration). Thus, FIG. **10A** illustrates that the top f2b after-bonding via **170** may be such that it is a blind via in the top IC structure, e.g., in the IC

structure **120**. Although not specifically shown in the cross-section of FIG. **10A**, the top f2b after-bonding via **170** may be electrically coupled to one or more components of the IC structure **120** to provide power, ground, and/or signal to these components. Furthermore, although also not specifically shown in the cross-section of FIG. **10A**, in some embodiments, the top f2b after-bonding via **170** that is a blind via in the IC structure **120**, e.g., as shown in FIG. **10A**, may be electrically coupled to one or more components of the IC structure **110** to provide power, ground, and/or signal to these components of the IC structure **110**, which may, e.g., be done by the top f2b after-bonding via **170** being electrically coupled to one or more interconnects **122**, and one of more of those interconnects **122** being electrically coupled to one or more components of the IC structure **110**.

(144) FIG. **10B** illustrates an embodiment where the top f2b after-bonding via **170** extends from the top of the microelectronic assembly **100** towards the bottom of the microelectronic assembly **100**, extends through all of the top IC structure of the microelectronic assembly **100** (e.g., the IC structure **120**), the bonding material **130**, and into the bottom IC structure of the microelectronic assembly **100** (e.g., the IC structure **110**), and stops before reaching the bottom of the microelectronic assembly **100** (i.e., FIG. **10B** is similar to FIG. **9B** except that FIG. **10B** illustrates the f2b bonding configuration). Thus, FIG. **10B** illustrates that the top f2b after-bonding via **170** may be such that it is a blind via extending through the top IC structure and into the bottom IC structure, e.g., from the IC structure **120** into the IC structure **110**. Although not specifically shown in the cross-section of FIG. **10B**, such top f2b after-bonding via **170** may be electrically coupled to one or more components of the IC structure **120** to provide power, ground, and/or signal to these components. Furthermore, although also not specifically shown in the cross-section of FIG. **10B**, the top f2b after-bonding via **170** that extends all through the IC structure **120** and forms a blind via in the IC structure **110** as shown in FIG. **10B** may further be electrically coupled to one or more components of the IC structure **110** to provide power, ground, and/or signal to these components of the IC structure **110**.

(145) FIG. **10C** illustrates an embodiment where the top f2b after-bonding via **170** extends from the top of the microelectronic assembly **100** all the way to the bottom of the microelectronic assembly **100** (i.e., FIG. **10C** is similar to FIG. **9C** except that FIG. **10C** illustrates the f2b bonding configuration). In this manner, the top f2b after-bonding via **170** is a TSV, extending between the opposing faces of the microelectronic assembly **100**. An electrical connection may then be made to the top f2b after-bonding via **170** from one or both of the top and the bottom of the microelectronic assembly **100**. Although not specifically shown in the cross-section of FIG. **10C**, the top f2b after-bonding via **170** that is a TSV in the microelectronic assembly **100** may be electrically coupled to one or more components of the IC structure **120** to provide power, ground, and/or signal to these components of the IC structure **120** and/or may be electrically coupled to one or more components of the IC structure **110** to provide power, ground, and/or signal to these components of the IC structure **110**.

(146) In some embodiments, one characteristic feature that is specific to the top f2b after-bonding via **170** extending from the top of the f2b-bonded microelectronic assembly as shown in FIGS. **10A-10C** may be that the cross-sectional trapezoidal shape of the top f2b after-bonding via **170** is oriented in the same manner (i.e., not inverted) as the trapezoidal shapes of the interconnects **122**. This is illustrated in FIGS. **10A-10C** with the short side of the trapezoidal cross-section of the top f2b after-bonding via **170** being closer to the bottom of the microelectronic assembly **100**, than its long side, and with the short sides of the trapezoidal cross-sections of the interconnects **122** being closer to the bottom of the microelectronic assembly **100** than their long sides. In some embodiments, another characteristic feature that is specific to the top f2b after-bonding via **170** extending from the top of the f2f-bonded microelectronic assembly as shown in FIGS. **10A-10C** may be that the cross-sectional trapezoidal shape of the top f2b after-bonding via **170** is also oriented in the same manner (i.e., not inverted) as the trapezoidal shapes of the interconnects **112**.

This is illustrated in FIGS. **10A-10C** with the short side of the trapezoidal cross-section of the top f2b after-bonding via **170** being closer to the bottom of the microelectronic assembly **100**, than its long side, and with the short sides of the trapezoidal cross-sections of the interconnects **112** being closer to the bottom of the microelectronic assembly **100** than their long sides.

(147) FIGS. **11A-11C** illustrate cross-sectional side views of a microelectronic assembly **100** that is a b2b-bonded assembly with modified via-last process and with an after-bonding via **170** formed from the top, in accordance with some embodiments. The microelectronic assembly **100** shown in FIGS. **11A-11C** is similar to that shown in FIG. **3A**, except that it further includes the after-bonding via **170** formed from the top of the microelectronic assembly **100**. Such an after-bonding via may also be formed in the b2b embodiments of the microelectronic assembly as shown in FIGS. **3B-3D**, as well as in any of the b2b embodiments described with reference to FIGS. **4-6**. Furthermore, an after-bonding via similar to that described with reference to FIGS. **11A-11C** may be formed from the bottom for any of the embodiments of the microelectronic assembly as shown in FIGS. **3-6**.

(148) For the embodiments of FIGS. **11A-11C**, the after-bonding via **170** may be referred to as a “top b2b after-bonding via **170**.” For the b2b embodiments shown in FIG. **11**, the “top” of the microelectronic assembly **100** is considered to be the frontside **134-2** of the IC structure **120**, while the “bottom” of the microelectronic assembly **100** is considered to be the frontside **134-2** of the IC structure **110**. Different ones of FIGS. **11A-11C** illustrate embodiments of the microelectronic assembly **100** with the top b2b after-bonding via **170** of different depths.

(149) FIG. **11A** illustrates an embodiment where the top b2b after-bonding via **170** extends from the top of the microelectronic assembly **100** towards the bottom of the microelectronic assembly **100** and stops before reaching the bonding material **130** (i.e., FIG. **11A** is similar to FIGS. **9A** and **10A** except that FIG. **11A** illustrates the b2b bonding configuration). Thus, FIG. **11A** illustrates that the top b2b after-bonding via **170** may be such that it is a blind via in the top IC structure, e.g., in the IC structure **120**. Although not specifically shown in the cross-section of FIG. **11A**, the top b2b after-bonding via **170** may be electrically coupled to one or more components of the IC structure **120** to provide power, ground, and/or signal to these components. Furthermore, although also not specifically shown in the cross-section of FIG. **11A**, in some embodiments, the top b2b after-bonding via **170** that is a blind via in the IC structure **120**, e.g., as shown in FIG. **11A**, may be electrically coupled to one or more components of the IC structure **110** to provide power, ground, and/or signal to these components of the IC structure **110**, which may, e.g., be done by the top b2b after-bonding via **170** being electrically coupled to one or more interconnects **122**, and one of more of those interconnects **122** being electrically coupled to one or more components of the IC structure **110**.

(150) FIG. **11B** illustrates an embodiment where the top b2b after-bonding via **170** extends from the top of the microelectronic assembly **100** towards the bottom of the microelectronic assembly **100**, extends through all of the top IC structure of the microelectronic assembly **100** (e.g., the IC structure **120**), the bonding material **130**, and into the bottom IC structure of the microelectronic assembly **100** (e.g., the IC structure **110**), and stops before reaching the bottom of the microelectronic assembly **100** (i.e., FIG. **11B** is similar to FIGS. **9B** and **10C** except that FIG. **11B** illustrates the f2b bonding configuration). Thus, FIG. **11B** illustrates that the top b2b after-bonding via **170** may be such that it is a blind via extending through the top IC structure and into the bottom IC structure, e.g., from the IC structure **120** into the IC structure **110**. Although not specifically shown in the cross-section of FIG. **11B**, such top b2b after-bonding via **170** may be electrically coupled to one or more components of the IC structure **120** to provide power, ground, and/or signal to these components. Furthermore, although also not specifically shown in the cross-section of FIG. **11B**, the top b2b after-bonding via **170** that extends all through the IC structure **120** and forms a blind via in the IC structure **110** as shown in FIG. **11B** may further be electrically coupled to one or more components of the IC structure **110** to provide power, ground, and/or signal to these components of the IC structure **110**.

(151) FIG. 11C illustrates an embodiment where the top b2b after-bonding via **170** extends from the top of the microelectronic assembly **100** all the way to the bottom of the microelectronic assembly **100** (i.e., FIG. 11C is similar to FIGS. 9C and 10C except that FIG. 11C illustrates the b2b bonding configuration). In this manner, the top b2b after-bonding via **170** is a TSV, extending between the opposing faces of the microelectronic assembly **100**. An electrical connection may then be made to the top b2b after-bonding via **170** from one or both of the top and the bottom of the microelectronic assembly **100**. Although not specifically shown in the cross-section of FIG. 11C, the top b2b after-bonding via **170** that is a TSV in the microelectronic assembly **100** may be electrically coupled to one or more components of the IC structure **120** to provide power, ground, and/or signal to these components of the IC structure **120** and/or may be electrically coupled to one or more components of the IC structure **110** to provide power, ground, and/or signal to these components of the IC structure **110**.

(152) In some embodiments, one characteristic feature that is specific to the top b2b after-bonding via **170** extending from the top of the b2b-bonded microelectronic assembly as shown in FIGS. 11A-11C may be that the cross-sectional trapezoidal shape of the top b2b after-bonding via **170** is oriented in the same manner (i.e., not inverted) as the trapezoidal shapes of the interconnects **122**. This is illustrated in FIGS. 11A-11C with the short side of the trapezoidal cross-section of the top b2b after-bonding via **170** being closer to the bottom of the microelectronic assembly **100**, than its long side, and with the short sides of the trapezoidal cross-sections of the interconnects **122** being closer to the bottom of the microelectronic assembly **100** than their long sides. In some embodiments, another characteristic feature that is specific to the top b2b after-bonding via **170** extending from the top of the f2f-bonded microelectronic assembly as shown in FIGS. 11A-11C may be that the cross-sectional trapezoidal shape of the top b2b after-bonding via **170** is inverted with respect to the trapezoidal shapes of the interconnects **112**. This is illustrated in FIGS. 11A-11C with the short side of the trapezoidal cross-section of the top b2b after-bonding via **170** being closer to the bottom of the microelectronic assembly **100**, than its long side, and with the long sides of the trapezoidal cross-sections of the frontside interconnects **112** being closer to the bottom of the microelectronic assembly **100** than their short sides.

(153) While each of FIGS. 9-11 illustrates a single top after-bonding via **170** extending from the top of the microelectronic assembly **100**, in other embodiments, the microelectronic assembly **100** may include any number of two or more top after-bonding vias **170**, which may be of same or different depths.

(154) Example Packages

(155) Any embodiments of one or more microelectronic assemblies **100** fabricated using hybrid manufacturing with modified via-last process as described herein may be included in an IC package. Some examples of such IC packages are shown in FIGS. 12A-12H, illustrating IC packages that may include one or more microelectronic assemblies **100** fabricated using hybrid manufacturing with modified via-last process, in accordance with some embodiments.

(156) FIG. 12A illustrates an IC package **200** that includes a package substrate **202**, a die **204-1**, and a die **204-2**, according to some embodiments.

(157) The package substrate **202** may include an insulating material (e.g., a dielectric material formed in multiple layers, as known in the art) and one or more conductive pathways through the dielectric material (e.g., including conductive trenches and/or conductive vias, as shown). In some embodiments, the insulating material of the package substrate **202** may be a dielectric material, such as an organic dielectric material, a fire retardant grade 4 material (FR-4), bismaleimide triazine (BT) resin, polyimide materials, glass reinforced epoxy matrix materials, or low-k and ultra-low-k dielectric (e.g., carbon-doped dielectrics, fluorine-doped dielectrics, porous dielectrics, and organic polymeric dielectrics). In particular, when the package substrate **202** is formed using standard printed circuit board (PCB) processes, the package substrate **202** may include FR-4, and the conductive pathways in the package substrate **202** may be formed by patterned sheets of copper

separated by build-up layers of the FR-4. The conductive pathways in the package substrate **202** may be bordered by liner materials, such as adhesion liners and/or barrier liners, as suitable.

(158) In some embodiments, the package substrate **202** may be a lower density medium and any of the dies **204** may be a higher density medium. As used herein, the term “lower density” and “higher density” are relative terms indicating that the conductive pathways (e.g., including conductive lines and conductive vias) in a lower density medium are larger and/or have a greater pitch than the conductive pathways in a higher density medium. In some embodiments, a higher density medium may be manufactured using a modified semi-additive process or a semi-additive build-up process with advanced lithography (with small vertical interconnect features formed by advanced laser or lithography processes), while a lower density medium may be a PCB manufactured using a standard PCB process (e.g., a standard subtractive process using etch chemistry to remove areas of unwanted copper, and with coarse vertical interconnect features formed by a standard laser process).

(159) In some embodiments, the die **204-1** may include any of the IC structures disclosed herein and the die **204-2** may include any other of the IC structures disclosed herein so that the IC structure of the die **204-1** is bonded to the IC structure of the die **204-2** using hybrid manufacturing as described herein, with the bonding material/interface **130** as described herein shown in the illustration of FIG. **12A**. For example, in some embodiments, the die **204-1** may include the IC structure **110** while the die **204-2** may include the IC structure **120**, or vice versa, with the IC structures **110** and **120** of these dies being bonded to form a microelectronic assembly **100** according to any embodiments of the present disclosure. In other embodiments, any of the dies **204-1** and **204-2** may include a stack of two or more IC structures bonded together as described herein, with the IC structures of these dies being bonded to form a microelectronic assembly **100** according to any embodiments of the present disclosure. For example, in some embodiments, the die **204-1** may include a stack of the IC structure **110-1** bonded to the IC structure **120-1**, while the die **204-2** may include a stack of the IC structure **110-2** bonded to the IC structure **120-2**, and the IC structure **120-1** of the stack of the die **204-1** may be bonded to the IC structure **110-2** of the stack of the die **204-2** with the bonding material/interface **130** that is shown in FIG. **12A**. In general, any of the dies **204-1** and **204-2** may include any number of one or more IC structures bonded with other IC structures, either within the same die or with the IC structures of the other die, to form the microelectronic assembly **100** according to any of the hybrid manufacturing embodiments described herein. Example structures that may be included in the dies **204** disclosed herein are discussed below with reference to FIG. **14**.

(160) In various embodiments of the IC package **200**, the different dies **204** (as shown in any of FIGS. **12A-12H**) may include any suitable circuitry. For example, in some embodiments, the die **204-1** may be an active or passive die, and the die **204-2** may include input/output circuitry, high-bandwidth memory, and/or enhanced/embedded dynamic random-access memory (EDRAM). Because the dies **204-1** and **204-2** include the IC structures **110** and/or **120** as described herein, descriptions of the circuits and devices that the IC structures **110**, **120** may include, or be a part of, provided above, are applicable to the dies **204-1** and **204-2**. For example, in some embodiments, the die **204-1** may include a power delivery circuitry and the die **204-2** may include a memory device, e.g., a high-bandwidth memory device, or, in other embodiments, the die **204-1** may include input/output circuitry and the die **204-2** may include a field programmable gate array logic. In some embodiments, any of the dies **204** may include one or more device layers including transistors (e.g., as discussed below with reference to FIG. **14**).

(161) In some embodiments, one of the dies **204-1** and **204-2** may have a smaller footprint than the other, e.g., shown in the example of FIG. **12A** with the die **204-2** being narrower than the die **204-1**. For example, in some embodiments, a width of the die **204-1** may be larger than a width of the die **204-2** by a distance **206**. In some embodiments, the distance **206** may be between about 0.5 millimeters and 5 millimeters (e.g., between about 0.75 millimeters and 2 millimeters, or

approximately 1 millimeter). Although the die **204-2** is shown to be aligned with the die **204-1** on the right side, as the IC structures **110** and **120** are shown to be aligned in various ones of FIGS. **1-11**, there does not have to be such alignment in various embodiments of the IC packages **200** and in various embodiments of the microelectronic assemblies **100** as described herein.

(162) As shown in FIG. **12A**, the package substrate **202** may be coupled to the die **204-1** by die-to-package substrate (DTPS) interconnects **210**. In particular, the top surface of the package substrate **202** may include a set of conductive contacts **212**, and the bottom surface of the die **204-1** may include a set of conductive contacts **214**; the conductive contacts **214** at the bottom surface of the die **204-1** may be electrically and mechanically coupled to the conductive contacts **212** at the top surface of the package substrate **202** by the DTPS interconnects **210**. Any of the conductive contacts disclosed herein (e.g., the conductive contacts **212**, **214**, **222**, **224**, **232**, **234**, etc.) may include bond pads, posts, or any other suitable conductive contacts, for example.

(163) In the embodiment of FIG. **12A**, the die **204-1** is not be disposed in a recess in the package substrate **202**, but, in other embodiments of the IC package **200**, the top surface of the package substrate **202** may include a recess **208** in which the die **204-1** is at least partially disposed and the conductive contacts **212** to which the die **204-1** is coupled may be located at the bottom of the recess **208**, as shown in FIG. **12B**. For example, in some embodiments, the recess **208** may be laser-drilled down to a planar metal stop in the package substrate **202** (not shown); once the metal stop is reached, the metal stop may be removed to expose the conductive contacts **212** at the bottom of the recess **208**. In some embodiments, the recess **208** may be formed by a mechanical drill. In some embodiments, a depth **216** of the recess **208** may be between about 10 microns and 200 microns (e.g., between about 10 microns and 30 microns, between about 30 microns and 100 microns, between about 60 microns and 80 microns, or approximately 75 microns). In some embodiments, the depth **216** may be equal to a certain number of layers of the dielectric material in the package substrate **202**. For example, the depth **216** may be approximately equal to between one and five layers of the dielectric material in the package substrate **202** (e.g., two or three layers of the dielectric material). In some embodiments, the depth **216** may be equal to the thickness of a solder resist material (not shown) on the top surface of the package substrate **202**. In some embodiments, the top surface of the die **204-1** may extend higher than the top surface of the package substrate **202**, as illustrated in FIG. **12B**. In other embodiments, the top surface of the die **204-1** may be substantially coplanar with the top surface of the package substrate **202** or may be recessed below the top surface of the package substrate **202**.

(164) The rest of the descriptions provided with respect to FIG. **12A** are applicable to, and may be combined with, the embodiment of FIG. **12B** (and vice versa), and, therefore, in the interests of brevity, are not repeated.

(165) The DTPS interconnects **210** disclosed herein may take any suitable form. In some embodiments, a set of DTPS interconnects **210** may include solder (e.g., solder bumps or balls that are subject to a thermal reflow to form the DTPS interconnects **210**). DTPS interconnects **210** that include solder may include any appropriate solder material, such as lead/tin, tin/bismuth, eutectic tin/silver, ternary tin/silver/copper, eutectic tin/copper, tin/nickel/copper, tin/bismuth/copper, tin/indium/copper, tin/zinc/indium/bismuth, or other alloys. In some embodiments, a set of DTPS interconnects **210** may include an anisotropic conductive material, such as an anisotropic conductive film or an anisotropic conductive paste. An anisotropic conductive material may include conductive materials dispersed in a non-conductive material. In some embodiments, an anisotropic conductive material may include microscopic conductive particles embedded in a binder or a thermoset adhesive film (e.g., a thermoset biphenyl-type epoxy resin, or an acrylic-based material). In some embodiments, the conductive particles may include a polymer and/or one or more metals (e.g., nickel or gold). For example, the conductive particles may include nickel-coated gold or silver-coated copper that is in turn coated with a polymer. In another example, the conductive particles may include nickel. When an anisotropic conductive material is uncompressed, there may

be no conductive pathway from one side of the material to the other. However, when the anisotropic conductive material is adequately compressed (e.g., by conductive contacts on either side of the anisotropic conductive material), the conductive materials near the region of compression may contact each other so as to form a conductive pathway from one side of the film to the other in the region of compression.

(166) FIG. 12C illustrates that, in some embodiments, the IC package **200** may also include a circuit board **218**. The package substrate **202** may be coupled to the circuit board **218** by second-level interconnects **220** at the bottom surface of the package substrate **202**. In particular, the package substrate **202** may include conductive contacts **222** at its bottom surface, and the circuit board **218** may include conductive contacts **224** at its top surface. The second-level interconnects **220** may electrically and mechanically couple the conductive contacts **224** and the conductive contacts **222**. The second-level interconnects **220** illustrated in FIG. 12C are solder balls (e.g., for a ball grid array arrangement), but any suitable second-level interconnects **220** may be used (e.g., pins in a pin grid array arrangement or lands in a land grid array arrangement) in other embodiments of the IC package **200**. The circuit board **218** may be a motherboard, for example, and may have other components attached to it (not shown). The circuit board **218** may include conductive pathways and other conductive contacts (not shown) for routing power, ground, and signals through the circuit board **218**, as known in the art. In some embodiments, the second-level interconnects **220** may not couple the package substrate **202** to a circuit board **218** but may instead couple the package substrate **202** to another IC package, an interposer, or any other suitable component.

(167) In some embodiments, one or more of the conductive pathways in the package substrate **202** may extend between one or more conductive contacts **212** at the top surface of the package substrate **202** and one or more conductive contacts **222** at the bottom surface of the package substrate **202**, as shown in FIG. 12C. In some embodiments, one or more of the conductive pathways in the package substrate **202** may extend between one or more conductive contacts **212** at the bottom of the recess **208** as shown in FIG. 12B and one or more conductive contacts **222** at the bottom surface of the package substrate **202**. In some embodiments, one or more of the conductive pathways in the package substrate **202** may extend between different conductive contacts **212** at the top surface of the package substrate **202**. In some embodiments, one or more of the conductive pathways in the package substrate **202** may extend between different conductive contacts **222** at the bottom surface of the package substrate **202**. The rest of the descriptions provided with respect to FIG. 12A and FIG. 12B are applicable to, and may be combined with, the embodiment of FIG. 12C (and vice versa), and, therefore, in the interests of brevity, are not repeated.

(168) In some embodiments, the die **204-1** of the IC package **200** as described herein may be a single-sided die (in the sense that the die **204-1** only has conductive contacts **214** on a single surface), as, e.g., shown in FIGS. 12A-12C. In other embodiments, the die **204-1** of the IC package **200** as described herein may be a double-sided (or “multi-level,” or “omni-directional”) die (in the sense that the die **204-1** has conductive contacts **214** on one surface and further conductive contacts on the opposing surface). One example of such implementation is shown in FIG. 12D.

(169) FIG. 12D illustrates that, in some embodiments, the IC package **200** may further include a die **204-3**. In some embodiments, the die **204-3** may be electrically and mechanically coupled to the die **204-1** by die-to-die (DTD) interconnects **230**. In particular, the top surface of the die **204-1** may include a set of conductive contacts **232**, and the bottom surface of the die **204-3** may include a set of conductive contacts **234**. One or more of the conductive contacts **234** at the bottom surface of the die **204-3** may be electrically and mechanically coupled to some of the conductive contacts **232** at the top surface of the die **204-1** by the DTD interconnects **230**. FIG. 12D illustrates that the pitch of the DTD interconnects **230** may be different from the pitch of the DTPS interconnects **210** in some embodiments (in other embodiments, these pitches may be substantially the same). In some embodiments, the die **204-3** of the IC package **200** may be a single-sided die (in the sense

that the die **204-3** only has conductive contacts **234** on a single surface), as shown in FIG. **12D**. In other embodiments (not specifically shown), the die **204-3** of the IC package **200** as described herein may be a double-sided die (in the sense that the die **204-3** has conductive contacts **234** on one surface and further conductive contacts on the opposing surface, the latter configured to couple the die **204-3** to further components). Although not specifically shown in FIG. **12D**, in some embodiments, the die **204-3** may be electrically and mechanically coupled to the package substrate **202** by DTPS interconnects similar to how the die **204-1** is coupled to the package substrate **202** by the DTPS interconnects **210**.

(170) The DTD interconnects **230** disclosed herein may take any suitable form. The DTD interconnects **230** may have a finer pitch than the DTPS interconnects **210** in an IC package. In some embodiments, the dies **204** on either side of a set of DTD interconnects **230** may be unpackaged dies, and/or the DTD interconnects **230** may include small conductive bumps or pillars (e.g., copper bumps or pillars) attached to the respective conductive contacts **232**, **234** by solder. The DTD interconnects **230** may have too fine a pitch to couple to the package substrate **202** directly (e.g., too fine to serve as DTPS interconnects **210**). In some embodiments, a set of DTD interconnects **230** may include solder. DTD interconnects **230** that include solder may include any appropriate solder material, such as any of the materials discussed above. In some embodiments, a set of DTD interconnects **230** may include an anisotropic conductive material, such as any of the materials discussed above. In some embodiments, the DTD interconnects **230** may be used as data transfer lanes, while the DTPS interconnects **210** may be used for power and ground lines, among others.

(171) In some embodiments, some or all of the DTD interconnects **230** in an IC package **200** may be metal-to-metal interconnects (e.g., copper-to-copper interconnects, or plated interconnects). In such embodiments, the conductive contacts **232**, **234** on either side of the DTD interconnect **230** may be bonded together (e.g., under elevated pressure and/or temperature) without the use of intervening solder or an anisotropic conductive material. In some embodiments, a thin cap of solder may be used in a metal-to-metal interconnect to accommodate planarity, and this solder may become an intermetallic compound during processing. In some metal-to-metal interconnects that utilize hybrid manufacturing, a dielectric material (e.g., silicon oxide, silicon nitride, silicon carbide, or an organic layer) may be present between the metals bonded together (e.g., between copper pads or posts that provide the associated conductive contacts **232**, **234**). In some embodiments, one side of a DTD interconnect **230** may include a metal pillar (e.g., a copper pillar), and the other side of the DTD interconnect may include a metal contact (e.g., a copper contact) recessed in a dielectric. In some embodiments, a metal-to-metal interconnect (e.g., a copper-to-copper interconnect) may include a noble metal (e.g., gold) or a metal whose oxides are conductive (e.g., silver). In some embodiments, a metal-to-metal interconnect may include metal nanostructures (e.g., nanorods) that may have a reduced melting point. Metal-to-metal interconnects may be capable of reliably conducting a higher current than other types of interconnects; for example, some solder interconnects may form brittle intermetallic compounds when current flows, and the maximum current provided through such interconnects may be constrained to mitigate mechanical failure.

(172) In some embodiments, some or all of the DTD interconnects **230** in an IC package **200** may be solder interconnects that include a solder with a higher melting point than a solder included in some or all of the DTPS interconnects **210**. For example, when the DTD interconnects **230** in a IC package **200** are formed before the DTPS interconnects **210** are formed, solder-based DTD interconnects **230** may use a higher-temperature solder (e.g., with a melting point above 200 degrees Celsius), while the DTPS interconnects **210** may use a lower-temperature solder (e.g., with a melting point below 200 degrees Celsius). In some embodiments, a higher-temperature solder may include tin; tin and gold; or tin, silver, and copper (e.g., 96.5% tin, 3% silver, and 0.5% copper). In some embodiments, a lower-temperature solder may include tin and bismuth (e.g.,

eutectic tin bismuth) or tin, silver, and bismuth. In some embodiments, a lower-temperature solder may include indium, indium and tin, or gallium.

(173) In the IC packages **200** disclosed herein, some or all of the DTPS interconnects **210** may have a larger pitch than some or all of the DTD interconnects **230**. DTD interconnects **230** may have a smaller pitch than DTPS interconnects **210** due to the greater similarity of materials in the different dies **204** on either side of a set of DTD interconnects **230** than between the die **204** and the package substrate **202** on either side of a set of DTPS interconnects **210**. In particular, the differences in the material composition of a die **204** and a package substrate **202** may result in differential expansion and contraction of the die **204** and the package substrate **202** due to heat generated during operation (as well as the heat applied during various manufacturing operations). To mitigate damage caused by this differential expansion and contraction (e.g., cracking, solder bridging, etc.), the DTPS interconnects **210** may be formed larger and farther apart than DTD interconnects **230**, which may experience less thermal stress due to the greater material similarity of the pair of dies **204** on either side of the DTD interconnects. In some embodiments, the DTPS interconnects **210** disclosed herein may have a pitch between about 80 microns and 300 microns, while the DTD interconnects **230** disclosed herein may have a pitch between about 7 microns and 100 microns.

(174) The rest of the descriptions provided with respect to FIG. **12A-12C** are applicable to, and may be combined with, the embodiment of FIG. **12D** (and vice versa), and, therefore, in the interests of brevity, are not repeated.

(175) FIG. **12E** illustrates that, in some embodiments, the IC package **200** may further include a die **204-4**, bonded to the die **204-3** using hybrid manufacturing as described herein. In some embodiments, the die **204-3** may include any of the IC structures disclosed herein and the die **204-4** may include any other of the IC structures disclosed herein so that the IC structure of the die **204-3** is bonded to the IC structure of the die **204-4** using hybrid manufacturing as described herein, with the bonding material/interface **130-2** as described herein shown in the illustration of FIG. **12E**. For example, in some embodiments, the die **204-3** may include the IC structure **110** while the die **204-4** may include the IC structure **120**, or vice versa, with the IC structures **110** and **120** of these dies being bonded to form a microelectronic assembly **100-2** according to any embodiments of the present disclosure. In other embodiments, any of the dies **204-3** and **204-4** may include a stack of two or more IC structures bonded together as described herein, with the IC structures of these dies being bonded to form a microelectronic assembly **100-2** according to any embodiments of the present disclosure. For example, in some embodiments, the die **204-3** may include a stack of the IC structure **110-1** bonded to the IC structure **120-1**, while the die **204-4** may include a stack of the IC structure **110-2** bonded to the IC structure **120-2**, and the IC structure **120-1** of the stack of the die **204-3** may be bonded to the IC structure **110-2** of the stack of the die **204-4** with the bonding material/interface **130-2** that is shown in FIG. **12E**. In general, any of the dies **204-3** and **204-4** may include any number of one or more IC structures bonded with other IC structures, either within the same die or with the IC structures of the other die, to form the microelectronic assembly **100-2** according to any of the hybrid manufacturing embodiments described herein. The microelectronic assembly **100** and the bonding material **130** as were shown in FIG. **12A** have been re-labeled in FIG. **12E** as, respectively, the microelectronic assembly **100-1** and the bonding material **130-1**, to differentiate between different instances of the microelectronic assembly **100** and the bonding material **130** as shown in FIG. **12E**.

(176) In some embodiments, one of the dies **204-3** and **204-4** may have a smaller footprint than the other, e.g., shown in the example of FIG. **12E** with the die **204-3** being narrower than the die **204-4**. For example, in some embodiments, a width of the die **204-4** may be larger than a width of the die **204-3** by a distance that may be between about 0.5 millimeters and 5 millimeters (e.g., between about 0.75 millimeters and 2 millimeters, or approximately 1 millimeter). Although the die **204-4** is shown to be aligned with the die **204-3** on the right side, there does not have to be such alignment

in various embodiments of the IC packages **200** as described herein. The rest of the descriptions provided with respect to FIG. **12A-12D** are applicable to, and may be combined with, the embodiment of FIG. **12E** (and vice versa), and, therefore, in the interests of brevity, are not repeated.

(177) FIG. **12F** illustrates that, in some embodiments, the IC package **200** may further include a die **204-5**. In some embodiments, the die **204-5** may be electrically and mechanically coupled to the die **204-2** by DTD interconnects **240**, similar to the DTD interconnects **230**, described above. In particular, the top surface of the die **204-2** may include a set of conductive contacts **242**, and the bottom surface of the die **204-5** may include a set of conductive contacts **244**. One or more of the conductive contacts **244** at the bottom surface of the die **204-5** may be electrically and mechanically coupled to some of the conductive contacts **242** at the top surface of the die **204-2** by the DTD interconnects **240**. FIG. **12F** illustrates that the pitch of the DTD interconnects **240** may be different from the pitch of the DTPS interconnects **210** in some embodiments (in other embodiments, these pitches may be substantially the same). In some embodiments, the die **204-5** of the IC package **200** may be a single-sided die (in the sense that the die **204-5** only has conductive contacts **244** on a single surface), as shown in FIG. **12F**. In other embodiments (not specifically shown), the die **204-5** of the IC package **200** as described herein may be a double-sided die (in the sense that the die **204-5** has conductive contacts **244** on one surface and further conductive contacts on the opposing surface, the latter configured to couple the die **204-5** to further components). Although not specifically shown in FIG. **12F**, in some embodiments, the die **204-5** may be electrically and mechanically coupled to the package substrate **202** by DTPS interconnects similar to how the die **204-1** is coupled to the package substrate **202** by the DTPS interconnects **210**. The rest of the descriptions provided with respect to FIG. **12A-12E** are applicable to, and may be combined with, the embodiment of FIG. **12F** (and vice versa), and, therefore, in the interests of brevity, are not repeated.

(178) FIG. **12G** illustrates an embodiment of the IC package **200** that includes the dies **204-3** and **204-4** as shown in FIG. **12E** and the die **204-5** as shown in FIG. **12F**. FIG. **12G** further illustrates after-bonding vias **170-1** provided in the microelectronic assembly **100-1** and after-bonding vias **170-2** provided in the microelectronic assembly **100-2**. Also a certain number and certain arrangements of the dies **204** and the after-bonding vias **170** are shown in FIG. **12G**, this is purely for illustrative purposes and in other embodiments, other arrangements of the dies **204** and the after-bonding vias **170** are possible and within the scope of the present disclosure. Any of the after-bonding vias **170** shown in FIG. **12G** may be implemented according to any of the embodiments of the after-bonding vias **170** as described herein, e.g., according to any of the embodiments of FIGS. **9-11**.

(179) FIG. **12G** is intended to illustrate two features that may be implemented in various embodiments of the IC package **200**.

(180) One feature is that, in general, any of the after-bonding vias **170** in the one or more of the microelectronic assemblies **100** included in the IC package **200** may, but do not have to, couple to conductive contacts that couple the dies to other components via, e.g., DTD interconnects or DTPS interconnects. For example, FIG. **12G** illustrates that at least some of the after-bonding vias **170-1** may be coupled to one or more of the conductive contacts **214** at the bottom of the microelectronic assembly **100-1** and to one or more of the conductive contacts **242** at the top of the microelectronic assembly **100-1**. Although not specifically shown, in further embodiments of FIG. **12G**, at least some of the after-bonding vias **170-1** may not be coupled to the conductive contacts **214** and/or to the conductive contacts **242**. FIG. **12G** further illustrates that at least some of the after-bonding vias **170-2** may not be coupled to any of the conductive contacts. Although not specifically shown, in further embodiments of FIG. **12G**, at least some of the after-bonding vias **170-2** may be coupled to one or more of the conductive contacts **234**.

(181) Another feature is that, in general, the pitch of various interconnects between different dies

204 of the IC package **200** may be any suitable pitch and, in some embodiments, some or all of the pitches may be different, or some or all of the pitches may be the same. For example, in various embodiments, the pitch of the after-bonding vias **170-1** may be 1) different or the same as the pitch of the DTPS interconnects **210**, and/or 2) different or the same as the pitch of the DTD interconnects **230** and/or **240**, and/or 3) different or the same as the pitch of the after-bonding vias **170-2**. In another example, in various embodiments, the pitch of the after-bonding vias **170-2** may be 1) different or the same as the pitch of the DTPS interconnects **210**, and/or 2) different or the same as the pitch of the DTD interconnects **230** and/or **240**, and/or 3) different or the same as the pitch of the after-bonding vias **170-1**. In yet another embodiment, the pitch of the DTPS interconnects **210** may be 1) different or the same as the pitch of the after-bonding vias **170-1**, and/or 2) different or the same as the pitch of the DTD interconnects **230** and/or **240**, and/or 3) different or the same as the pitch of the after-bonding vias **170-2**. Although not specifically shown in FIG. 12, in various further embodiments of the IC package **200**, any of the dies **204** may be a mixed-pitch die (in the sense that any of the dies **204** may have multiple sets of conductive contacts and/or multiple sets of the after-bonding vias **170** with different pitch). The rest of the descriptions provided with respect to FIG. 12A-12F are applicable to, and may be combined with, the embodiment of FIG. 12G (and vice versa), and, therefore, in the interests of brevity, are not repeated.

(182) FIG. 12H illustrates an embodiment of the IC package **200** that may further include one or more of a mold material **250**, a thermal interface material (TIM) **252**, and a heat spreader **254**. Although the mold material **250**, the TIM **252**, and the heat spreader **254** are shown for the example of the IC package **200** as shown in FIG. 12D (where not all of the reference numerals shown in FIG. 12D are shown in FIG. 12H in order to not clutter the drawing), one or more of the mold material **250**, the TIM **252**, and the heat spreader **254** may be used with any other embodiments of the IC package **200** as described herein.

(183) The mold material **250** may extend around one or more of the dies **204** on the package substrate **202**. In some embodiments, the mold material **250** may extend above one or more of the dies **204** on the package substrate **202**. In some embodiments, the mold material **250** may extend between one or more of the dies **204** and the package substrate **202** around the associated DTPS interconnects **210**. In such embodiments, the mold material **250** may serve as an underfill material. In some embodiments, the mold material **250** may extend between different ones of the dies **204** around the associated DTD interconnects **230**. In such embodiments, the mold material **250** may serve as an underfill material. The mold material **250** may include multiple different mold materials (e.g., an underfill material, and a different overmold material). The mold material **250** may be an insulating material, such as an appropriate epoxy material. In some embodiments, the mold material **250** may include an underfill material that is an epoxy flux that assists with soldering the die **204-1** to the package substrate **202** when forming the DTPS interconnects **210**, and then polymerizes and encapsulates the DTPS interconnects **210**. The mold material **250** may be selected to have a coefficient of thermal expansion (CTE) that may mitigate or minimize the stress between the dies **204** and the package substrate **202** arising from uneven thermal expansion in the IC package **200**. In some embodiments, the CTE of the mold material **250** may have a value that is intermediate to the CTE of the package substrate **202** (e.g., the CTE of the dielectric material of the package substrate **202**) and a CTE of the dies **204**.

(184) The TIM **252** may include a thermally conductive material (e.g., metal particles) in a polymer or other binder. The TIM **252** may be a thermal interface material paste or a thermally conductive epoxy (which may be a fluid when applied and may harden upon curing, as known in the art). The TIM **252** may provide a path for heat generated by the dies **204** to readily flow to the heat spreader **254**, where it may be spread and/or dissipated. Some embodiments of the IC package **200** of FIG. 12H may include a sputtered backside metallization (not shown) across the mold material **250** and the dies **204**. In such embodiments, the TIM **252** (e.g., a solder TIM) may be disposed on this

backside metallization.

(185) The heat spreader **254** may be used to move heat away from the dies **204** (e.g., so that the heat may be more readily dissipated by a heat sink or other thermal management device). The heat spreader **254** may include any suitable thermally conductive material (e.g., metal, appropriate ceramics, etc.), and may include any suitable features (e.g., fins). In some embodiments, the heat spreader **254** may be an integrated heat spreader.

(186) The rest of the descriptions provided with respect to FIG. **12A-12G** are applicable to, and may be combined with, the embodiment of FIG. **12H** (and vice versa), and, therefore, in the interests of brevity, are not repeated.

(187) In some embodiments of FIG. **12**, the die **204-1** may provide high density interconnect routing in a localized area of the IC package **200**. In some embodiments, the presence of the die **204-1** may support direct chip attach of fine-pitch semiconductor dies (e.g., the dies **204-2** and **204-3**) that cannot be attached entirely directly to the package substrate **202**. In particular, as discussed above, the die **204-1** may support trench widths and spacings that are not achievable in the package substrate **202**. The proliferation of wearable and mobile electronics, as well as Internet of Things (IoT) applications, are driving reductions in the size of electronic systems, but limitations of the PCB manufacturing process and the mechanical consequences of thermal expansion during use have meant that chips having fine interconnect pitch cannot be directly mounted to a PCB. Various embodiments of the IC packages **200** disclosed herein may be capable of supporting chips with high density interconnects and chips with low-density interconnects without sacrificing performance or manufacturability.

(188) In various embodiments of FIG. **12**, any of the dies **204-3** and **204-5** may be a single-sided, single-pitch die; in other embodiments, any of the dies **204-3** and **204-5** may be a double-sided die, and additional components may be disposed on the top surface of any of the dies **204-3** and **204-5**. Additional passive components, such as surface-mount resistors, capacitors, and/or inductors, may be disposed on the top surface or the bottom surface of the package substrate **202**, or embedded in the package substrate **202**. More generally, any suitable number of the dies **204** in an IC package **200** may be double-sided dies **204**.

(189) The elements of the IC package **200** may have any suitable dimensions. Only a subset of the accompanying figures are labeled with reference numerals representing dimensions, but this is simply for clarity of illustration, and any of the IC packages **200** disclosed herein may have components having the dimensions discussed herein. For example, in some embodiments, a thickness **226** (labeled in FIG. **12H**) of the package substrate **202** may be between about 0.1 millimeters and 1.4 millimeters (e.g., between about 0.1 millimeters and 0.35 millimeters, between about 0.25 millimeters and 0.8 millimeters, or approximately 1 millimeter).

(190) Example Devices

(191) The microelectronic assemblies manufactured according to hybrid manufacturing with modified via-last process and IC packages that include such microelectronic assemblies, disclosed herein, may be included in any suitable electronic device. FIGS. **13-16** illustrate various examples of apparatuses that may include one or more of the microelectronic assemblies and IC packages disclosed herein.

(192) FIGS. **13A** and **13B** are top views of, respectively, a wafer and dies that may be included in one or more of the microelectronic assemblies **100** fabricated using hybrid manufacturing with modified via-last process in accordance with any of the embodiments disclosed herein. In some embodiments, the dies **1502** may be included in an IC package (e.g., the IC package **200** as shown in FIG. **12**) and/or in an IC device (e.g., the IC device **1600** as shown in FIG. **14**), in accordance with any of the embodiments disclosed herein. For example, any of the dies **1502** may serve as the die **204** shown in FIG. **12** and/or any of the dies **1502** may serve as the IC device **1600** shown in FIG. **14**. The wafer **1500** may be composed of semiconductor material and may include one or more dies **1502** having IC structures formed on a surface of the wafer **1500**. Each of the dies **1502**

may be a repeating unit of a semiconductor product that includes any suitable IC (e.g., ICs including IC structures to be included in microelectronic assemblies fabricated using hybrid manufacturing with modified via-last process as described herein, e.g., ICs including the IC structures **110** and/or **120** as described herein). After the fabrication of the semiconductor product is complete (e.g., after manufacture of one or more of the IC structures to be included in microelectronic assemblies fabricated using hybrid manufacturing as described herein, e.g., ICs including the IC structures **110** and/or **120** as described herein), the wafer **1500** may undergo a singulation process in which each of the dies **1502** is separated from one another to provide discrete “chips” of the semiconductor product. In particular, devices that include one or more IC structures to be included in microelectronic assemblies fabricated using hybrid manufacturing as described herein, e.g., ICs including the IC structures **110** and/or **120** as disclosed herein may take the form of the wafer **1500** (e.g., not singulated) or the form of the die **1502** (e.g., singulated). The die **1502** may include one or more transistors (e.g., some of the transistors **1640** of FIG. **14**, discussed below), supporting circuitry to route electrical signals to the transistors, passive components (e.g., signal trenches, resistors, capacitors, or inductors), and/or any other IC components. The die **1502** may include supporting circuitry to route electrical signals to various memory cells, transistors, capacitors, as well as any other IC components. In some embodiments, the wafer **1500** or the die **1502** may implement or include a memory device (e.g., a static random-access memory (SRAM) device), a logic device (e.g., an AND, OR, NAND, or NOR gate), or any other suitable circuit element. Multiple ones of these devices may be combined on a single die **1502**. For example, a memory array formed by multiple memory devices may be formed on a same die **1502** as a processing device (e.g., the processing device **1802** of FIG. **16**) or other logic that is configured to store information in the memory devices or execute instructions stored in the memory array.

(193) FIG. **14** is a cross-sectional side view of an IC device **1600** that may be included in any of the microelectronic assemblies **100** disclosed herein (e.g., in any of the IC structures **110** and/or **120**). One or more of the IC devices **1600** may be included in one or more dies **1502** (FIG. **13**). One or more of the IC devices **1600** may be included in one or more of the IC structures **110** and/or the IC structures **120** as described herein, e.g., in the IC package **200** (FIG. **12**). One or more of the IC devices **1600** may be included in one or more dies **1502** (FIG. **13**). The IC device **1600** may be formed on a die substrate **1602** (e.g., the wafer **1500** of FIG. **13**) and may be included in a die (e.g., the die **1502** of FIG. **13**). The die substrate **1602** may be a semiconductor substrate composed of semiconductor material systems including, for example, N-type or P-type materials systems (or a combination of both). The die substrate **1602** may include, for example, a crystalline substrate formed using a bulk silicon or a silicon-on-insulator (SOI) substructure. In some embodiments, the die substrate **1602** may be formed using alternative materials, which may or may not be combined with silicon, that include, but are not limited to, germanium, indium antimonide, lead telluride, indium arsenide, indium phosphide, gallium arsenide, or gallium antimonide. Further materials classified as group II-VI, III-V, or IV may also be used to form the die substrate **1602**. Although a few examples of materials from which the die substrate **1602** may be formed are described here, any material that may serve as a foundation for an IC device **1600** may be used. The die substrate **1602** may be part of a singulated die (e.g., the dies **1502** of FIG. **13**) or a wafer (e.g., the wafer **1500** of FIG. **13**).

(194) The IC device **1600** may include one or more device layers **1604** disposed on the die substrate **1602**. The device layer **1604** may include features of one or more transistors **1640** (e.g., metal oxide semiconductor field-effect transistors (MOSFETs)) formed on the die substrate **1602**. The device layer **1604** may include, for example, one or more source and/or drain (S/D) regions **1620**, a gate **1622** to control current flow in the transistors **1640** between the S/D regions **1620**, and one or more S/D contacts **1624** to route electrical signals to/from the S/D regions **1620**. The transistors **1640** may include additional features not depicted for the sake of clarity, such as device isolation regions, gate contacts, and the like. The transistors **1640** are not limited to the type and

configuration depicted in FIG. 14 and may include a wide variety of other types and configurations such as, for example, planar transistors, non-planar transistors, or a combination of both. Non-planar transistors may include fin-based field-effect transistors (FinFET transistors), such as double-gate transistors or tri-gate transistors, and wrap-around or all-around gate transistors, such as nanoribbon and nanowire transistors.

(195) Each transistor **1640** may include a gate **1622** formed of at least two layers, a gate dielectric and a gate electrode. The gate dielectric may include one layer or a stack of layers. The one or more layers may include silicon oxide, silicon dioxide, silicon carbide, and/or a high-k dielectric material. The high-k dielectric material may include elements such as hafnium, silicon, oxygen, titanium, tantalum, lanthanum, aluminum, zirconium, barium, strontium, yttrium, lead, scandium, niobium, and zinc. Examples of high-k materials that may be used in the gate dielectric include, but are not limited to, hafnium oxide, hafnium silicon oxide, lanthanum oxide, lanthanum aluminum oxide, zirconium oxide, zirconium silicon oxide, tantalum oxide, titanium oxide, barium strontium titanium oxide, barium titanium oxide, strontium titanium oxide, yttrium oxide, aluminum oxide, lead scandium tantalum oxide, and lead zinc niobate. In some embodiments, an annealing process may be carried out on the gate dielectric to improve its quality when a high-k material is used.

(196) The gate electrode may be formed on the gate dielectric and may include at least one P-type work function metal or N-type work function metal, depending on whether the transistor **1640** is to be a p-type metal oxide semiconductor (PMOS) or an n-type metal oxide semiconductor (NMOS) transistor. In some implementations, the gate electrode may consist of a stack of two or more metal layers, where one or more metal layers are work function metal layers and at least one metal layer is a fill metal layer. Further metal layers may be included for other purposes, such as a barrier layer. For a PMOS transistor, metals that may be used for the gate electrode include, but are not limited to, ruthenium, palladium, platinum, cobalt, nickel, conductive metal oxides (e.g., ruthenium oxide), and any of the metals discussed below with reference to an NMOS transistor (e.g., for work function tuning). For an NMOS transistor, metals that may be used for the gate electrode include, but are not limited to, hafnium, zirconium, titanium, tantalum, aluminum, alloys of these metals, carbides of these metals (e.g., hafnium carbide, zirconium carbide, titanium carbide, tantalum carbide, and aluminum carbide), and any of the metals discussed above with reference to a PMOS transistor (e.g., for work function tuning).

(197) In some embodiments, when viewed as a cross-section of the transistor **1640** along the source-channel-drain direction, the gate electrode may consist of a U-shaped structure that includes a bottom portion substantially parallel to the surface of the die substrate **1602** and two sidewall portions that are substantially perpendicular to the top surface of the die substrate **1602**. In other embodiments, at least one of the metal layers that form the gate electrode may simply be a planar layer that is substantially parallel to the top surface of the die substrate **1602** and does not include sidewall portions substantially perpendicular to the top surface of the die substrate **1602**. In other embodiments, the gate electrode may consist of a combination of U-shaped structures and planar, non-U-shaped structures. For example, the gate electrode may consist of one or more U-shaped metal layers formed atop one or more planar, non-U-shaped layers.

(198) In some embodiments, a pair of sidewall spacers may be formed on opposing sides of the gate stack to bracket the gate stack. The sidewall spacers may be formed from materials such as silicon nitride, silicon oxide, silicon carbide, silicon nitride doped with carbon, and silicon oxynitride. Processes for forming sidewall spacers are well known in the art and generally include deposition and etching process steps. In some embodiments, a plurality of spacer pairs may be used; for instance, two pairs, three pairs, or four pairs of sidewall spacers may be formed on opposing sides of the gate stack.

(199) The S/D regions **1620** may be formed within the die substrate **1602** adjacent to the gate **1622** of each transistor **1640**. The S/D regions **1620** may be formed using an implantation/diffusion process or an etching/deposition process, for example. In the former process, dopants such as

boron, aluminum, antimony, phosphorous, or arsenic may be ion-implanted into the die substrate **1602** to form the S/D regions **1620**. An annealing process that activates the dopants and causes them to diffuse farther into the die substrate **1602** may follow the ion-implantation process. In the latter process, the die substrate **1602** may first be etched to form recesses at the locations of the S/D regions **1620**. An epitaxial deposition process may then be carried out to fill the recesses with material that is used to fabricate the S/D regions **1620**. In some implementations, the S/D regions **1620** may be fabricated using a silicon alloy such as silicon germanium or silicon carbide. In some embodiments, the epitaxially deposited silicon alloy may be doped in situ with dopants such as boron, arsenic, or phosphorous. In some embodiments, the S/D regions **1620** may be formed using one or more alternate semiconductor materials such as germanium or a group III-V material or alloy. In further embodiments, one or more layers of metal and/or metal alloys may be used to form the S/D regions **1620**.

(200) Electrical signals, such as power and/or input/output (I/O) signals, may be routed to and/or from the devices (e.g., transistors **1640**) of the device layer **1604** through one or more interconnect layers disposed on the device layer **1604** (illustrated in FIG. **14** as interconnect layers **1606-1610**). For example, electrically conductive features of the device layer **1604** (e.g., the gate **1622** and the S/D contacts **1624**) may be electrically coupled with the interconnect structures **1628** of the interconnect layers **1606-1610**. The one or more interconnect layers **1606-1610** may form a metallization stack (also referred to as an “ILD stack”) **1619** of the IC device **1600**.

(201) The interconnect structures **1628** may be arranged within the interconnect layers **1606-1610** to route electrical signals according to a wide variety of designs; in particular, the arrangement is not limited to the particular configuration of interconnect structures **1628** depicted in FIG. **14**. Although a particular number of interconnect layers **1606-1610** is depicted in FIG. **14**, embodiments of the present disclosure include IC devices having more or fewer interconnect layers than depicted.

(202) In some embodiments, the interconnect structures **1628** may include lines (or trenches) **1628a** and/or vias **1628b** filled with an electrically conductive material such as a metal. The lines **1628a** may be arranged to route electrical signals in a direction of a plane that is substantially parallel with a surface of the die substrate **1602** upon which the device layer **1604** is formed. For example, the lines **1628a** may route electrical signals in a direction in and out of the page from the perspective of FIG. **14**. The vias **1628b** may be arranged to route electrical signals in a direction of a plane that is substantially perpendicular to the surface of the die substrate **1602** upon which the device layer **1604** is formed. In some embodiments, the vias **1628b** may electrically couple lines **1628a** of different interconnect layers **1606-1610** together.

(203) The interconnect layers **1606-1610** may include a dielectric material **1626** disposed between the interconnect structures **1628**, as shown in FIG. **14**. In some embodiments, the dielectric material **1626** disposed between the interconnect structures **1628** in different ones of the interconnect layers **1606-1610** may have different compositions; in other embodiments, the composition of the dielectric material **1626** between different interconnect layers **1606-1610** may be the same.

(204) A first interconnect layer **1606** (referred to as Metal **1** or “M1”) may be formed directly on the device layer **1604**. In some embodiments, the first interconnect layer **1606** may include lines **1628a** and/or vias **1628b**, as shown. The lines **1628a** of the first interconnect layer **1606** may be coupled with contacts (e.g., the S/D contacts **1624**) of the device layer **1604**.

(205) A second interconnect layer **1608** (referred to as Metal **2** or “M2”) may be formed directly on the first interconnect layer **1606**. In some embodiments, the second interconnect layer **1608** may include vias **1628b** to couple the lines **1628a** of the second interconnect layer **1608** with the lines **1628a** of the first interconnect layer **1606**. Although the lines **1628a** and the vias **1628b** are structurally delineated with a line within each interconnect layer (e.g., within the second interconnect layer **1608**) for the sake of clarity, the lines **1628a** and the vias **1628b** may be

structurally and/or materially contiguous (e.g., simultaneously filled during a dual-damascene process) in some embodiments.

(206) A third interconnect layer **1610** (referred to as Metal 3 or “M3”) (and additional interconnect layers, as desired) may be formed in succession on the second interconnect layer **1608** according to similar techniques and configurations described in connection with the second interconnect layer **1608** or the first interconnect layer **1606**. In some embodiments, the interconnect layers that are “higher up” in the metallization stack **1619** in the IC device **1600** (i.e., farther away from the device layer **1604**) may be thicker.

(207) The IC device **1600** may include a solder resist material **1634** (e.g., polyimide or similar material) and one or more conductive contacts **1636** formed on the interconnect layers **1606-1610**. In FIG. **14**, the conductive contacts **1636** are illustrated as taking the form of bond pads. The conductive contacts **1636** may be electrically coupled with the interconnect structures **1628** and configured to route the electrical signals of the transistor(s) **1640** to other external devices. For example, solder bonds may be formed on the one or more conductive contacts **1636** to mechanically and/or electrically couple a chip including the IC device **1600** with another component (e.g., a circuit board). The IC device **1600** may include additional or alternate structures to route the electrical signals from the interconnect layers **1606-1610**; for example, the conductive contacts **1636** may include other analogous features (e.g., posts) that route the electrical signals to external components. The conductive contacts **1636** may serve as the conductive contacts **122** or **124**, as appropriate.

(208) In some embodiments in which the IC device **1600** is a double-sided die (e.g., like the die **114-1**), the IC device **1600** may include another metallization stack (not shown) on the opposite side of the device layer(s) **1604**. This metallization stack may include multiple interconnect layers as discussed above with reference to the interconnect layers **1606-1610**, to provide conductive pathways (e.g., including conductive lines and vias) between the device layer(s) **1604** and additional conductive contacts (not shown) on the opposite side of the IC device **1600** from the conductive contacts **1636**. These additional conductive contacts may serve as the conductive contacts **122** or **124**, as appropriate.

(209) In other embodiments in which the IC device **1600** is a double-sided die (e.g., like the die **114-1**), the IC device **1600** may include one or more TSVs through the die substrate **1602**; these TSVs may make contact with the device layer(s) **1604**, and may provide conductive pathways between the device layer(s) **1604** and additional conductive contacts (not shown) on the opposite side of the IC device **1600** from the conductive contacts **1636**. These additional conductive contacts may serve as the conductive contacts **122** or **124**, as appropriate.

(210) FIG. **15** is a cross-sectional side view of an IC device assembly **1700** that may include components having one or more microelectronic assemblies fabricated using hybrid manufacturing with modified via-last process in accordance with any of the embodiments disclosed herein. The IC device assembly **1700** includes a number of components disposed on a circuit board **1702** (which may be, e.g., a motherboard). The IC device assembly **1700** includes components disposed on a first face **1740** of the circuit board **1702** and an opposing second face **1742** of the circuit board **1702**; generally, components may be disposed on one or both faces **1740** and **1742**. In particular, any suitable ones of the components of the IC device assembly **1700** may include any of one or more microelectronic assemblies fabricated using hybrid manufacturing with modified via-last process in accordance with any of the embodiments disclosed herein.

(211) In some embodiments, the circuit board **1702** may be a PCB including multiple metal layers separated from one another by layers of dielectric material and interconnected by electrically conductive vias. Any one or more of the metal layers may be formed in a desired circuit pattern to route electrical signals (optionally in conjunction with other metal layers) between the components coupled to the circuit board **1702**. In other embodiments, the circuit board **1702** may be a non-PCB substrate.

(212) The IC device assembly **1700** illustrated in FIG. **15** includes a package-on-interposer structure **1736** coupled to the first face **1740** of the circuit board **1702** by coupling components **1716**. The coupling components **1716** may electrically and mechanically couple the package-on-interposer structure **1736** to the circuit board **1702**, and may include solder balls (e.g., as shown in FIG. **15**), male and female portions of a socket, an adhesive, an underfill material, and/or any other suitable electrical and/or mechanical coupling structure.

(213) The package-on-interposer structure **1736** may include an IC package **1720** coupled to an interposer **1704** by coupling components **1718**. The coupling components **1718** may take any suitable form for the application, such as the forms discussed above with reference to the coupling components **1716**. The IC package **1720** may be or include, for example, a die (the die **1502** of FIG. **13B**), an IC device, or any other suitable component. In particular, the IC package **1720** may include one or more microelectronic assemblies **100** fabricated using hybrid manufacturing and/or one or more IC packages **200** as described herein. Although a single IC package **1720** is shown in FIG. **15**, multiple IC packages may be coupled to the interposer **1704**; indeed, additional interposers may be coupled to the interposer **1704**. The interposer **1704** may provide an intervening substrate used to bridge the circuit board **1702** and the IC package **1720**. Generally, the interposer **1704** may spread a connection to a wider pitch or reroute a connection to a different connection. For example, the interposer **1704** may couple the IC package **1720** (e.g., a die) to a ball grid array (BGA) of the coupling components **1716** for coupling to the circuit board **1702**. In the embodiment illustrated in FIG. **15**, the IC package **1720** and the circuit board **1702** are attached to opposing sides of the interposer **1704**; in other embodiments, the IC package **1720** and the circuit board **1702** may be attached to a same side of the interposer **1704**. In some embodiments, three or more components may be interconnected by way of the interposer **1704**.

(214) The interposer **1704** may be formed of an epoxy resin, a fiberglass-reinforced epoxy resin, a ceramic material, or a polymer material such as polyimide. In some implementations, the interposer **1704** may be formed of alternate rigid or flexible materials that may include the same materials described above for use in a semiconductor substrate, such as silicon, germanium, and other group III-V and group IV materials. The interposer **1704** may include any number of metal lines **1710**, vias **1708**, and TSVs **1706**. The interposer **1704** may further include embedded devices **1714**, including both passive and active devices. Such devices may include, but are not limited to, capacitors, decoupling capacitors, resistors, inductors, fuses, diodes, transformers, sensors, electrostatic discharge (ESD) protection devices, and memory devices. More complex devices such as RF devices, power amplifiers, power management devices, antennas, arrays, sensors, and microelectromechanical systems (MEMS) devices may also be formed on the interposer **1704**. The package-on-interposer structure **1736** may take the form of any of the package-on-interposer structures known in the art.

(215) The IC device assembly **1700** may include an IC package **1724** coupled to the first face **1740** of the circuit board **1702** by coupling components **1722**. The coupling components **1722** may take the form of any of the embodiments discussed above with reference to the coupling components **1716**, and the IC package **1724** may take the form of any of the embodiments discussed above with reference to the IC package **1720**.

(216) The IC device assembly **1700** illustrated in FIG. **15** includes a package-on-package structure **1734** coupled to the second face **1742** of the circuit board **1702** by coupling components **1728**. The package-on-package structure **1734** may include an IC package **1726** and an IC package **1732** coupled together by coupling components **1730** such that the IC package **1726** is disposed between the circuit board **1702** and the IC package **1732**. The coupling components **1728** and **1730** may take the form of any of the embodiments of the coupling components **1716** discussed above, and the IC packages **1726** and **1732** may take the form of any of the embodiments of the IC package **1720** discussed above. The package-on-package structure **1734** may be configured in accordance with any of the package-on-package structures known in the art.

(217) FIG. 16 is a block diagram of an example computing device **1800** that may include one or more components with one or more microelectronic assemblies **100** fabricated using hybrid manufacturing with modified via-last process in accordance with any of the embodiments disclosed herein. For example, any suitable ones of the components of the computing device **1800** may include an IC package (e.g., the IC package **1720**, shown in FIG. 15 and/or the IC package **200** shown in FIG. 12) including one or more microelectronic assemblies **100** fabricated using hybrid manufacturing with modified via-last process in accordance with any of the embodiments disclosed herein. Any of the components of the computing device **1800** may include an IC device **1400** (e.g., as shown in FIG. 14). Any of the components of the computing device **1800** may include an IC device assembly **1700** (e.g., as shown in FIG. 15).

(218) A number of components are illustrated in FIG. 16 as included in the computing device **1800**, but any one or more of these components may be omitted or duplicated, as suitable for the application. In some embodiments, some or all of the components included in the computing device **1800** may be attached to one or more motherboards. In some embodiments, some or all of these components are fabricated onto a single system on a chip (SoC) die.

(219) Additionally, in various embodiments, the computing device **1800** may not include one or more of the components illustrated in FIG. 16, but the computing device **1800** may include interface circuitry for coupling to the one or more components. For example, the computing device **1800** may not include a display device **1806**, but may include display device interface circuitry (e.g., a connector and driver circuitry) to which a display device **1806** may be coupled. In another set of examples, the computing device **1800** may not include an audio input device **1818** or an audio output device **1808**, but may include audio input or output device interface circuitry (e.g., connectors and supporting circuitry) to which an audio input device **1818** or audio output device **1808** may be coupled.

(220) The computing device **1800** may include a processing device **1802** (e.g., one or more processing devices). As used herein, the term “processing device” or “processor” may refer to any device or portion of a device that processes electronic data from registers and/or memory to transform that electronic data into other electronic data that may be stored in registers and/or memory. The processing device **1802** may include one or more digital signal processors (DSPs), application-specific ICs (ASICs), central processing units (CPUs), graphics processing units (GPUs), cryptoprocessors (specialized processors that execute cryptographic algorithms within hardware), server processors, or any other suitable processing devices. The computing device **1800** may include a memory **1804**, which may itself include one or more memory devices such as volatile memory (e.g., DRAM), nonvolatile memory (e.g., read-only memory (ROM)), flash memory, solid state memory, and/or a hard drive. In some embodiments, the memory **1804** may include memory that shares a die with the processing device **1802**.

(221) In some embodiments, the computing device **1800** may include a communication chip **1812** (e.g., one or more communication chips). For example, the communication chip **1812** may be configured for managing wireless communications for the transfer of data to and from the computing device **1800**. The term “wireless” and its derivatives may be used to describe circuits, devices, systems, methods, techniques, communications channels, etc., that may communicate data through the use of modulated electromagnetic radiation through a nonsolid medium. The term does not imply that the associated devices do not contain any wires, although in some embodiments they might not.

(222) The communication chip **1812** may implement any of a number of wireless standards or protocols, including but not limited to Institute for Electrical and Electronic Engineers (IEEE) standards including Wi-Fi (IEEE 802.11 family), IEEE 802.16 standards (e.g., IEEE 802.16-2005 Amendment), Long-Term Evolution (LTE) project along with any amendments, updates, and/or revisions (e.g., advanced LTE project, ultramobile broadband (UMB) project (also referred to as “3GPP2”), etc.). IEEE 802.16 compatible Broadband Wireless Access (BWA) networks are

generally referred to as WiMAX networks, an acronym that stands for Worldwide Interoperability for Microwave Access, which is a certification mark for products that pass conformity and interoperability tests for the IEEE 802.16 standards. The communication chip **1812** may operate in accordance with a Global System for Mobile Communication (GSM), General Packet Radio Service (GPRS), Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Evolved HSPA (E-HSPA), or LTE network. The communication chip **1812** may operate in accordance with Enhanced Data for GSM Evolution (EDGE), GSM EDGE Radio Access Network (GERAN), Universal Terrestrial Radio Access Network (UTRAN), or Evolved UTRAN (E-UTRAN). The communication chip **1812** may operate in accordance with Code Division Multiple Access (CDMA), Time Division Multiple Access (TDMA), Digital Enhanced Cordless Telecommunications (DECT), Evolution-Data Optimized (EV-DO), and derivatives thereof, as well as any other wireless protocols that are designated as 3G, 4G, 5G, and beyond. The communication chip **1812** may operate in accordance with other wireless protocols in other embodiments. The computing device **1800** may include an antenna **1822** to facilitate wireless communications and/or to receive other wireless communications (such as AM or FM radio transmissions).

(223) In some embodiments, the communication chip **1812** may manage wired communications, such as electrical, optical, or any other suitable communication protocols (e.g., the Ethernet). As noted above, the communication chip **1812** may include multiple communication chips. For instance, a first communication chip **1812** may be dedicated to shorter-range wireless communications such as Wi-Fi or Bluetooth, and a second communication chip **1812** may be dedicated to longer-range wireless communications such as global positioning system (GPS), EDGE, GPRS, CDMA, WiMAX, LTE, EV-DO, or others. In some embodiments, a first communication chip **1812** may be dedicated to wireless communications, and a second communication chip **1812** may be dedicated to wired communications.

(224) The computing device **1800** may include battery/power circuitry **1814**. The battery/power circuitry **1814** may include one or more energy storage devices (e.g., batteries or capacitors) and/or circuitry for coupling components of the computing device **1800** to an energy source separate from the computing device **1800** (e.g., AC line power).

(225) The computing device **1800** may include a display device **1806** (or corresponding interface circuitry, as discussed above). The display device **1806** may include any visual indicators, such as a heads-up display, a computer monitor, a projector, a touchscreen display, a liquid crystal display (LCD), a light-emitting diode display, or a flat panel display, for example.

(226) The computing device **1800** may include an audio output device **1808** (or corresponding interface circuitry, as discussed above). The audio output device **1808** may include any device that generates an audible indicator, such as speakers, headsets, or earbuds, for example.

(227) The computing device **1800** may include an audio input device **1818** (or corresponding interface circuitry, as discussed above). The audio input device **1818** may include any device that generates a signal representative of a sound, such as microphones, microphone arrays, or digital instruments (e.g., instruments having a musical instrument digital interface (MIDI) output).

(228) The computing device **1800** may include a GPS device **1816** (or corresponding interface circuitry, as discussed above). The GPS device **1816** may be in communication with a satellite-based system and may receive a location of the computing device **1800**, as known in the art.

(229) The computing device **1800** may include an other output device **1810** (or corresponding interface circuitry, as discussed above). Examples of the other output device **1810** may include an audio codec, a video codec, a printer, a wired or wireless transmitter for providing information to other devices, or an additional storage device.

(230) The computing device **1800** may include an other input device **1820** (or corresponding interface circuitry, as discussed above). Examples of the other input device **1820** may include an accelerometer, a gyroscope, a compass, an image capture device, a keyboard, a cursor control device such as a mouse, a stylus, a touchpad, a bar code reader, a Quick Response (QR) code

reader, any sensor, or a radio frequency identification (RFID) reader.

(231) The computing device **1800** may have any desired form factor, such as a handheld or mobile computing device (e.g., a cell phone, a smart phone, a mobile internet device, a music player, a tablet computer, a laptop computer, a netbook computer, an ultrabook computer, a personal digital assistant (PDA), an ultramobile personal computer, etc.), a desktop computing device, a server or other networked computing component, a printer, a scanner, a monitor, a set-top box, an entertainment control unit, a vehicle control unit, a digital camera, a digital video recorder, or a wearable computing device. In some embodiments, the computing device **1800** may be any other electronic device that processes data.

SELECT EXAMPLES

(232) The following paragraphs provide various examples of the embodiments disclosed herein.

(233) Example 1 provides a microelectronic assembly that includes a first IC structure (e.g., IC structure **110**); a second IC structure (e.g., IC structure **120**), where a first face of the second IC structure is bonded to a first face of the first IC structure; a bonding interface (e.g., bonding interface **130**) between at least portions of the first face of the first IC structure and the first face of the second IC structure (i.e., between faces that are bonded together); and a conductive via structure (e.g., conductive via structure **150**), extending from a second face of the second IC structure, through the bonding interface, and at least partially into the first IC structure. The conductive via structure includes a first portion (e.g., portion **152-1**) extending from the second face of the second IC structure to the bonding interface, a second portion (e.g., portion **152-2**) extending through the bonding interface, a third portion (e.g., portion **152-3**), extending, in the first IC structure, from the bonding interface to a portion of the conductive via structure within the first IC structure at a first distance from the bonding interface, and a fourth portion (e.g., portion **152-4**), extending, in the first IC structure, from the first distance to a portion of the conductive via structure within the first IC structure at a second distance from the bonding interface, the second distance being greater than the first distance.

(234) Example 2 provides the microelectronic assembly according to example 1, where each of the first portion, the second portion, and the third portion includes a first electrically conductive fill material (e.g., material **154**); the fourth portion includes a second electrically conductive fill material (e.g., material **144**); and a material composition of the second electrically conductive fill material is different from a material composition of the first electrically conductive fill material. For example, the first electrically conductive fill material may include copper (Cu), while the second electrically conductive fill material may include tungsten (W), aluminum (Al), ruthenium (Ru), cobalt (Co), or AlCu (e.g., in proportions of between 1:1 to 1:100), or vice versa.

(235) Example 3 provides the microelectronic assembly according to examples 1 or 2, where the conductive via structure includes a grain boundary between an electrically conductive fill material in the third portion and an electrically conductive fill material in the fourth portion.

(236) Example 4 provides the microelectronic assembly according to any one of examples 1-3, where the third portion includes a first liner and the fourth portion includes a second liner, where a material composition of the second liner is different from a material composition of the first liner, or a thickness of the second liner is different from a thickness of the first liner.

(237) Example 5 provides the microelectronic assembly according to example 4, where each of the first portion and the second portion includes the first liner.

(238) Example 6 provides the microelectronic assembly according to example 5, where the first liner is continuous between the first portion, the second portion, and the third portion.

(239) Example 7 provides the microelectronic assembly according to any one of examples 4-6, where each of the first portion, the second portion, and the third portion includes a first electrically conductive fill material (e.g., material **154**), the fourth portion includes a second electrically conductive fill material (e.g., material **144**), and the first liner separates the first electrically conductive fill material and the second electrically conductive fill material.

(240) In a further example of the microelectronic assembly according to any one of examples 4-7, the first liner is in contact with the second liner where the third portion meets the fourth portion.

(241) Example 8 provides the microelectronic assembly according to any one of examples 1-3, where the first portion includes a first liner and each of the third portion and the fourth portion includes a second liner, where a material composition of the second liner is different from a material composition of the first liner, or a thickness of the second liner is different from a thickness of the first liner. For example, the first and second liners may be as described with reference to example 4.

(242) Example 9 provides the microelectronic assembly according to example 8, where the second portion does not include a liner.

(243) Example 10 provides the microelectronic assembly according to example 8, where the second portion includes the first liner.

(244) Example 11 provides the microelectronic assembly according to example 8, where the second portion includes the second liner.

(245) Example 12 provides the microelectronic assembly according to any one of examples 1-3, where the third portion includes a liner, and the fourth portion does not include a liner.

(246) 13. Example 13 provides the microelectronic assembly according to example 12, where each of the first portion and the second portion includes a liner having a material composition that is substantially same as the liner of the third portion.

(247) Example 14 provides the microelectronic assembly according to example 13, where the liner is continuous between the first portion, the second portion, and the third portion.

(248) Example 15 provides the microelectronic assembly according to any one of examples 1-3, where the fourth portion includes a liner, and the third portion does not include a liner.

(249) Example 16 provides the microelectronic assembly according to example 15, where each of the first portion and the second portion does not include a liner.

(250) Example 17 provides the microelectronic assembly according to any one of examples 1-16, where, in a plane that is substantially perpendicular to the bonding interface (or in a plane that is substantially perpendicular to a support structure providing mechanical stability for the microelectronic assembly), a cross-section of each of the first portion and the fourth portion is a trapezoid that includes two parallel sides, one of which is a short side and another one of which is a long side, for the trapezoid of the first portion, the long side is closer to the bonding interface than the short side, and, for the trapezoid of the fourth portion, the long side is closer to the bonding interface than the short side.

(251) Example 18 provides the microelectronic assembly according to example 17, where the first portion includes a liner at sidewalls of the conductive via structure and an electrically conductive fill material enclosed by the liner, and at the second face of the second IC structure the liner is only at the sidewalls of the conductive via structure (and not extending at the top of the second face of the second IC structure).

(252) Example 19 provides the microelectronic assembly according to any one of examples 1-16, where, in a plane that is substantially perpendicular to the bonding interface (or in a plane that is substantially perpendicular to a support structure providing mechanical stability for the microelectronic assembly), a cross-section of each of the first portion and the fourth portion is a trapezoid that includes two parallel sides, one of which is a short side and another one of which is a long side, for the trapezoid of the first portion, the short side is closer to the bonding interface than the long side, and, for the trapezoid of the fourth portion, the long side is closer to the bonding interface than the short side.

(253) Example 20 provides the microelectronic assembly according to example 19, where the first portion includes a liner at sidewalls of the conductive via structure and an electrically conductive fill material enclosed by the liner, the liner is absent in at least about 70% (e.g., in at least about 80% or in at least about 90%) of each of an interface between the first portion and the second

portion, and an interface between the second portion and the third portion.

(254) Example 21 provides the microelectronic assembly according to any one of examples 1-16, where, in a plane that is substantially perpendicular to the bonding interface (or in a plane that is substantially perpendicular to a support structure providing mechanical stability for the microelectronic assembly), a cross-section of each of the first portion and the fourth portion is a trapezoid that includes two parallel sides, one of which is a short side and another one of which is a long side, for the trapezoid of the first portion, the short side is closer to the bonding interface than the long side, and, for the trapezoid of the fourth portion, the short side is closer to the bonding interface than the long side.

(255) Example 22 provides the microelectronic assembly according to example 21, where the first portion includes a liner at sidewalls of the conductive via structure and an electrically conductive fill material enclosed by the liner, the liner is absent in at least about 70% (e.g., in at least about 80% or in at least about 90%) of each of an interface between the first portion and the second portion, and an interface between the second portion and the third portion.

(256) Example 23 provides the microelectronic assembly according to examples 21 or 22, where the third portion includes a liner at sidewalls of the conductive via structure and an electrically conductive fill material enclosed by the liner, the liner is absent in at least about 70% (e.g., in at least about 80% or in at least about 90%) of each of an interface between the third portion and the second portion, and an interface between the second portion and the first portion.

(257) Example 24 provides the microelectronic assembly according to any one of the preceding examples, further including an after-bonding interconnect extending from a second face of the second IC structure to the first face of the second IC structure, through the bonding interface, and into the first IC structure, where the second face of the second IC structure is opposite the first face of the second IC structure.

(258) Example 25 provides the microelectronic assembly according to example 24, where the after-bonding interconnect extends to a second face of the first IC structure, where the second face of the first IC structure is opposite the first face of the first IC structure.

(259) Example 26 provides the microelectronic assembly according to examples 24 or 25, where, in a plane that is substantially perpendicular to the bonding interface (or in a plane that is substantially perpendicular to a support structure providing mechanical stability for the microelectronic assembly), a cross-section of the after-bonding interconnect is a trapezoid that includes two parallel sides, one of which is a short side and another one of which is a long side, and, for the trapezoid of the after-bonding interconnect, the long side is closer to the second face of the second IC structure than the short side.

(260) Example 27 provides the microelectronic assembly according to any one of the preceding examples, further including an after-bonding interconnect extending from a second face of the first IC structure to the first face of the first IC structure, through the bonding interface, and into the second IC structure, where the second face of the first IC structure is opposite the first face of the first IC structure.

(261) Example 28 provides the microelectronic assembly according to example 27, where the after-bonding interconnect extends to a second face of the second IC structure, where the second face of the second IC structure is opposite the first face of the second IC structure.

(262) Example 29 provides the microelectronic assembly according to examples 27 or 28, where, in a plane that is substantially perpendicular to the bonding interface (or in a plane that is substantially perpendicular to a support structure providing mechanical stability for the microelectronic assembly), a cross-section of the after-bonding interconnect is a trapezoid that includes two parallel sides, one of which is a short side and another one of which is a long side, and, for the trapezoid of the after-bonding interconnect, the long side is closer to the second face of the first IC structure than the short side.

(263) Example 30 provides a microelectronic assembly that includes a first IC structure (e.g., IC

structure **110**); a second IC structure (e.g., IC structure **120**), where a first face of the second IC structure is bonded to a first face of the first IC structure; a bonding interface (e.g., bonding interface **130**) between at least portions of the first face of the first IC structure and the first face of the second IC structure (i.e., between faces that are bonded together); and a conductive via structure (e.g., conductive via structure **150**), extending from a second face of the second IC structure, through the bonding interface, and at least partially into the first IC structure. The conductive via structure includes a first portion (e.g., a combination of portions **152-1**, **152-2** and **152-3**) extending from the second face of the second IC structure to the bonding interface, through the bonding interface, and further extending in the first IC structure to a first distance from the bonding interface, and a second portion (e.g., portion **152-4**), extending, in the first IC structure, from the first distance to a portion of the conductive via structure within the first IC structure at a second distance from the bonding interface, the second distance being greater than the first distance, where the first portion includes a first electrically conductive fill material (e.g., material **154**), the second portion includes a second electrically conductive fill material (e.g., material **144**), and a liner separates the first electrically conductive fill material and the second electrically conductive fill material.

(264) Example 31 provides the microelectronic assembly according to example 30, where a material composition of the second electrically conductive fill material is different from a material composition of the first electrically conductive fill material.

(265) Example 32 provides the microelectronic assembly according to example 30, where a material composition of the second electrically conductive fill material is substantially same as a material composition of the first electrically conductive fill material.

(266) Example 33 provides the microelectronic assembly according to any one of examples 30-32, where the liner is on inner sidewalls and a bottom of the first portion.

(267) Example 34 provides the microelectronic assembly according to any one of examples 30-33, where, in the conductive via structure, the liner is continuous between the first IC structure and the second IC structure.

(268) Example 35 provides the microelectronic assembly according to any one of examples 30-34, where the liner is a first liner, and the second portion includes a second liner on sidewalls of the second portion.

(269) Example 36 provides the microelectronic assembly according to example 35, where the second liner is further on a bottom of the second portion.

(270) Example 37 provides the microelectronic assembly according to example 35 or 36, where a material composition of the second liner is different from a material composition of the first liner.

(271) Example 38 provides the microelectronic assembly according to any one of example 35-37, where a thickness of the second liner is different from a thickness of the first liner.

(272) Example 39 provides the microelectronic assembly according to any one of example 35-38, where the first liner is in contact with the second liner where the first portion meets the second portion.

(273) Example 40 provides the microelectronic assembly according to any one of the preceding examples, where the second IC structure is bonded to the first IC structure by an adhesive material between at least portions of the first IC structure and the second IC structure.

(274) Example 41 provides the microelectronic assembly according to any one of the preceding examples, further including an etch-stop material in at least portions where the second IC structure is bonded to the first IC structure, the etch-stop material including silicon, nitrogen, and carbon.

(275) Example 42 provides the microelectronic assembly according to any one of the preceding examples, where at least one of the first IC structure and the second IC structure further includes or is a part of a central processing unit.

(276) Example 43 provides the microelectronic assembly according to any one of the preceding examples, where at least one of the first IC structure and the second IC structure further includes or

is a part of a memory device, e.g., a high-bandwidth memory device.

(277) Example 44 provides the microelectronic assembly according to any one of the preceding examples, where at least one of the first IC structure and the second IC structure further includes or is a part of a logic circuit.

(278) Example 45 provides the microelectronic assembly according to any one of the preceding examples, where at least one of the first IC structure and the second IC structure further includes or is a part of input/output circuitry.

(279) Example 46 provides the microelectronic assembly according to any one of the preceding examples, where at least one of the first IC structure and the second IC structure further includes or is a part of a field programmable gate array transceiver.

(280) Example 47 provides the microelectronic assembly according to any one of the preceding examples, where at least one of the first IC structure and the second IC structure further includes or is a part of a field programmable gate array logic.

(281) Example 48 provides the microelectronic assembly according to any one of the preceding examples, where at least one of the first IC structure and the second IC structure further includes or is a part of a power delivery circuitry.

(282) Example 49 provides the microelectronic assembly according to any one of the preceding examples, where at least one of the first IC structure and the second IC structure further includes or is a part of a III-V amplifier.

(283) Example 50 provides the microelectronic assembly according to any one of the preceding examples, where at least one of the first IC structure and the second IC structure further includes or is a part of Peripheral Component Interconnect Express circuitry or Double Data Rate transfer circuitry.

(284) Example 51 provides the microelectronic assembly according to any one of the preceding examples, where at least one of the first IC structure and the second IC structure further includes frontend transistors.

(285) Example 52 provides the microelectronic assembly according to any one of the preceding examples, where at least one of the first IC structure and the second IC structure further includes backend transistors.

(286) Example 53 provides the microelectronic assembly according to any one of the preceding examples, where at least one of the first IC structure and the second IC structure includes TFTs.

(287) Example 54 provides an IC package that includes a microelectronic assembly according to any one of the preceding examples; and a further IC component, coupled to the IC die.

(288) Example 55 provides the IC package according to example 54, where the further IC component includes one of a package substrate, a circuit board, an interposer, or a further IC die.

(289) Example 56 provides a computing device that includes a carrier substrate and a microelectronic assembly, coupled to the carrier substrate, where the microelectronic assembly is a microelectronic assembly according to any one of the preceding examples or the microelectronic assembly is included in the IC package according to any one of examples.

(290) Example 57 provides the computing device according to example 56, where the computing device is a wearable or handheld computing device.

(291) Example 58 provides the computing device according to examples 56 or 57, where the computing device further includes one or more communication chips and an antenna.

(292) Example 59 provides the computing device according to any one of examples 56-58, where the carrier substrate is a motherboard.

(293) Example 60 provides a method of manufacturing a microelectronic assembly, the method including providing the microelectronic assembly according to any one of the preceding examples.

(294) Example 61 provides a method of manufacturing a microelectronic assembly, the method including providing a first IC structure (e.g., IC structure **110**), the first IC structure including a first face and a second face, opposite the first face, and further including a first via extending from

the first face of the first IC structure towards the second face of the first IC structure, where at least a portion of the first via that extends from the first face of the first IC structure to a first distance from the first face of the first IC structure includes a first sacrificial material; providing a second IC structure (e.g., IC structure **120**), the second IC structure including a first face and a second face, opposite the first face, and further including a second via extending between the first face of the second IC structure and the second face of the second IC structure, where the second via includes a second sacrificial material; bonding the first face of the first IC structure and the first face of the second IC structure so that at least a portion of the first sacrificial material is opposite at least a portion of the second sacrificial material; after the bonding, performing an etch, from the second face of the second IC structure, to remove the second sacrificial material and the first sacrificial material; and depositing an electrically conductive fill material in an opening formed by removing the second sacrificial material and the first sacrificial material during the etch.

(295) Example 62 provides the method according to example 61, further including depositing a liner material on sidewalls of the opening prior to depositing the electrically conductive fill material.

(296) Example 63 provides the method according to examples 61 or 62, where the portion of the first via that extends from the first face of the first IC structure to the first distance from the first face of the first IC structure is a first portion of the first via, the first via further includes a second portion, extending from the first portion of the first via to a second distance from the first face of the first IC structure, the second distance being greater than the first distance, the electrically conductive fill material is a first electrically conductive fill material, and the second portion includes a second electrically conductive fill material.

(297) Example 64 provides the method according to example 63, where a material composition of the second electrically conductive fill material is different from a material composition of the first electrically conductive fill material.

(298) Example 65 provides the method according to examples 63 or 64, where the method further includes depositing a first liner material on sidewalls of the opening prior to depositing the first electrically conductive fill material, and the second portion includes a second liner on sidewalls of the second portion.

(299) Example 66 provides the method according to example 65, where a material composition of the second liner is different from a material composition of the first liner.

(300) Example 67 provides the method according to examples 65 or 66, where a thickness of the second liner is different from a thickness of the first liner.

(301) Example 68 provides the method according to any one of examples 61-67, where the bonding includes providing a bonding material over at least one of the first face of the first IC structure and the first face of the second IC structure so that, after the bonding, the bonding material is over at least a portion of the first sacrificial material at the first face of the first IC structure or at least a portion of the second sacrificial material at the first face of the second IC structure.

(302) Example 69 provides the method according to any one of examples 61-67, where the bonding includes providing a bonding material over at least one of the first face of the first IC structure and the first face of the second IC structure so that, after the bonding, none of the bonding material is over the first sacrificial material at the first face of the first IC structure and over the second sacrificial material at the first face of the second IC structure.

(303) Example 70 provides the method according to any one of examples 61-69, further including processes for providing the microelectronic assembly according to any one of the preceding examples.

(304) The above description of illustrated implementations of the disclosure, including what is described in the Abstract, is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. While specific implementations of, and examples for, the disclosure are described herein for illustrative purposes, various equivalent modifications are possible within the scope of

the disclosure, as those skilled in the relevant art will recognize. These modifications may be made to the disclosure in light of the above detailed description.

Claims

1. A microelectronic assembly, comprising: a first integrated circuit (IC) structure; a second IC structure, where a first face of the second IC structure is bonded to a first face of the first IC structure; a bonding interface between at least portions of the first face of the first IC structure and the first face of the second IC structure; and a conductive via structure, extending from a second face of the second IC structure, through the bonding interface, and at least partially into the first IC structure, the conductive via structure having: a first portion extending from the second face of the second IC structure to the bonding interface, a second portion extending through the bonding interface, a third portion extending in the first IC structure, from the bonding interface to a first distance from the bonding interface, and a fourth portion extending in the first IC structure, from the first distance to a second distance from the bonding interface, wherein: the second distance is greater than the first distance, each of the first portion, the second portion, and the third portion includes a first electrically conductive fill material, the fourth portion includes a second electrically conductive fill material, and a material composition of the second electrically conductive fill material is different from a material composition of the first electrically conductive fill material.
2. The microelectronic assembly according to claim 1, wherein the conductive via structure includes a grain boundary between an electrically conductive fill material in the third portion and an electrically conductive fill material in the fourth portion.
3. The microelectronic assembly according to claim 1, wherein the third portion includes a first liner and the fourth portion includes a second liner, wherein: a material composition of the second liner is different from a material composition of the first liner, or a thickness of the second liner is different from a thickness of the first liner.
4. The microelectronic assembly according to claim 3, wherein: each of the first portion, the second portion, and the third portion includes a first electrically conductive fill material, the fourth portion includes a second electrically conductive fill material, and the first liner separates the first electrically conductive fill material and the second electrically conductive fill material.
5. The microelectronic assembly according to claim 1, wherein the first portion includes a first liner and each of the third portion and the fourth portion includes a second liner, wherein: a material composition of the second liner is different from a material composition of the first liner, or a thickness of the second liner is different from a thickness of the first liner.
6. The microelectronic assembly according to claim 1, wherein: in a plane that is substantially perpendicular to the bonding interface, a cross-section of each of the first portion and the fourth portion is substantially a trapezoid that includes a first side and a second side substantially parallel to the first side, wherein the first side is shorter than the second side, for the cross-section of the first portion, the second side is closer to the bonding interface than the first side, and for the cross-section of the fourth portion, the second side is closer to the bonding interface than the first side.
7. The microelectronic assembly according to claim 6, wherein: the first portion includes a liner at sidewalls of the conductive via structure and an electrically conductive fill material enclosed by the liner, and at the second face of the second IC structure the liner is only at the sidewalls of the conductive via structure.
8. The microelectronic assembly according to claim 1, wherein: in a plane that is substantially perpendicular to the bonding interface, a cross-section of each of the first portion and the fourth portion is substantially a trapezoid that includes a first side and a second side substantially parallel to the first side, wherein the first side is shorter than the second side, for the cross-section of the first portion, the first side is closer to the bonding interface than the second side, and for the cross-section of the fourth portion, the second side is closer to the bonding interface than the first side.

9. The microelectronic assembly according to claim 8, wherein: the first portion includes a liner at sidewalls of the conductive via structure and an electrically conductive fill material enclosed by the liner, and the liner is absent in at least about 70% of each of: an interface between the first portion and the second portion, and an interface between the second portion and the third portion.

10. The microelectronic assembly according to claim 1, wherein: in a plane that is substantially perpendicular to the bonding interface, a cross-section of each of the first portion and the fourth portion is substantially a trapezoid that includes a first side and a second side substantially parallel to the first side, wherein the first side is shorter than the second side, for the cross-section of the first portion, the first side is closer to the bonding interface than the second side, and for the cross-section of the fourth portion, the first side is closer to the bonding interface than the second side.

11. The microelectronic assembly according to claim 10, wherein: the first portion includes a liner at sidewalls of the conductive via structure and an electrically conductive fill material enclosed by the liner, and the liner is absent in at least about 70% of each of: an interface between the first portion and the second portion, and an interface between the second portion and the third portion.

12. The microelectronic assembly according to claim 10, wherein: the third portion includes a liner at sidewalls of the conductive via structure and an electrically conductive fill material enclosed by the liner, and the liner is absent in at least about 70% of each of: an interface between the third portion and the second portion, and an interface between the second portion and the first portion.

13. A microelectronic assembly, comprising: a first integrated circuit (IC) structure; a second IC structure, where a first face of the second IC structure and a first face of the first IC structure are attached to one another; a bonding interface between at least a portion of the first face of the first IC structure and at least a portion of the first face of the second IC structure; and a conductive via extending from a second face of the second IC structure, through the bonding interface, and at least partially into the first IC structure, the conductive via having: a first portion extending from the second face of the second IC structure to the bonding interface, a second portion extending through the bonding interface, a third portion extending in the first IC structure, from the bonding interface to a first distance from the bonding interface, and a fourth portion extending in the first IC structure, from the first distance to a second distance from the bonding interface, wherein: the second distance is greater than the first distance, in a plane that is substantially perpendicular to the bonding interface, a cross-section of each of the first portion and the fourth portion includes a first side and a second side substantially parallel to the first side, wherein the first side is shorter than the second side, for the cross-section of the first portion, the second side is closer to the bonding interface than the first side, and for the cross-section of the fourth portion, the second side is closer to the bonding interface than the first side.

14. The microelectronic assembly according to claim 13, wherein: the first portion includes a liner at sidewalls of the conductive via and an electrically conductive fill material enclosed by the liner, and at the second face of the second IC structure the liner is only at the sidewalls of the conductive via.

15. The microelectronic assembly according to claim 13, wherein the conductive via includes a grain boundary between an electrically conductive fill material in the third portion and an electrically conductive fill material in the fourth portion.

16. A microelectronic assembly, comprising: a first integrated circuit (IC) structure; a second IC structure, where a first face of the second IC structure and a first face of the first IC structure are attached to one another; a bonding interface between at least a portion of the first face of the first IC structure and at least a portion of the first face of the second IC structure; and a conductive via extending from a second face of the second IC structure, through the bonding interface, and at least partially into the first IC structure, the conductive via having: a first portion extending from the second face of the second IC structure to the bonding interface, a second portion extending through the bonding interface, a third portion extending in the first IC structure, from the bonding interface to a first distance from the bonding interface, and a fourth portion extending in the first IC

structure, from the first distance to a second distance from the bonding interface, wherein: the second distance is greater than the first distance, in a plane that is substantially perpendicular to the bonding interface, a cross-section of each of the first portion and the fourth portion includes a first side and a second side substantially parallel to the first side, wherein the first side is shorter than the second side, and for the cross-section of the first portion, the first side is closer to the bonding interface than the second side.

17. The microelectronic assembly according to claim 16, wherein: for the cross-section of the fourth portion, the second side is closer to the bonding interface than the first side.

18. The microelectronic assembly according to claim 16, wherein: for the cross-section of the fourth portion, the first side is closer to the bonding interface than the second side.

19. The microelectronic assembly according to claim 16, wherein the conductive via includes a grain boundary between an electrically conductive fill material in the third portion and an electrically conductive fill material in the fourth portion.

20. The microelectronic assembly according to claim 16, wherein: each of the first portion, the second portion, and the third portion includes a first electrically conductive fill material, the fourth portion includes a second electrically conductive fill material, and a material composition of the second electrically conductive fill material is different from a material composition of the first electrically conductive fill material.

21. A microelectronic assembly, comprising: a first integrated circuit (IC) structure; a second IC structure; a bonding interface between at least portions of a first face of the first IC structure and a first face of the second IC structure; and a conductive via structure, extending from a second face of the second IC structure, through the bonding interface, and at least partially into the first IC structure, wherein the conductive via structure includes: a first portion extending from the second face of the second IC structure to the bonding interface, a second portion extending through the bonding interface, a third portion extending in the first IC structure from the bonding interface to a first distance from the bonding interface, and a fourth portion extending in the first IC structure from the first distance to a second distance from the bonding interface, the second distance being greater than the first distance, wherein the conductive via structure includes a grain boundary between an electrically conductive material in the third portion and an electrically conductive material in the fourth portion.

22. The microelectronic assembly according to claim 21, wherein a material composition of the electrically conductive material in the third portion is different from a material composition of the electrically conductive material in the fourth portion.

23. The microelectronic assembly according to claim 21, wherein the third portion includes a first liner and the fourth portion includes a second liner, wherein: a material composition of the second liner is different from a material composition of the first liner, or a thickness of the second liner is different from a thickness of the first liner.

24. The microelectronic assembly according to claim 21, wherein the first portion includes a first liner and each of the third portion and the fourth portion includes a second liner, wherein: a material composition of the second liner is different from a material composition of the first liner, or a thickness of the second liner is different from a thickness of the first liner.

25. The microelectronic assembly according to claim 21, wherein: in a plane that is substantially perpendicular to the bonding interface, a cross-section of each of the first portion and the fourth portion includes a first side and a second side substantially parallel to the first side, wherein the first side is shorter than the second side, for the cross-section of the first portion, the second side is closer to the bonding interface than the first side, and for the cross-section of the fourth portion, the second side is closer to the bonding interface than the first side.

26. The microelectronic assembly according to claim 21, wherein: in a plane that is substantially perpendicular to the bonding interface, a cross-section of each of the first portion and the fourth portion includes a first side and a second side substantially parallel to the first side, wherein the

first side is shorter than the second side, and for the cross-section of the first portion, the first side is closer to the bonding interface than the second side.

27. The microelectronic assembly according to claim 26, wherein: for the cross-section of the fourth portion, the second side is closer to the bonding interface than the first side.

28. The microelectronic assembly according to claim 26, wherein: for the cross-section of the fourth portion, the first side is closer to the bonding interface than the second side.

29. A microelectronic assembly, comprising: a first integrated circuit (IC) structure; a second IC structure; a bonding interface between at least portions of a first face of the first IC structure and a first face of the second IC structure; and a conductive via structure, extending from a second face of the second IC structure, through the bonding interface, and at least partially into the first IC structure, wherein the conductive via structure includes: a first portion extending from the second face of the second IC structure to the bonding interface, a second portion extending through the bonding interface, a third portion extending in the first IC structure from the bonding interface to a first distance from the bonding interface, wherein the third portion includes a first liner, and a fourth portion extending in the first IC structure from the first distance to a second distance from the bonding interface, wherein the second distance is greater than the first distance, the fourth portion includes a second liner, and wherein: a material composition of the second liner is different from a material composition of the first liner, or a thickness of the second liner is different from a thickness of the first liner.

30. The microelectronic assembly according to claim 29, wherein a material composition of an electrically conductive fill material in the third portion is different from a material composition of an electrically conductive fill material in the fourth portion.

31. The microelectronic assembly according to claim 29, wherein the conductive via structure includes a grain boundary between an electrically conductive fill material in the third portion and an electrically conductive fill material in the fourth portion.

32. The microelectronic assembly according to claim 29, wherein: in a plane that is substantially perpendicular to the bonding interface, a cross-section of each of the first portion and the fourth portion includes a first side and a second side substantially parallel to the first side, wherein the first side is shorter than the second side, for the cross-section of the first portion, the second side is closer to the bonding interface than the first side, and for the cross-section of the fourth portion, the second side is closer to the bonding interface than the first side.

33. The microelectronic assembly according to claim 29, wherein: in a plane that is substantially perpendicular to the bonding interface, a cross-section of each of the first portion and the fourth portion includes a first side and a second side substantially parallel to the first side, wherein the first side is shorter than the second side, and for the cross-section of the first portion, the first side is closer to the bonding interface than the second side.

34. The microelectronic assembly according to claim 33, wherein: for the cross-section of the fourth portion, the second side is closer to the bonding interface than the first side.

35. The microelectronic assembly according to claim 33, wherein: for the cross-section of the fourth portion, the first side is closer to the bonding interface than the second side.

36. A microelectronic assembly, comprising: a first integrated circuit (IC) structure; a second IC structure; a bonding interface between at least portions of a first face of the first IC structure and a first face of the second IC structure; and a conductive via structure, extending from a second face of the second IC structure, through the bonding interface, and at least partially into the first IC structure, wherein the conductive via structure includes: a first portion extending from the second face of the second IC structure to the bonding interface, wherein the first portion includes a first liner, a second portion extending through the bonding interface, a third portion extending in the first IC structure from the bonding interface to a first distance from the bonding interface, wherein the third portion includes a second liner, and a fourth portion extending in the first IC structure from the first distance to a second distance from the bonding interface, wherein the second distance

is greater than the first distance, the fourth portion includes the second liner, and wherein: a material composition of the second liner is different from a material composition of the first liner, or a thickness of the second liner is different from a thickness of the first liner.

37. The microelectronic assembly according to claim 36, wherein a material composition of an electrically conductive fill material in the third portion is different from a material composition of an electrically conductive fill material in the fourth portion.

38. The microelectronic assembly according to claim 36, wherein the conductive via structure includes a grain boundary between an electrically conductive fill material in the third portion and an electrically conductive fill material in the fourth portion.

39. The microelectronic assembly according to claim 36, wherein: in a plane that is substantially perpendicular to the bonding interface, a cross-section of each of the first portion and the fourth portion includes a first side and a second side substantially parallel to the first side, wherein the first side is shorter than the second side, for the cross-section of the first portion, the second side is closer to the bonding interface than the first side, and for the cross-section of the fourth portion, the second side is closer to the bonding interface than the first side.

40. The microelectronic assembly according to claim 36, wherein: in a plane that is substantially perpendicular to the bonding interface, a cross-section of each of the first portion and the fourth portion includes a first side and a second side substantially parallel to the first side, wherein the first side is shorter than the second side, and for the cross-section of the first portion, the first side is closer to the bonding interface than the second side.

41. The microelectronic assembly according to claim 40, wherein: for the cross-section of the fourth portion, the second side is closer to the bonding interface than the first side.

42. The microelectronic assembly according to claim 40, wherein: for the cross-section of the fourth portion, the first side is closer to the bonding interface than the second side.
